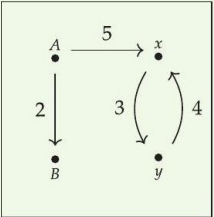


fong-spivak-invitation — formula report

4061 inline formulas (section omitted; 1 did not render), 262 display equations, 68 tables, 505 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_EQ0202 [conf 0.001]	291	0.001 U +70	G_{1}=\begin{array}{ c } \hline \text{ 李 } \\ \hline f=g \\ \end{array} \quad G_{2}=\begin{array}{ c } \hline f \\ \stackrel{\mathcal{Q}}{\downarrow} \\ a \\ \hline f=a \\ \end{array} \quad G_{3}=\begin{array}{ ccc } \bullet & \xrightarrow{f} & \bullet \\ g\downarrow & & \downarrow h \\ \bullet & \xrightarrow{i} & \bullet \\ f\circ h=g\circ i \\ \hline \end{array} \quad G_{4}=\begin{array}{ ccc } \bullet & \xrightarrow{f} & \bullet \\ g\downarrow & & \downarrow h \\ \bullet & & \bullet \\ \hline \text{no equations} \\ \end{array}	(not rendered)	
fong-spivak-invitation_EQ0062 [conf 0.001]	113	0.001 C +8	I=\left\ \begin{array}{ccc} 1 & a & 2 \\ \downarrow & & \downarrow \\ & \triangleright & 3 \\ & & 4 \\ & & b \end{array}\right.		
fong-spivak-invitation_EQ0180 [conf 0.013]	273	0.013 C +7	G=\left \begin{array}{ccc} 1 & & 2 \\ & a^{\downarrow}e^{\downarrow}_c & \\ & & v_3 \end{array}\right.		
fong-spivak-invitation_EQ0135 [conf 0.019]	215	0.019 C +8	\begin{aligned} \mu_X &:= \left(X + X \xrightarrow{\mathrm{id}_X, \mathrm{id}_X} X \xleftarrow{\mathrm{id}_X} X\right) \\ \eta_X &:= \left(\emptyset \xrightarrow{!_X} X \xleftarrow{\mathrm{id}_X} X\right) \\ \delta_X &:= \left(X \xrightarrow{\mathrm{id}_X} X \xrightarrow[\mathrm{id}_X, \mathrm{id}_X]{} X + X\right) \\ \epsilon_X &:= \left(X \xrightarrow{\mathrm{id}_X} X \xleftarrow{!_X} \emptyset\right) \end{aligned}	$\mu_X := \left(X + X \xrightarrow{[\mathrm{id}_X, \mathrm{id}_X]} X \xleftarrow{\mathrm{id}_X} X\right)$ $\eta_X := \left(\emptyset \xrightarrow{!_X} X \xleftarrow{\mathrm{id}_X} X\right)$ $\delta_X := \left(X \xrightarrow{\mathrm{id}_X} X \xrightarrow[\mathrm{id}_X, \mathrm{id}_X]{} X + X\right)$ $\epsilon_X := \left(X \xrightarrow{\mathrm{id}_X} X \xleftarrow{!_X} \emptyset\right)$	$\begin{aligned} \mu_X &:= \left(X + X \xrightarrow{[\mathrm{id}_X, \mathrm{id}_X]} X \xleftarrow{\mathrm{id}_X} X\right) \\ \eta_X &:= \left(\emptyset \xrightarrow{!_X} X \xleftarrow{\mathrm{id}_X} X\right) \\ \delta_X &:= \left(X \xrightarrow{\mathrm{id}_X} X \xleftarrow{[\mathrm{id}_X, \mathrm{id}_X]} X + X\right) \\ \epsilon_X &:= \left(X \xrightarrow{\mathrm{id}_X} X \xleftarrow{!_X} \emptyset\right) \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation-EQ0096 [conf 0.022]	146	■ 0.022 ● W -2	$\backslashoperatorname{Col}(\backslashPhi)=\backslashbegin{array}{ cc } \& \backslashstackrel{5}{\backslashlongrightarrow} \backslash \backslash 2 \backslashdownarrow \& 3 \backslashleft(\backslashbigcup_{\backslashdot{y}}^x 4 \backslashright. \backslash \backslash \backslashstackrel{\backslashbullet}{\backslashdot{a}} \& \backslash \backslash \backslashunderset{\backslashdot{y}}{a} \backslashend{array}$	$\operatorname{Col}(\Phi) = \left \begin{array}{c} 2 \downarrow \\ \bullet \\ a \\ y \end{array} \quad \begin{array}{c} \xrightarrow{5} \\ 3 \left(\bigcup_y^x 4 \right. \end{array} \right $	
fong-spivak-invitation-EQ0199 [conf 0.051]	291	■ 0.051 ● S +0	$A, \quad A \text{ : } f, \quad A \text{ : } g, \quad A \text{ : } f \text{ : } h, \quad A \text{ : } g \text{ : } i, \quad B, \quad B \text{ : } h, \quad C, \quad C \text{ : } i, \quad D$	$A, \quad A : f, \quad A : g, \quad A : f : h, \quad A : g : i, \quad B, \quad B : h, \quad C, \quad C : i, \quad D$	$A, \quad A \circ f, \quad A \circ g, \quad A \circ f \circ h, \quad A \circ g \circ i, \quad B, \quad B \circ h, \quad C, \quad C \circ i, \quad D$
fong-spivak-invitation-EQ0257 (A.2) [conf 0.064]	323	■ 0.064 ● S -1	$\backslashmultimap \backslashbecause N \backslashrightarrow{\backslashmathrm{id}} X \backslashstackrel{\backslashtext{[id,id]}}{\backslashlongleftarrow} X+X \backslashstackrel{\backslashtext{[id,id]}}{\backslashlongrightarrow} X \backslashstackrel{\backslashtext{id}}{\backslashlongleftarrow} X$	$\multimap \cdot N \xrightarrow{\text{id}} X \xleftarrow{[\text{id}, \text{id}]} X + X \xrightarrow{[\text{id}, \text{id}]} X \xleftarrow{\text{id}} X$	$\multimap \cdot \bigcirc = X \xrightarrow{\text{id}} X \xleftarrow{[\text{id}, \text{id}]} X + X \xrightarrow{[\text{id}, \text{id}]} X \xleftarrow{\text{id}} X$
fong-spivak-invitation-EQ0187 [conf 0.087]	287	■ 0.087 ● C -5	$\backslashbegin{aligned} (\backslashmathcal{X} \& \backslashtimes \backslashmathcal{Y}) \backslashleft(\backslashleft(x_{\backslash1}, y_{\backslash1} \backslashright), \backslashleft(x_{\backslash2}, y_{\backslash2} \backslashright) \backslashright) \backslashotimes (\backslashmathcal{X} \backslashtimes \backslashmathcal{Y}) \backslashleft(\backslashleft(x_{\backslash2}, y_{\backslash2} \backslashright), \backslashleft(x_{\backslash3}, y_{\backslash3} \backslashright) \backslashright) \backslash \& = \backslashmathcal{X} \backslashleft(x_{\backslash1}, x_{\backslash2} \backslashright) \backslashotimes y \backslashleft(y_{\backslash1}, y_{\backslash2} \backslashright) \backslashotimes \backslashmathcal{X} \backslashleft(x_{\backslash2}, x_{\backslash3} \backslashright) \backslashotimes y \backslashleft(y_{\backslash2}, y_{\backslash3} \backslashright) \backslash \& = \backslashmathcal{X} \backslashleft(x_{\backslash1}, x_{\backslash2} \backslashright) \backslashotimes \backslashmathcal{X} \backslashleft(x_{\backslash2}, x_{\backslash3} \backslashright) \backslashotimes y \backslashleft(y_{\backslash1}, y_{\backslash2} \backslashright) \backslashotimes y \backslashleft(y_{\backslash2}, y_{\backslash3} \backslashright) \backslash \& \leq \backslashmathcal{X} \backslashleft(x_{\backslash1}, x_{\backslash3} \backslashright) \backslashotimes y \backslashleft(y_{\backslash1}, y_{\backslash3} \backslashright) \backslash \& = (\backslashmathcal{X} \backslashtimes y) \backslashleft(\backslashleft(x_{\backslash1}, y_{\backslash1} \backslashright), \backslashleft(x_{\backslash3}, y_{\backslash3} \backslashright) \backslashright) . \backslashend{aligned}$	$(\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_2, y_2)) \otimes (\mathcal{X} \times \mathcal{Y})((x_2, y_2), (x_3, y_3)) \\ = \mathcal{X}(x_1, x_2) \otimes y(y_1, y_2) \otimes \mathcal{X}(x_2, x_3) \otimes y(y_2, y_3) \\ = \mathcal{X}(x_1, x_2) \otimes \mathcal{X}(x_2, x_3) \otimes y(y_1, y_2) \otimes y(y_2, y_3) \\ \leq \mathcal{X}(x_1, x_3) \otimes y(y_1, y_3) \\ = (\mathcal{X} \times y)((x_1, y_1), (x_3, y_3)).$	$(\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_2, y_2)) \otimes (\mathcal{X} \times \mathcal{Y})((x_2, y_2), (x_3, y_3)) \\ = \mathcal{X}(x_1, x_2) \otimes \mathcal{Y}(y_1, y_2) \otimes \mathcal{X}(x_2, x_3) \otimes \mathcal{Y}(y_2, y_3) \\ = \mathcal{X}(x_1, x_2) \otimes \mathcal{X}(x_2, x_3) \otimes \mathcal{Y}(y_1, y_2) \otimes \mathcal{Y}(y_2, y_3) \\ \leq \mathcal{X}(x_1, x_3) \otimes \mathcal{Y}(y_1, y_3) \\ = (\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_3, y_3)).$
fong-spivak-invitation-EQ0095	146	■ 0.151 ● C +37	$\backslashoperatorname{Col}(\backslashPhi)(a, b) := \backslashbegin{cases} \backslashmathcal{X}(a, b) \& \backslashtext{if } a, b \backslashin \backslashmathcal{X} \\ \backslash \backslash \backslashPhi(a, b) \& \backslashtext{if } a \backslashin \backslashmathcal{X}, b \backslashin \backslashmathcal{Y} \\ \backslash \backslash \backslashvarnothing \& \backslashtext{if } a \backslashin \backslashmathcal{X}, b \backslashin \backslashmathfrak{X} \\ y(a, b) \& \backslashtext{if } a, b \backslashin \backslashmathcal{Y} \backslashend{cases}$	$\operatorname{Col}(\Phi)(a, b) := \begin{cases} \mathcal{X}(a, b) & \text{if } a, b \in \mathcal{X} \\ \Phi(a, b) & \text{if } a \in \mathcal{X}, b \in \mathcal{Y} \\ \emptyset & \text{if } a \in \mathcal{Y}, b \in \mathfrak{X} \\ y(a, b) & \text{if } a, b \in \mathcal{Y} \end{cases}$	$\operatorname{Col}(\Phi)(a, b) := \begin{cases} \mathcal{X}(a, b) & \text{if } a, b \in \mathcal{X}, \\ \Phi(a, b) & \text{if } a \in \mathcal{X}, b \in \mathcal{Y}, \\ \emptyset & \text{if } a \in \mathcal{Y}, b \in \mathcal{X}, \\ \mathcal{Y}(a, b) & \text{if } a, b \in \mathcal{Y}. \end{cases}$
fong-spivak-invitation-EQ0085	138	■ 0.156 ● K +0	$\backslashtext{Chassis: } (\backslashtext{load} \times \backslashtext{velocity}) \backslashrightarrow (\backslashtext{torque} \times \backslashtext{speed} \times \$) \backslashtimes \$ \backslashtext{.}$	$\text{Chassis: } (\text{load} \times \text{velocity}) \rightarrow (\text{torque} \times \text{speed} \times \$).$	$\text{Chassis: } (\text{load} \times \text{velocity}) \rightarrow (\text{torque} \times \text{speed} \times \$).$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_E00201	291	■ 0.169 ● S +1	$\mathcal{D}:=\left\lvert\!\!\!\begin{array}{c} s \\ \bullet \\ z \end{array}\right\rvert$	$\mathcal{D} := \left\lvert \begin{array}{c} s \\ \bullet \\ z \end{array} \right\rvert$	$\mathcal{D} := \left\lvert \begin{array}{c} s \\ \bullet \\ z \end{array} \right\rvert$
fong-spivak-invitation_E00020 (2.3)	056	■ 0.184 ● C +25	$\begin{array}{l} h_1 := \{2\oslash, 3\oslash, 10 \\ \&, J\bullet, Q\bullet\}, \\ h_2 := \{2\oslash, 3\oslash, 4\oslash, K \\ \&, A \end{array}$	$h_1 := \{2\oslash, 3\oslash, 10 \\ \&, J\bullet, Q\bullet\}, \quad i_1 := \{4\bullet, 4\bullet, 6\oslash, 6\oslash, 10\oslash\}, \\ h_2 := \{2\oslash, 3\oslash, 4\oslash, K \\ \&, A$	$h_1 := \{2\oslash, 3\oslash, 10\clubsuit, J\clubsuit, Q\clubsuit\}, \quad i_1 := \{4\clubsuit, 4\clubsuit, 6\heartsuit, 6\heartsuit, 10\heartsuit\}, \\ h_2 := \{2\heartsuit, 3\heartsuit, 4\heartsuit, K\clubsuit, A\clubsuit\}, \quad i_2 := \{5\clubsuit, 5\heartsuit, 7\heartsuit, J\heartsuit, Q\heartsuit\}.$
fong-spivak-invitation_E00242	313	■ 0.193 ● C +12	$a_m := (1 + \cdots + 1 + \prec); \cdots; (1 + 1 + \prec);$	$a_m := (1 + \cdots + 1 + \prec); \cdots; (1 + 1 + \prec);$	$a_m := (1 + \cdots + 1 + \succ) \circ \cdots \circ (1 + 1 + \succ) \circ (1 + \succ) \circ \succ: m \rightarrow 1.$
fong-spivak-invitation_E00223	305	■ 0.204 ● N -1	$\mathcal{X}(x, z) = \mathcal{X}(x, z) \otimes I \leq \mathcal{X}(x, z) \otimes \mathcal{X}(z, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z),$	$\mathcal{X}(x, z) = \mathcal{X}(x, z) \otimes I \leq \mathcal{X}(x, z) \otimes \mathcal{X}(z, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z),$	$\mathcal{X}(x, z) = \mathcal{X}(x, z) \otimes I \leq \mathcal{X}(x, z) \otimes \mathcal{X}(z, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z),$
fong-spivak-invitation_E00070	118	■ 0.219 ● C +9	$C \underset{R}{\stackrel{L}{\Rightarrow}} D$	$C \underset{R}{\stackrel{L}{\Rightarrow}} D$	$\mathcal{C} \underset{R}{\stackrel{L}{\Rightarrow}} \mathcal{D}$
fong-spivak-invitation_E00185	282	■ 0.226 ● C -5	$\begin{array}{l} t+u \leq (v+w)+u \quad (\text{monotonicity, } t \leq v+w, u \leq u) \\ = v+(w+u) \quad (\text{associativity}) \\ \leq v+(x+z) \quad (\text{monotonicity, } v \leq v, w+u \leq x+v) \\ = (v+x)+z \quad (\text{associativity}) \\ \leq y+z. \quad (\text{monotonicity, } v+x \leq y, z \leq z) \end{array}$	$\begin{array}{l} t+u \leq (v+w)+u \quad (\text{monotonicity, } t \leq v+w, u \leq u) \\ = v+(w+u) \quad (\text{associativity}) \\ \leq v+(x+z) \quad (\text{monotonicity, } v \leq v, w+u \leq x+v) \\ = (v+x)+z \quad (\text{associativity}) \\ \leq y+z. \quad (\text{monotonicity, } v+x \leq y, z \leq z) \end{array}$	$\begin{array}{l} t+u \leq (v+w)+u \quad (\text{monotonicity, } t \leq v+w, u \leq u) \\ = v+(w+u) \quad (\text{associativity}) \\ \leq v+(x+z) \quad (\text{monotonicity, } v \leq v, w+u \leq x+v) \\ = (v+x)+z \quad (\text{associativity}) \\ \leq y+z. \quad (\text{monotonicity, } v+x \leq y, z \leq z) \end{array}$
fong-spivak-invitation_E00222	302	■ 0.229 ● W +0	$\begin{aligned} & ((\Phi; \Psi); \Upsilon)(p, s) = \bigvee_{r \in \mathcal{R}} \left(\bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \right) \otimes \Upsilon(r, s) \\ &= \bigvee_{r \in \mathcal{R}, q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \otimes \Upsilon(r, s) \\ &= \bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \bigvee_{r \in \mathcal{R}} (\Psi(q, r) \otimes \Upsilon(r, s)) \\ &= (\Phi; (\Psi; \Upsilon))(p, s). \end{aligned}$	$\begin{aligned} & ((\Phi; \Psi); \Upsilon)(p, s) = \bigvee_{r \in \mathcal{R}} \left(\bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \right) \otimes \Upsilon(r, s) \\ &= \bigvee_{r \in \mathcal{R}, q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \otimes \Upsilon(r, s) \\ &= \bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \bigvee_{r \in \mathcal{R}} (\Psi(q, r) \otimes \Upsilon(r, s)) \\ &= (\Phi; (\Psi; \Upsilon))(p, s). \end{aligned}$	$\begin{aligned} & ((\Phi; \Psi); \Upsilon)(p, s) = \bigvee_{r \in \mathcal{R}} \left(\bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \right) \otimes \Upsilon(r, s) \\ &= \bigvee_{r \in \mathcal{R}, q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \otimes \Upsilon(r, s) \\ &= \bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \bigvee_{r \in \mathcal{R}} (\Psi(q, r) \otimes \Upsilon(r, s)) \\ &= (\Phi; (\Psi; \Upsilon))(p, s). \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_E00134	213	■ 0.235 ● C +91	$\begin{array}{rrrr} \operatorname{in}(\mu) & := 2, & \operatorname{in}(\eta) & := 0, \\ \operatorname{in}(\delta) & := 1, & \operatorname{in}(\epsilon) & := 1, \\ \operatorname{out}(\mu) & := 1, & \operatorname{out}(\eta) & := 1, \\ \operatorname{out}(\delta) & := 2, & \operatorname{out}(\epsilon) & := 0. \end{array}$	$\operatorname{in}(\mu) := 2, \quad \operatorname{in}(\eta) := 0, \quad \operatorname{in}(\delta) := 1, \quad \operatorname{in}(\epsilon) := 1,$	$\begin{array}{rrrr} \operatorname{in}(\mu) & := 2, & \operatorname{in}(\eta) & := 0, \\ \operatorname{in}(\delta) & := 1, & \operatorname{in}(\epsilon) & := 1, \\ \operatorname{out}(\mu) & := 1, & \operatorname{out}(\eta) & := 1, \\ \operatorname{out}(\delta) & := 2, & \operatorname{out}(\epsilon) & := 0. \end{array}$
fong-spivak-invitation_E00116	180	■ 0.238 ● C -3	$\text{and } X \mapsto \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right) \text{ and } -a- \mapsto (a).$	$- \mapsto (1) \quad \text{and} \quad X \mapsto \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right) \quad \text{and} \quad -a- \mapsto (a).$	$\text{---} \mapsto (1) \quad \text{and} \quad \text{X} \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \text{---} \overline{a} \mapsto (a).$
fong-spivak-invitation_E00023	058	■ 0.254 ● S +0	$\overline{\frac{3.14}{-1}} = \frac{3.14}{-1} \quad \overline{0} = \text{nothing}$	$\overline{\frac{3.14}{-1}} = \frac{3.14}{-1} \quad \overline{0} = \text{nothing}$	$\frac{3.14}{-1} = \frac{3.14}{-1} \quad \frac{0}{0} = \text{nothing}$
fong-spivak-invitation_E00227	306	■ 0.259 ● C +43	$\begin{aligned} & \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x, (a, 1)) \otimes \mathrm{U}_{\mathcal{X}}(a, b) \otimes \eta_{\mathcal{X}}(1, c, d) \\ & \otimes (\epsilon_{\mathcal{X}} \times \mathrm{U}_{\mathcal{X}})((b, c, d), (1, e)) \otimes \alpha((1, e), y) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x, (a, 1)) \otimes \mathrm{U}_{\mathcal{X}}(a, b) \otimes \eta_{\mathcal{X}}(1, c, d) \\ & \otimes \epsilon_{\mathcal{X}}(b, c, 1) \otimes \mathrm{U}_{\mathcal{X}}(d, e) \otimes \alpha((1, e), y) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} \mathcal{X}(x, a) \otimes \mathcal{X}(a, b) \otimes \mathcal{X}(c, d) \otimes \mathcal{X}(b, c) \otimes \mathcal{X}(d, e) \otimes \mathcal{X}(e, y) \\ & = \mathcal{X}(x, y), \end{aligned}$	$\begin{aligned} & \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x, (a, 1)) \otimes (\mathrm{U}_{\mathcal{X}} \times \eta_{\mathcal{X}})((a, 1), (b, c, d)) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x, (a, 1)) \otimes \mathrm{U}_{\mathcal{X}}(a, b) \otimes \eta_{\mathcal{X}}(1, c, d) \otimes (\epsilon_{\mathcal{X}} \times \mathrm{U}_{\mathcal{X}})((b, c, d), (1, e)) \otimes \alpha((1, e), y) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} x(x, a) \otimes \epsilon_{\mathcal{X}}(b, c, 1) \otimes \mathrm{U}_{\mathcal{X}}(d, e) \otimes \alpha((1, e), y) \\ & = x(x, y), \end{aligned}$	$\begin{aligned} & \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x, (a, 1)) \otimes (\mathrm{U}_{\mathcal{X}} \times \eta_{\mathcal{X}})((a, 1), (b, c, d)) \\ & \otimes (\epsilon_{\mathcal{X}} \times \mathrm{U}_{\mathcal{X}})((b, c, d), (1, e)) \otimes \alpha((1, e), y) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x, (a, 1)) \otimes \mathrm{U}_{\mathcal{X}}(a, b) \otimes \eta_{\mathcal{X}}(1, c, d) \\ & \otimes \epsilon_{\mathcal{X}}(b, c, 1) \otimes \mathrm{U}_{\mathcal{X}}(d, e) \otimes \alpha((1, e), y) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} \mathcal{X}(x, a) \otimes \mathcal{X}(a, b) \otimes \mathcal{X}(c, d) \otimes \mathcal{X}(b, c) \otimes \mathcal{X}(d, e) \otimes \mathcal{X}(e, y) \\ & = \mathcal{X}(x, y), \end{aligned}$
fong-spivak-invitation_E00040	078	■ 0.267 ● W +1	$\begin{aligned} & \mathcal{C}_{\mathcal{C}}(c, d) \otimes_W \mathcal{C}_{\mathcal{C}}(d, e) = f(\mathcal{C}(c, d)) \otimes_W f(\mathcal{C}(d, e)) \\ & \leq f(\mathcal{C}(c, d) \otimes_V \mathcal{C}(d, e)) \quad (f \text{ is monoidal monotone}) \\ & \leq f(\mathcal{C}(c, e)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ & = \mathcal{C}_{\mathcal{C}}(c, e) \quad (\text{definition of } \mathcal{C}_{\mathcal{C}}). \end{aligned}$	$\begin{aligned} \mathcal{C}_f(c, d) \otimes_W \mathcal{C}_f(d, e) &= f(\mathcal{C}(c, d)) \otimes_W f(\mathcal{C}(d, e)) && (\text{definition of } \mathcal{C}_f) \\ &\leq f(\mathcal{C}(c, d) \otimes_V \mathcal{C}(d, e)) && (f \text{ is monoidal monotone}) \\ &\leq f(\mathcal{C}(c, e)) && (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ &= \mathcal{C}_f(c, e) && (\text{definition of } \mathcal{C}_f). \end{aligned}$	$\mathcal{C}_f(c, d) \otimes_W \mathcal{C}_f(d, e) = f(\mathcal{C}(c, d)) \otimes_W f(\mathcal{C}(d, e)) \quad (\text{definition of } \mathcal{C}_f)$ $\leq f(\mathcal{C}(c, d) \otimes_V \mathcal{C}(d, e)) \quad (f \text{ is monoidal monotone})$ $\leq f(\mathcal{C}(c, e)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category})$ $= \mathcal{C}_f(c, e) \quad (\text{definition of } \mathcal{C}_f).$
fong-spivak-invitation_E00256	320	■ 0.281 ● C +4	$\left[\operatorname{id}_A, !_A \right] \circ \iota_A = \left[\operatorname{id}_A \circ \iota_A, !_A \circ \iota_A \right] = [\iota_A, !_A \circ \iota_A] = [\iota_A, \iota_{\emptyset}] = \operatorname{id}_{A+\emptyset}.$	$[\operatorname{id}_A, !_A] \circ \iota_A = [\operatorname{id}_A \circ \iota_A, !_A \circ \iota_A] = [\iota_A, !_A \circ \iota_A] = [\iota_A, \iota_{\emptyset}] = \operatorname{id}_{A+\emptyset}.$	$[\operatorname{id}_A, !_A] \circ \iota_A = [\operatorname{id}_A \circ \iota_A, !_A \circ \iota_A] = [\iota_A, !_A \circ \iota_A] = [\iota_A, \iota_{\emptyset}] = \operatorname{id}_{A+\emptyset}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_ EQ0043 (2.22)	080	0.305 N +0	$(\mathcal{X} \times \mathcal{Y}) \left((x_1, y_1), (x_2, y_2) \right) \otimes (\mathcal{X} \times \mathcal{Y}) \left((x_2, y_2), (x_3, y_3) \right) \leq (\mathcal{X} \times \mathcal{Y}) \left((x_1, y_1), (x_3, y_3) \right).$	$(\mathcal{X} \times \mathcal{Y}) \left((x_1, y_1), (x_2, y_2) \right) \otimes (\mathcal{X} \times \mathcal{Y}) \left((x_2, y_2), (x_3, y_3) \right) \leq (\mathcal{X} \times \mathcal{Y}) \left((x_1, y_1), (x_3, y_3) \right).$	$(\mathcal{X} \times \mathcal{Y}) \left((x_1, y_1), (x_2, y_2) \right) \otimes (\mathcal{X} \times \mathcal{Y}) \left((x_2, y_2), (x_3, y_3) \right) \leq (\mathcal{X} \times \mathcal{Y}) \left((x_1, y_1), (x_3, y_3) \right).$
fong-spivak-invitation_ EQ0112 (5.6)	177	0.315 C +5	$G_{\{R\}} := \{\bigcirc -, \circ -, \subset, \cdots\} \cup \{-a- \mid a \in R\},$	$G_R := \{\bigcirc -, \circ -, \subset, \cdots\} \cup \{-a- \mid a \in R\},$	$G_R := \left\{ \bigcirc -, \circ -, \subset, \cdots \right\} \cup \left\{ -\boxed{a}- \mid a \in R \right\},$
fong-spivak-invitation_ EQ0059 (3.8)	108	0.323 N +0	$\mathcal{C} := \begin{array}{ c } \hline s \\ \hline Q \\ \bullet \\ z \\ \hline s; s = s \end{array}$	$\mathcal{C} := \begin{array}{ c } \hline s \\ \hline Q \\ \bullet \\ z \\ \hline s; s = s \end{array}$	$\mathcal{C} := \begin{array}{ c } \hline s \\ \hline Q \\ \bullet \\ z \\ \hline s \circ s = s \end{array}$
fong-spivak-invitation_ EQ0118	184	0.327 C +10	$G_{\{R\}} := \{\supset -, \circ -, \subset, \cdots\} \cup \{-a- \mid a \in R\},$	$G_R := \{\supset -, \circ -, \subset, \cdots\} \cup \{-a- \mid a \in R\},$	$G_R := \left\{ \supset -, \circ -, \subset, \cdots \right\} \cup \left\{ -\boxed{a}- \mid a \in R \right\},$
fong-spivak-invitation_ EQ0037	074	0.341 N +0	$X := \operatorname{Ob}(\mathcal{X}), \quad d(x, y) := \mathcal{X}(x, y).$	$X := \operatorname{Ob}(\mathcal{X}), \quad d(x, y) := \mathcal{X}(x, y).$	$X := \operatorname{Ob}(\mathcal{X}), \quad d(x, y) := \mathcal{X}(x, y).$
fong-spivak-invitation_ EQ0260	332	0.372 K +0	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} (B_i \cap Y) = \left(\bigcup_{i \in I} B_i \right) \cap Y \in \mathbf{Op}_{? \cap Y}.$	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} (B_i \cap Y) = \left(\bigcup_{i \in I} B_i \right) \cap Y \in \mathbf{Op}_{? \cap Y}.$	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} (B_i \cap Y) = \left(\bigcup_{i \in I} B_i \right) \cap Y \in \mathbf{Op}_{? \cap Y}.$
fong-spivak-invitation_ EQ0075 (3.15)	121	0.380 U +40	$\mathcal{G} := \mid \stackrel{\text{Email}}{\text{received_by}} \stackrel{\text{sent_by}}{\text{Address}}$	$(not\ rendered)$	$\mathcal{G} := \begin{array}{c} \text{Email} \xrightarrow{\text{sent_by}} \text{Address} \\ \text{received_by} \\ \text{no equations} \end{array}$
fong-spivak-invitation_ EQ0141	221	0.382 W +1	$\begin{aligned} \operatorname{Circ}(f) : \operatorname{Circ}(V) &\longrightarrow \operatorname{Circ}(V') \\ \operatorname{Circ}(V) &\longrightarrow \operatorname{Circ}(V') \\ \operatorname{Circ}(V) &\longrightarrow \operatorname{Circ}(V') \end{aligned}$	$\operatorname{Circ}(f) : \operatorname{Circ}(V) \longrightarrow \operatorname{Circ}(V')$ $(V, A, s, t, \ell) \mapsto (V', A, (s \div f), (t; f), \ell).$	$\operatorname{Circ}(f) : \operatorname{Circ}(V) \longrightarrow \operatorname{Circ}(V');$ $(V, A, s, t, \ell) \mapsto (V', A, (s \circ f), (t \circ f), \ell).$
fong-spivak-invitation_ EQ0214	299	0.401 W -2	$\begin{aligned} \Lambda(\text{monoid}, \text{nothing}) &= \Lambda(\text{monoid}, \text{this book}) \\ \Lambda(\text{preorder}, \text{nothing}) &= \Lambda(\text{preorder}, \text{this book}) \\ \Lambda(\text{category}, \text{nothing}) &= \Lambda(\text{category}, \text{this book}) \end{aligned}$	$\Lambda(\text{monoid}, \text{nothing}) = \Lambda(\text{monoid}, \text{this book})$ $= \Lambda(\text{preorder}, \text{this book}) = \Lambda(\text{category}, \text{this book}) = \text{true}$	$\Lambda(\text{monoid}, \text{nothing}) = \Lambda(\text{monoid}, \text{this book})$ $= \Lambda(\text{preorder}, \text{this book}) = \Lambda(\text{category}, \text{this book}) = \text{true}$ $\Lambda(\text{preorder}, \text{nothing}) = \Lambda(\text{category}, \text{nothing}) = \text{false}.$
fong-spivak-invitation_ EQ0024	059	0.473 N +2	$\left(\frac{x}{y} \right) = \frac{x \otimes y}{z} = \left(\frac{x}{y \otimes z} \right) = \left(\frac{y}{z} \right)$	$\left(\frac{x}{y} \right) = \frac{x \otimes y}{z} = \left\{ \frac{x}{y \otimes z} \right\} = \left\{ \frac{y}{z} \right\}$	$\left(\frac{x}{y} \right) = \frac{x \otimes y}{z} = \left\{ \frac{x}{y \otimes z} \right\} = \left\{ \frac{y}{z} \right\}$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean • not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_EQ0255	320	0.480 N +1	$\mathrm{id}_{\{A\}}+\mathrm{id}_{\{B\}}=\left[\mathrm{id}_{\{A\}}\{\circ\}^{\circ}\mathrm{id}_{\{B\}}\right]=\left[\mathrm{id}_{\{A\}}\{\circ\}^{\circ}\mathrm{id}_{\{B\}}\right]=\mathrm{id}_{\{A+B\}}.$	$\mathrm{id}_A + \mathrm{id}_B = [\mathrm{id}_A \circ \iota_A, \mathrm{id}_B \circ \iota_B] = [\iota_A, \iota_B] = \mathrm{id}_{A+B}.$	$\mathrm{id}_A + \mathrm{id}_B = [\mathrm{id}_A \circ \iota_A, \mathrm{id}_B \circ \iota_B] = [\iota_A, \iota_B] = \mathrm{id}_{A+B}.$
fong-spivak-invitation_EQ0210	296	0.483 W -2	$H:=\stackrel{4}{\underset{c}{\leftarrow}}\stackrel{d}{\rightarrow}\stackrel{5}{\rightarrow}\circ e$	$H := \stackrel{4}{\underset{c}{\leftarrow}}\stackrel{d}{\rightarrow}\stackrel{5}{\rightarrow}\circ e$	$H := \left[\begin{array}{c} 4 \\ \bullet \end{array} \xrightarrow[c]{d} \begin{array}{c} 5 \\ \bullet \end{array} \curvearrowright e$
fong-spivak-invitation_EQ0186	284	0.486 C +6	$\begin{gathered} g(\text{ false })+g(\text{ false })=\infty+\infty\geq\infty=g(\text{ false }\wedge\text{ false }), \\ g(\text{ false })+g(\text{ true })=\infty+0\geq\infty=g(\text{ false }\wedge\text{ true }), \\ g(\text{ true })+g(\text{ false })=0+\infty\geq\infty=g(\text{ true }\wedge\text{ false }), \\ g(\text{ true })+g(\text{ true })=0+0\geq0=g(\text{ true }\wedge\text{ true }) \end{gathered}$	$g(\text{ false }) + g(\text{ false }) = \infty + \infty \geq \infty = g(\text{ false } \wedge \text{ false }),$ $g(\text{ false }) + g(\text{ true }) = \infty + 0 \geq \infty = g(\text{ false } \wedge \text{ true }),$ $g(\text{ true }) + g(\text{ false }) = 0 + \infty \geq \infty = g(\text{ true } \wedge \text{ false }),$ $g(\text{ true }) + g(\text{ true }) = 0 + 0 \geq 0 = g(\text{ true } \wedge \text{ true })$	$g(\text{false}) + g(\text{false}) = \infty + \infty \geq \infty = g(\text{false} \wedge \text{false}),$ $g(\text{false}) + g(\text{true}) = \infty + 0 \geq \infty = g(\text{false} \wedge \text{true}),$ $g(\text{true}) + g(\text{false}) = 0 + \infty \geq \infty = g(\text{true} \wedge \text{false}),$ $g(\text{true}) + g(\text{true}) = 0 + 0 \geq 0 = g(\text{true} \wedge \text{true}).$
fong-spivak-invitation_EQ0092	143	0.504 C +3	$(\Phi;\Psi);\Upsilon=\Phi;(\Psi;\Upsilon).$	$(\Phi;\Psi);\Upsilon=\Phi;(\Psi;\Upsilon).$	$(\Phi;\Psi);\Upsilon=\Phi;(\Psi;\Upsilon).$
fong-spivak-invitation_EQ0077	122	0.507 U -7	$\left.\frac{\mathrm{Em}_6}{\mathrm{Em}_6}\right\lvert,$	<i>(not rendered)</i>	$\frac{\mathbf{1}}{\mathrm{Em}_6}$
fong-spivak-invitation_EQ0071	118	0.519 W +2	$\text{Set}\underset{(-)^B}{\overset{-\times B}{\rightleftarrows}}\text{Set}\quad\text{Set}(A\times B,C)\cong\text{Set}\left(A,C^B\right).$	$\text{Set}\overset{-\times B}{\underset{(-)^B}{\rightleftarrows}}\text{Set}\quad\text{Set}(A\times B,C)\cong\text{Set}\left(A,C^B\right).$	$\text{Set}\overset{-\times B}{\underset{(-)^B}{\rightleftarrows}}\text{Set}\quad\text{Set}(A\times B,C)\cong\text{Set}\left(A,C^B\right).$
fong-spivak-invitation_EQ0206	293	0.529 N +0	$F(a):=a^{\prime},\quad G(a):=a^{\prime},\quad F(b):=b^{\prime},\quad G(b):=b^{\prime},\quad F(f):=f_1,\quad\text{and}\quad G(f):=f_2.$	$F(a):=a',\quad G(a):=a',\quad F(b):=b',\quad G(b):=b',\quad F(f):=f_1,\quad\text{and}\quad G(f):=f_2.$	$F(a):=a',\quad G(a):=a',\quad F(b):=b',\quad G(b):=b',\quad F(f):=f_1,\text{ and }G(f):=f_2.$
fong-spivak-invitation_EQ0074	121	0.559 N +0	$\Sigma_!:\mathcal{C}-\text{Inst}\rightarrow\text{Set}\quad\text{and}\quad\Pi_!:\mathcal{C}-\text{Inst}\rightarrow\text{Set}.$	$\Sigma_!:\mathcal{C}-\text{Inst}\rightarrow\text{Set}\quad\text{and}\quad\Pi_!:\mathcal{C}-\text{Inst}\rightarrow\text{Set}.$	$\Sigma_!:\mathcal{C}\text{-Inst}\rightarrow\text{Set}\quad\text{and}\quad\Pi_!:\mathcal{C}\text{-Inst}\rightarrow\text{Set}.$
fong-spivak-invitation_EQ0239	313	0.561 C -4	$c_n:=-c:(1+{-c}):(1+1+{-c}):\ldots:(1+\cdots+1+<c):1\rightarrow n.$	$c_n:=-c:(1+{-c}):(1+1+{-c}):\ldots:(1+\cdots+1+<c):1\rightarrow n.$	$c_n:=-c\circ(1+<)\circ(1+1+<)\circ\ldots\circ(1+\cdots+1+<):1\rightarrow n.$
fong-spivak-invitation_EQ0073 (3.14)	121	0.576 K +0	$!:\mathcal{C}\rightarrow\mathbf{1}.$	$!:\mathcal{C}\rightarrow\mathbf{1}.$	$!:\mathcal{C}\rightarrow\mathbf{1}.$
fong-spivak-invitation_EQ0025	059	0.583 W +0	$\frac{\frac{x}{y}}{z}=\frac{\frac{x}{y}}{\frac{z}{z}}$	$\frac{\frac{x}{y}}{z}=\frac{\frac{x}{y}}{\frac{z}{z}}$	$\frac{\frac{x}{y}}{z}=\frac{\frac{x}{y}}{\frac{z}{z}}$
fong-spivak-invitation_EQ0132	206	0.584 K +0	$\left\{(v,d)\mid d\in D(v),\text{ where }v=v_1\text{ or }v=v_2\right\}.$	$\{(v,d)\mid d\in D(v),\text{ where }v=v_1\text{ or }v=v_2\}.$	$\{(v,d)\mid d\in D(v),\text{ where }v=v_1\text{ or }v=v_2\}.$
fong-spivak-invitation_EQ0057	098	0.592 N +0	$\{A,B,C,D,f,g,h,i,f;h\}.$	$\{A,B,C,D,f,g,h,i,f;h\}.$	$\{A,B,C,D,f,g,h,i,f\circ h\}.$
fong-spivak-invitation_EQ0123 (5.14)	189	0.602 K +0	$\mathbf{SFG}_R^+:=\text{Free}\left(G_R\sqcup G_R^{\text{op}}\right).$	$\mathbf{SFG}_R^+:=\text{Free}\left(G_R\sqcup G_R^{\text{op}}\right).$	$\mathbf{SFG}_R^+:=\text{Free}\left(G_R\sqcup G_R^{\text{op}}\right).$

Conf. MathPix confidence:

≥0.9

0.5–0.9

<0.5

— no value

Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_ EQ0221	301	0.622 W +0	$\begin{aligned} \Phi: \Psi &= (M_X^3 * M_\Phi * M_Y^2) * (M_Y^2 * M_\Psi * M_Z^3) \\ &= M_X^3 * M_\Phi * M_Y^4 * M_\Psi * M_Z^3 \\ &= (M_X^3 * M_\Phi * M_Y^2) * M_\Psi * M_Z^3 \\ &= \Phi * M_\Psi * M_Z^3 \end{aligned}$	$\Phi: \Psi = (M_X^3 * M_\Phi * M_Y^2) * (M_Y^2 * M_\Psi * M_Z^3) \\ = M_X^3 * M_\Phi * M_Y^4 * M_\Psi * M_Z^3 \\ = (M_X^3 * M_\Phi * M_Y^2) * M_\Psi * M_Z^3 \\ = \Phi * M_\Psi * M_Z^3$ $= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty & \infty \\ \infty & \infty & 0 & \infty \\ 4 & \infty & \infty & 4 \end{pmatrix} \begin{pmatrix} 0 & 2 & 4 & 5 \\ 4 & 0 & 2 & 3 \\ 2 & 4 & 0 & 1 \\ 1 & 3 & 5 & 0 \end{pmatrix}$ $= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty & \infty \\ 2 & 4 & 0 & 1 \\ 4 & 6 & 8 & 4 \end{pmatrix}$ $= \begin{pmatrix} 22 & 24 & 20 & 21 \\ 16 & 18 & 14 & 15 \\ 19 & 21 & 17 & 18 \\ 11 & 13 & 9 & 10 \end{pmatrix}.$	$\Phi \circ \Psi = (M_X^3 * M_\Phi * M_Y^2) * (M_Y^2 * M_\Psi * M_Z^3) \\ = M_X^3 * M_\Phi * M_Y^4 * M_\Psi * M_Z^3 \\ = (M_X^3 * M_\Phi * M_Y^2) * M_\Psi * M_Z^3 \\ = \Phi * M_\Psi * M_Z^3$ $= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty & \infty \\ \infty & \infty & 0 & \infty \\ 4 & \infty & \infty & 4 \end{pmatrix} \begin{pmatrix} 0 & 2 & 4 & 5 \\ 4 & 0 & 2 & 3 \\ 2 & 4 & 0 & 1 \\ 1 & 3 & 5 & 0 \end{pmatrix}$ $= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty & \infty \\ 2 & 4 & 0 & 1 \\ 4 & 6 & 8 & 4 \end{pmatrix}$ $= \begin{pmatrix} 22 & 24 & 20 & 21 \\ 16 & 18 & 14 & 15 \\ 19 & 21 & 17 & 18 \\ 11 & 13 & 9 & 10 \end{pmatrix}.$
fong-spivak-invitation_ EQ0010 (1.7)	044	0.623 C +21	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$
fong-spivak-invitation_ EQ0155 (7.7)	242	0.623 N +2	$X = \{(\text{true}, \text{true}), (\text{true}, \text{false}), (\text{false}, \text{true})\} \subseteq \mathbb{B} \times \mathbb{B}.$	$X = \{(\text{true}, \text{true}), (\text{true}, \text{false}), (\text{false}, \text{true})\} \subseteq \mathbb{B} \times \mathbb{B}.$	$X = \{(\text{true}, \text{true}), (\text{true}, \text{false}), (\text{false}, \text{true})\} \subseteq \mathbb{B} \times \mathbb{B}.$
fong-spivak-invitation_ EQ0156	247	0.628 N +1	$B(p; \epsilon) := \{p' \in \mathbb{R}^2 \mid d(p, p') < \epsilon\}.$	$B(p; \epsilon) := \{p' \in \mathbb{R}^2 \mid d(p, p') < \epsilon\}.$	$B(p; \epsilon) := \{p' \in \mathbb{R}^2 \mid d(p, p') < \epsilon\}.$
fong-spivak-invitation_ EQ0124	190	0.641 N +2	$\mathrm{B} \circ (h^{\mathrm{op}}) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$	$\mathrm{B} \circ (h^{\mathrm{op}}) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$	$\mathrm{B} \circ (h^{\mathrm{op}}) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$
fong-spivak-invitation_ EQ0080 (4.2)	134	0.641 C +11	$\Phi: X^{\mathrm{op}} \times y \rightarrow \mathbf{Bool}.$	$\Phi: X^{\mathrm{op}} \times y \rightarrow \mathbf{Bool}.$	$\Phi: \mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \rightarrow \mathbf{Bool}.$
fong-spivak-invitation_ EQ0097	150	0.644 C +3	$S \times (T \times U) := \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \\ = (S \times T) \times U := \{((s, t), u) \mid s \in S, t \in T, u \in U\}.$	$S \times (T \times U) := \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \\ = (S \times T) \times U := \{((s, t), u) \mid s \in S, t \in T, u \in U\}.$	$S \times (T \times U) := \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \\ = (S \times T) \times U := \{((s, t), u) \mid s \in S, t \in T, u \in U\}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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fong-spivak-invitation_EQ0224	305	0.649 N -1	$\bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, z) = \mathcal{X}(x, z).$	$\bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, z) = \mathcal{X}(x, z).$	$\bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, z) = \mathcal{X}(x, z).$
fong-spivak-invitation_EQ0055 (3.5)	096	0.649 W +1	$\mathbf{2} := \text{Free} \left(\begin{array}{ ccc } \hline v_1 & f_1 & v_2 \\ \hline \bullet & & \bullet \\ \hline \end{array} \right)$	$\mathbf{2} := \text{Free} \left(\begin{array}{ ccc } \hline v_1 & f_1 & v_2 \\ \hline \bullet & & \bullet \\ \hline \end{array} \right)$	$\mathbf{2} := \text{Free} \left(\begin{array}{ ccc } \hline v_1 & f_1 & v_2 \\ \hline \bullet & & \bullet \\ \hline \end{array} \right)$
fong-spivak-invitation_EQ0107 (5.4)	167	0.670 N +0	$G + G' := ((V \sqcup V'), [\text{in}, \text{in'}], [\text{out}, \text{out'}], (\iota \sqcup \iota')).$	$G + G' := ((V \sqcup V'), [\text{in}, \text{in'}], [\text{out}, \text{out'}], (\iota \sqcup \iota')).$	$G + G' := ((V \sqcup V'), [\text{in}, \text{in'}], [\text{out}, \text{out'}], (\iota \sqcup \iota')).$
fong-spivak-invitation_EQ0151 (7.3)	239	0.683 N +0	$\mathcal{C}(A \times C, D) \cong \mathcal{C}(A, D^C).$	$\mathcal{C}(A \times C, D) \cong \mathcal{C}(A, D^C).$	$\mathcal{C}(A \times C, D) \cong \mathcal{C}(A, D^C).$
fong-spivak-invitation_EQ0191	288	0.683 N +0	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a \right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0.$	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a \right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0.$	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a \right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0.$
fong-spivak-invitation_EQ0200	291	0.687 S +0	$A, \quad A: f, \quad A: g, \quad A: j, \quad B, \quad B: h, \quad C, \quad C: i, \quad D$	$A, \quad A: f, \quad A: g, \quad A: j, \quad B, \quad B: h, \quad C, \quad C: i, \quad D$	$A, \quad A \circ f, \quad A \circ g, \quad A \circ j, \quad B, \quad B \circ h, \quad C, \quad C \circ i, \quad D$
fong-spivak-invitation_EQ0009	142	0.688 C +7	$\mathrm{U}_{\mathcal{P}}; \Phi = \Phi = \Phi; \mathrm{U}_{\mathcal{Q}}.$	$\mathrm{U}_{\mathcal{P}}; \Phi = \Phi = \Phi; \mathrm{U}_{\mathcal{Q}}.$	$\mathrm{U}_{\mathcal{P}} \circ \Phi = \Phi = \Phi \circ \mathrm{U}_{\mathcal{Q}}.$
fong-spivak-invitation_EQ0170 (7.18)	263	0.706 K +0	$\forall (t : T). \exists (s : S). f(s) = t$	$\forall (t : T). \exists (s : S). f(s) = t$	$\forall (t : T). \exists (s : S). f(s) = t$
fong-spivak-invitation_EQ0061 (3.10)	113	0.708 C +13	$\mathbf{Gr} := \mid \stackrel{\text{Arrow}}{\text{source}} \stackrel{\text{Vertex}}{\text{target}} \bullet$	$\mathbf{Gr} := \mid \stackrel{\text{Arrow}}{\text{source}} \stackrel{\text{Vertex}}{\text{target}} \bullet$	$\mathbf{Gr} := \begin{array}{ c } \hline \begin{array}{c} \text{Arrow} \\ \bullet \end{array} \xrightarrow[\text{target}]{\text{source}} \begin{array}{c} \text{Vertex} \\ \bullet \end{array} \\ \hline \text{no equations} \end{array}$
fong-spivak-invitation_EQ0018	049	0.726 N -1	$\text{fix}_j := \{p \in P \mid j(p) \cong p\}.$	$\text{fix}_j := \{p \in P \mid j(p) \cong p\}.$	$\text{fix}_j := \{p \in P \mid j(p) \cong p\}.$
fong-spivak-invitation_EQ0149 (7.2)	234	0.727 K +0	$((A = \text{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \text{stop}) \Rightarrow \ddot{S} = 0).$	$((A = \text{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \text{stop}) \Rightarrow \ddot{S} = 0).$	$((A = \text{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \text{stop}) \Rightarrow \ddot{S} = 0).$
fong-spivak-invitation_EQ0082	135	0.730 C +8	$\Phi : x^{\text{op}} \times y \rightarrow \mathcal{V}.$	$\Phi : x^{\text{op}} \times y \rightarrow \mathcal{V}.$	$\Phi : \mathcal{X}^{\text{op}} \times \mathcal{Y} \rightarrow \mathcal{V}.$
fong-spivak-invitation_EQ0248	316	0.734 N +0	$\mathbf{B}(g_9^\circ(h^{\text{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$	$\mathbf{B}(g_9^\circ(h^{\text{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$	$\mathbf{B}(g \circ (h^{\text{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$
fong-spivak-invitation_EQ0114	178	0.740 N +2	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$
fong-spivak-invitation_EQ0138	220	0.748 N +1	$C : -\{\text{light, switch, battery}\} \sqcup \{x\Omega \mid x \in \mathbb{R}^+\}.$	$C : -\{\text{light, switch, battery}\} \sqcup \{x\Omega \mid x \in \mathbb{R}^+\}.$	$C := \{\text{light, switch, battery}\} \sqcup \{x\Omega \mid x \in \mathbb{R}^+\}.$
fong-spivak-invitation_EQ0143 (6.21)	226	0.754 C +20	$\circ_i : \mathcal{O}(s_1, \dots, s_m; t_i) \times \mathcal{O}(t_1, \dots, t_n; t) \rightarrow \mathcal{O}(t_1, \dots, t_{i-1}, s_1, \dots, s_m, t_{i+1}, \dots, t_n; t);$	$\circ_i : \mathcal{O}(s_1, \dots, s_m; t_i) \times \mathcal{O}(t_1, \dots, t_n; t) \rightarrow \mathcal{O}(t_1, \dots, t_{i-1}, s_1, \dots, s_m, t_{i+1}, \dots, t_n; t);$	$\circ_i : \mathcal{O}(s_1, \dots, s_m; t_i) \times \mathcal{O}(t_1, \dots, t_n; t) \rightarrow \mathcal{O}(t_1, \dots, t_{i-1}, s_1, \dots, s_m, t_{i+1}, \dots, t_n; t);$
Conf. MathPix confidence: 0.5-0.9 0.5-0.9 <0.5 no value					
Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_EQ0198	290	0.762 N +0	$\stackrel{\bullet}{\text{v}}_1 \bullet \xrightarrow{\text{f}_1} \stackrel{\bullet}{\text{v}}_2 \bullet \xrightarrow{\text{f}_2} \dots \xrightarrow{\text{f}_{n-1}} \stackrel{\bullet}{\text{v}}_n \bullet$	$\bullet \xrightarrow{v_1 \text{ f}_1} \bullet \xrightarrow{v_2 \text{ f}_2} \dots \xrightarrow{v_{n-1} \text{ f}_{n-1}} \bullet$	$v_1 \xrightarrow{f_1} v_2 \xrightarrow{f_2} \dots \xrightarrow{f_{n-1}} v_n$
fong-spivak-invitation_EQ0209	296	0.772 C +3	$G := \stackrel{\bullet}{1} \xrightarrow{a} \stackrel{\bullet}{2} \xrightarrow{b} \bullet$	$G := \left \begin{array}{c} \bullet \xrightarrow{1 \text{ a}} \bullet \xrightarrow{2 \text{ b}} \bullet \end{array} \right.$	$G := \left \begin{array}{c} \bullet \xrightarrow{1 \text{ a}} \bullet \xrightarrow{2 \text{ b}} \bullet \end{array} \right.$
fong-spivak-invitation_EQ0160	253	0.775 S +2	$\operatorname{Sec}_f(U) := \{g: U \rightarrow X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\}.$	$\operatorname{Sec}_f(U) := \{g: U \rightarrow X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\}.$	$\operatorname{Sec}_f(U) := \{g: U \rightarrow X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\}.$
fong-spivak-invitation_EQ0217	300	0.776 N -1	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes \Psi(y, y') \leq \Phi(x', y').$	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes \Psi(y, y') \leq \Phi(x', y').$	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes \Psi(y, y') \leq \Phi(x', y').$
fong-spivak-invitation_EQ0022 (nothing)	057	0.778 N +0	$\bar{I}$	$\bar{I}$	$\bar{I}$
fong-spivak-invitation_EQ0128	198	0.782 N +2	$\mathcal{C} := \stackrel{\bullet}{A} \xrightarrow{f} \stackrel{\bullet}{B}$	$\mathcal{C} := \stackrel{\bullet}{A} \xrightarrow{f} \stackrel{\bullet}{B}$	$\mathcal{C} := \stackrel{\bullet}{A} \xrightarrow{f} \stackrel{\bullet}{B}$
fong-spivak-invitation_EQ0148	231	0.787 N +2	$\operatorname{Circ}(\varphi): \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0).$	$\operatorname{Circ}(\varphi): \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0).$	$\operatorname{Circ}(\varphi): \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0).$
fong-spivak-invitation_EQ0003	026	0.790 C +5	$S := \begin{array}{ c c c } \hline 11 & 12 & 13 \\ \hline \bullet & \bullet & \\ \hline 21 & 22 & 23 \\ \hline \bullet & \bullet & \bullet \\ \hline \end{array}$	$S := \begin{array}{ c c c } \hline 11 & 12 & 13 \\ \hline \bullet & \bullet & \\ \hline 21 & 22 & 23 \\ \hline \bullet & \bullet & \bullet \\ \hline \end{array}$	$S := \begin{array}{ c c c } \hline 11 & 12 & 13 \\ \hline \bullet & \bullet & \\ \hline 21 & 22 & 23 \\ \hline \bullet & \bullet & \bullet \\ \hline \end{array}$
fong-spivak-invitation_EQ0066	114	0.798 W -1	$\begin{array}{ c c c } \hline \text{Arrow} & \text{source} & \text{target} \\ \hline a & 1 & 2 \\ b & 2 & 3 \\ \hline \end{array}$	$G := \begin{array}{ c c c } \hline \text{Arrow} & \text{source} & \text{target} \\ \hline a & 1 & 2 \\ b & 2 & 3 \\ \hline \end{array}$	$G := \begin{array}{ c c c } \hline \text{Arrow} & \text{source} & \text{target} \\ \hline a & 1 & 2 \\ b & 2 & 3 \\ \hline \end{array}$
fong-spivak-invitation_EQ0159 (7.10)	251	0.806 N +0	$\operatorname{Sec}_f(U) := \{s: U \rightarrow X \mid (s \circ f)(u) = u \text{ for all } u \in U\}.$	$\operatorname{Sec}_f(U) := \{s: U \rightarrow X \mid (s \circ f)(u) = u \text{ for all } u \in U\}.$	$\operatorname{Sec}_f(U) := \{s: U \rightarrow X \mid (s \circ f)(u) = u \text{ for all } u \in U\}.$
fong-spivak-invitation_EQ0139	221	0.813 N +0	$\operatorname{Circ}: (\operatorname{FinSet}, +) \longrightarrow (\operatorname{Set}, \times)$	$\operatorname{Circ}: (\operatorname{FinSet}, +) \longrightarrow (\operatorname{Set}, \times)$	$\operatorname{Circ}: (\operatorname{FinSet}, +) \longrightarrow (\operatorname{Set}, \times)$
fong-spivak-invitation_EQ0225	305	0.820 C -4	$\left(\alpha^{-1}; \alpha\right)(x, z) = \bigvee_{(y, 1) \in \mathcal{X} \times \mathbf{1}} \alpha(x, (y, 1)) \otimes \alpha^{-1}((y, 1), z) = \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) = \mathcal{X}(x, z).$	$\left(\alpha^{-1}; \alpha\right)(x, z) = \bigvee_{(y, 1) \in \mathcal{X} \times \mathbf{1}} \alpha(x, (y, 1)) \otimes \alpha^{-1}((y, 1), z) = \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) = \mathcal{X}(x, z).$	$(\alpha^{-1} \circ \alpha)(x, z) = \bigvee_{(y, 1) \in \mathcal{X} \times \mathbf{1}} \alpha(x, (y, 1)) \otimes \alpha^{-1}((y, 1), z) = \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) = \mathcal{X}(x, z).$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_EQ0100	157	0.831 N +0	$\alpha\!:\!\beta\!:=\iota_{A\sqcup C}^*\!\left(\!\left(\iota_{A\sqcup B}\right)!\!\left(\alpha\right)\vee\!\left(\iota_{B\sqcup C}\right)!\!\left(\beta\right)\!\right).$	$\alpha:\beta:=\iota_{A\sqcup C}^*((\iota_{A\sqcup B})!(\alpha)\vee(\iota_{B\sqcup C})!(\beta)).$	$\alpha\mathbin{\circ}\beta:=\iota_{A\sqcup C}^*((\iota_{A\sqcup B})!(\alpha)\vee(\iota_{B\sqcup C})!(\beta)).$
fong-spivak-invitation_EQ0083	135	0.833 N +0	$\mathcal{X}\!\left(x^{\prime},x\right)\otimes\Phi(x,y)\otimes y\!\left(y,y^{\prime}\right)\leq\Phi\!\left(x^{\prime},y^{\prime}\right).$	$\mathcal{X}(x',x)\otimes\Phi(x,y)\otimes y(y,y')\leq\Phi(x',y').$	$\mathcal{X}(x',x)\otimes\Phi(x,y)\otimes\mathcal{Y}(y,y')\leq\Phi(x',y').$
fong-spivak-invitation_EQ0021	057	0.839 W -2	$\frac{\overline{x}}{\overline{y}}\quad\overline{x\otimes y}$	$\frac{\bar{x}}{\frac{\underline{y}}{y}}\quad\overline{x\otimes y}$	$\frac{x}{\frac{y}{y}}\quad\overline{x\otimes y}$
fong-spivak-invitation_EQ0150	236	0.840 U +8	$\{p\in P\mid\text{likes_cats}(p)\}.$	<i>(not rendered)</i>	$\{p\in P\mid\text{likes_cats}(p)\}.$
fong-spivak-invitation_EQ0047	083	0.847 K +0	$2\,\mathrm{H}_{\mathrm{2}}\,\mathrm{H}_{\mathrm{0}}\rightarrow\!\!\left(2\,\mathrm{Na}\multimap(2\,\mathrm{NaOH}\!+\!\mathrm{H}_{\mathrm{2}})\right).$	$2\mathrm{H}_2\mathrm{O}\rightarrow(2\mathrm{Na}\multimap(2\mathrm{NaOH}+\mathrm{H}_2)).$	$2\mathrm{H}_2\mathrm{O}\rightarrow(2\mathrm{Na}\multimap(2\mathrm{NaOH}+\mathrm{H}_2)).$
fong-spivak-invitation_EQ0088 (4.8)	142	0.851 N +0	$\mathrm{U}_{\mathcal{X}}\!\left(\mathrm{U}_{\mathcal{X}}\right)(x,y)\!:=\!\mathfrak{X}(x,y).$	$\mathrm{U}_{\mathcal{X}}(x,y):=\mathfrak{X}(x,y).$	$\mathrm{U}_{\mathcal{X}}(x,y):=\mathfrak{X}(x,y).$
fong-spivak-invitation_EQ0090 (4.9)	142	0.868 W +1	$\begin{aligned} \Phi(p,q) &= I \otimes \Phi(p,q) \leq I \otimes \Phi(p,q) \\ &\leq \bigvee_{p_1 \in P} (\mathcal{P}(p,p_1) \otimes \Phi(p_1,q)) = (\mathrm{U}_{\mathcal{P}};\Phi)(p,q). \end{aligned}$	$\Phi(p,q) = I \otimes \Phi(p,q) \leq \mathcal{P}(p,p) \otimes \Phi(p,q) \\ \leq \bigvee_{p_1 \in P} (\mathcal{P}(p,p_1) \otimes \Phi(p_1,q)) = (\mathrm{U}_{\mathcal{P}};\Phi)(p,q).$	$\Phi(p,q) = I \otimes \Phi(p,q) \leq \mathcal{P}(p,p) \otimes \Phi(p,q) \\ \leq \bigvee_{p_1 \in P} (\mathcal{P}(p,p_1) \otimes \Phi(p_1,q)) = (\mathrm{U}_{\mathcal{P}};\Phi)(p,q).$
fong-spivak-invitation_EQ0099	152	0.880 N +0	$(f\times g);\!\left(f^{\prime}\times g^{\prime}\right)=(f\;;\;g)\times\!\left(f^{\prime}\times g^{\prime}\right).$	$(f\times g);(f'\times g')=(f;g)\times(f';g').$	$(f\times g)\mathbin{\circ}(f'\times g')=(f\mathbin{\circ}g)\times(f'\mathbin{\circ}g').$
fong-spivak-invitation_EQ0167	260	0.881 K +0	$s:S\mid p(s)\vdash q(s)\quad\text{or}\quad p(s)\vdash_{s:S}q(s).$	$s:S\mid p(s)\vdash q(s)\quad\text{or}\quad p(s)\vdash_{s:S}q(s).$	$s:S\mid p(s)\vdash q(s)\quad\text{or}\quad p(s)\vdash_{s:S}q(s).$
fong-spivak-invitation_EQ0129 (6.2)	200	0.882 S +0	$[f,g](x)=\begin{cases} f(x) & \text{if } x=\iota_A(a) \text{ for some } a\in A, \\ g(x) & \text{if } x=\iota_B(b) \text{ for some } b\in B. \end{cases}$	$[f,g](x)=\begin{cases} f(x) & \text{if } x=\iota_A(a) \text{ for some } a\in A, \\ g(x) & \text{if } x=\iota_B(b) \text{ for some } b\in B. \end{cases}$	$[f,g](x)=\begin{cases} f(x) & \text{if } x=\iota_A(a) \text{ for some } a\in A, \\ g(x) & \text{if } x=\iota_B(b) \text{ for some } b\in B. \end{cases}$
fong-spivak-invitation_EQ0144	228	0.883 K +0	$\begin{aligned} (g\circ i f)\!\left(x_{\mathrm{1}},\ldots,x_{\mathrm{i-1}},\ldots,w_{\mathrm{1}},\ldots,w_{\mathrm{m}},x_{\mathrm{i+1}},\ldots,x_{\mathrm{n}}\right) \\ =g\!\left(x_{\mathrm{1}},\ldots,x_{\mathrm{i-1}},f\!\left(w_{\mathrm{1}},\ldots,w_{\mathrm{m}}\right),x_{\mathrm{i+1}},\ldots,x_{\mathrm{n}}\right) \end{aligned}$	$(g\circ i f)(x_1,\ldots,x_{i-1},w_1,\ldots,w_m,x_{i+1},\ldots,x_n) \\ =g(x_1,\ldots,x_{i-1},f(w_1,\ldots,w_m),x_{i+1},\ldots,x_n)$	$(g\circ_i f)(x_1,\ldots,x_{i-1},w_1,\ldots,w_m,x_{i+1},\ldots,x_n) \\ =g(x_1,\ldots,x_{i-1},f(w_1,\ldots,w_m),x_{i+1},\ldots,x_n)$
fong-spivak-invitation_EQ0119	188	0.887 N -1	$\mathrm{B}(-\mathrm{C})=\{(x,(x,x))\mid x\in R\}.$	$\mathrm{B}(-\mathrm{C})=\{(x,(x,x))\mid x\in R\}.$	$\mathrm{B}(\neg) = \{(x,(x,x))\mid x\in R\}.$
fong-spivak-invitation_EQ0165 (7.16)	257	0.892 N +0	$(U\Rightarrow V)\!:=\!\bigcup_{\{R\in\mathbf{Op}\mid R\cap U\subseteq V\}}R.$	$(U\Rightarrow V):=\bigcup_{\{R\in\mathbf{Op}\mid R\cap U\subseteq V\}}R.$	$(U\Rightarrow V):=\bigcup_{\{R\in\mathbf{Op}\mid R\cap U\subseteq V\}}R.$
fong-spivak-invitation_EQ0153	241	0.910 W -1	$\ulcorner m\urcorner(y)\!:=\!\begin{cases} \text{true} & \text{if } m(x)=y \text{ for some } x\in X, \\ \text{false} & \text{otherwise.} \end{cases}$	$\ulcorner m\urcorner(y):=\begin{cases} \text{true} & \text{if } m(x)=y \text{ for some } x\in X, \\ \text{false} & \text{otherwise.} \end{cases}$	$\ulcorner m\urcorner(y):=\begin{cases} \text{true} & \text{if } m(x)=y \text{ for some } x\in X, \\ \text{false} & \text{otherwise.} \end{cases}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_ E00122 (5.13)	189	■ 0.912 ● N -2	$B_{\{1\}}; B_{\{2\}} := \left\{ (x, z) \mid \text{there exists } y \in R^n \text{ such that } (x, y) \in B_{\{1\}} \text{ and } (y, z) \in B_{\{2\}} \right\}.$	$B_1; B_2 := \{(x, z) \mid \text{there exists } y \in R^n \text{ such that } (x, y) \in B_1 \text{ and } (y, z) \in B_2\}.$	$B_1 \circ B_2 := \{(x, z) \mid \text{there exists } y \in R^n \text{ such that } (x, y) \in B_1 \text{ and } (y, z) \in B_2\}.$
fong-spivak-invitation_ E00235	312	■ 0.913 ● K +0	$A+B=\left(\begin{array}{cccccc} 3 & 3 & 1 & 0 & 0 & 0 & 0 \\ 0 & \backslash \backslash 2 & 0 & 4 & 0 & 0 & 0 & 0 & \backslash \backslash 0 & 0 & 0 & 0 & 2 & 5 & 6 & 1 \\ \end{array}\right).$	$A+B=\left(\begin{array}{cccccc} 3 & 3 & 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 & 1 \end{array}\right).$	$A+B=\left(\begin{array}{cccccc} 3 & 3 & 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 & 1 \end{array}\right).$
fong-spivak-invitation_ E00005	036	■ 0.924 ● C +5	$p=g(f(p)) \quad \text{and} \quad q=f(g(q)).$	$p = g(f(p)) \quad \text{and} \quad q = f(g(q)).$	$p = g(f(p)) \quad \text{and} \quad q = f(g(q)).$
fong-spivak-invitation_ E00193	289	■ 0.936 ● W -2	$\begin{aligned} ((M * N) * P)(w, z) &= \bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y) \right) \otimes P(y, z) \\ &\cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z) \\ &\cong \bigvee_{x \in X} M(w, x) \otimes \left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z) \right) \\ &= (M * (N * P))(w, z) \end{aligned}$	$\begin{aligned} ((M * N) * P)(w, z) &= \bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y) \right) \otimes P(y, z) \\ &\cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z) \\ &\cong \bigvee_{x \in X} M(w, x) \otimes \left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z) \right) \\ &= (M * (N * P))(w, z) \end{aligned}$	$\begin{aligned} ((M * N) * P)(w, z) &= \bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y) \right) \otimes P(y, z) \\ &\cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z) \\ &\cong \bigvee_{x \in X} M(w, x) \otimes \left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z) \right) \\ &= (M * (N * P))(w, z). \end{aligned}$
fong-spivak-invitation_ E00093	144	■ 0.937 ● C +19	$\widehat{F}(p, q) := \mathcal{Q}(F(p), q) \quad \text{and} \quad \check{F}(q, p) := \mathcal{Q}(q, F(p)).$	$\widehat{F}(p, q) := \mathcal{Q}(F(p), q) \quad \text{and} \quad \check{F}(q, p) := \mathcal{Q}(q, F(p)).$	$\widehat{F}(p, q) := \mathcal{Q}(F(p), q) \quad \text{and} \quad \check{F}(q, p) := \mathcal{Q}(q, F(p)).$
fong-spivak-invitation_ E00174 (7.19)	268	■ 0.937 ● W +0	$\begin{aligned} &\forall (t : \mathbb{R}). @_t(\text{bad_pos}(D)) \Rightarrow \\ &\exists (r : \mathbb{R}). (0 < r < 1) \wedge \forall (r' : \mathbb{R}). 0 \leq r' \leq 5 \Rightarrow @_{t+r+r'}(\text{engaged}(T)). \end{aligned}$	$\forall (t : \mathbb{R}). @_t(\text{bad_pos}(D)) \Rightarrow \\ \exists (r : \mathbb{R}). (0 < r < 1) \wedge \forall (r' : \mathbb{R}). 0 \leq r' \leq 5 \Rightarrow @_{t+r+r'}(\text{engaged}(T)).$	$\forall (t : \mathbb{R}). @_t(\text{bad_pos}(D)) \Rightarrow \\ \exists (r : \mathbb{R}). (0 < r < 1) \wedge \forall (r' : \mathbb{R}). 0 \leq r' \leq 5 \Rightarrow @_{t+r+r'}(\text{engaged}(T)).$
fong-spivak-invitation_ E00079	127	■ 0.943 ● C +3	$\begin{aligned} \lim_{\mathcal{J}} D := \{ (d_1, \dots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and} \\ \text{for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j \} \end{aligned}$	$\lim_{\mathcal{J}} D := \{(d_1, \dots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and} \\ \text{for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j\}$	$\lim_{\mathcal{J}} D := \{(d_1, \dots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and} \\ \text{for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j\}$
fong-spivak-invitation_ E00226	305	■ 0.948 ● W +1	$x \xrightarrow{\alpha^{-1}} x \times \mathbf{1} \xrightarrow{U_x \times \eta_x} x \times x^{\text{op}} \times x \xrightarrow{\epsilon_x \times U_x} \mathbf{1} \times x \xrightarrow{\alpha} x$	$x \xrightarrow{\alpha^{-1}} x \times \mathbf{1} \xrightarrow{U_x \times \eta_x} x \times x^{\text{op}} \times x \xrightarrow{\epsilon_x \times U_x} \mathbf{1} \times x \xrightarrow{\alpha} x$	$\mathcal{X} \xrightarrow{\alpha^{-1}} \mathcal{X} \times \mathbf{1} \xrightarrow{U_{\mathcal{X}} \times \eta_{\mathcal{X}}} \mathcal{X} \times \mathcal{X}^{\text{op}} \times \mathcal{X} \xrightarrow{\epsilon_{\mathcal{X}} \times U_{\mathcal{X}}} \mathbf{1} \times \mathcal{X} \xrightarrow{\alpha} \mathcal{X}$
fong-spivak-invitation_ E00145 (6.22)	228	■ 0.950 ● N +0	$(g \circ i f) : X_1 \times \cdots \times X_{i-1} \times W_1 \times \cdots \times W_m \times X_{i+1} \times \cdots \times X_n \longrightarrow Y,$	$(g \circ i f) : X_1 \times \cdots \times X_{i-1} \times W_1 \times \cdots \times W_m \times X_{i+1} \times \cdots \times X_n \longrightarrow Y,$	$(g \circ i f) : X_1 \times \cdots \times X_{i-1} \times W_1 \times \cdots \times W_m \times X_{i+1} \times \cdots \times X_n \longrightarrow Y,$
fong-spivak-invitation_ E00207	293	■ 0.954 ● S +0	$(F; G)(f; g) = G(F(f); F(g)) = G(F(f)); G(F(g))$	$(F; G)(f; g) = G(F(f); F(g)) = G(F(f)); G(F(g))$	$(F \circ G)(f \circ g) = G(F(f) \circ F(g)) = G(F(f)) \circ G(F(g))$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_EQ0048 (2.24)	084	0.954 C +13	$\left(v\otimes\bigvee_{a\in A}a\right)\cong\bigvee_{a\in A}(v\otimes a)$.	$\left(v\otimes\bigvee_{a\in A}a\right)\cong\bigvee_{a\in A}(v\otimes a).$	$\left(v\otimes\bigvee_{a\in A}a\right)\cong\bigvee_{a\in A}(v\otimes a).$
fong-spivak-invitation_EQ0219	300	0.957 W +0	$\begin{aligned} \Phi &= M_X^3 * M_\Phi * M_Y^2 = \begin{pmatrix} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty \\ 11 & \infty & \infty \\ \infty & \infty & \infty \\ \infty & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 17 & 20 & \infty \\ 11 & 14 & \infty \\ 14 & 17 & \infty \\ 20 & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix} \end{aligned}$	$\Phi = M_X^3 * M_\Phi * M_Y^2 = \begin{pmatrix} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty \\ 11 & \infty & \infty \\ \infty & \infty & \infty \\ \infty & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix} \\ = \begin{pmatrix} 17 & 20 & \infty \\ 11 & 14 & \infty \\ 14 & 17 & \infty \\ 20 & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix}$	$\Phi = M_X^3 * M_\Phi * M_Y^2 = \begin{pmatrix} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty \\ 11 & \infty & \infty \\ \infty & \infty & \infty \\ \infty & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix} \\ = \begin{pmatrix} 17 & 20 & \infty \\ 11 & 14 & \infty \\ 14 & 17 & \infty \\ 20 & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix}$
fong-spivak-invitation_EQ0072	120	0.959 N -2	$\Sigma_F(I)(\text{Airline Seat})=I(\text{Economy})\sqcup I(\text{First Class})$.	$\Sigma_F(I)(\text{Airline Seat})=I(\text{Economy})\sqcup I(\text{First Class})$.	$\Sigma_F(I)(\text{Airline Seat})=I(\text{Economy})\sqcup I(\text{First Class})$.
fong-spivak-invitation_EQ0103	159	0.962 N +0	$\eta_X\colon\mathbf{1}\times\mathcal{X}^{\mathrm{op}}\times\mathcal{X}\rightarrow\mathcal{V};$ $\mathcal{X}\times\mathcal{X}\rightarrow\mathcal{V};$	$\eta_X\colon\mathbf{1}\times\mathcal{X}^{\mathrm{op}}\times\mathcal{X}\rightarrow\mathcal{V};$	$\eta_X\colon\mathbf{1}\times\mathcal{X}^{\mathrm{op}}\times\mathcal{X}\rightarrow\mathcal{V};$
fong-spivak-invitation_EQ0046	083	0.964 N +0	$a+c\rightarrow d\quad\text{iff}\quad a\rightarrow(c\multimap d)$.	$a+c\rightarrow d\quad\text{iff}\quad a\rightarrow(c\multimap d).$	$a+c\rightarrow d\quad\text{iff}\quad a\rightarrow(c\multimap d).$
fong-spivak-invitation_EQ0110	176	0.965 N +0	$\begin{aligned} \dot{x}+3\ddot{y}-2z&=0,\\ \ddot{y}+5\dot{z}&=0. \end{aligned}$	$\begin{aligned} \dot{x}+3\ddot{y}-2z&=0,\\ \ddot{y}+5\dot{z}&=0. \end{aligned}$	$\begin{aligned} \dot{x}+3\ddot{y}-2z&=0,\\ \ddot{y}+5\dot{z}&=0. \end{aligned}$
fong-spivak-invitation_EQ0172	265	0.965 N +0	$\mathbb{R}\cong\{[d,u]\in\mathbb{R}\mid d=u\}$.	$\mathbb{R}\cong\{[d,u]\in\mathbb{R}\mid d=u\}.$	$\mathbb{R}\cong\{[d,u]\in\mathbb{R}\mid d=u\}.$
fong-spivak-invitation_EQ0216	300	0.965 N -1	$X(x',x)\otimes y(y,y')\leq\Phi(x,y)\multimap\Phi(x',y')$.	$X(x',x)\otimes y(y,y')\leq\Phi(x,y)\multimap\Phi(x',y')$.	$\mathcal{X}(x',x)\otimes\mathcal{Y}(y,y')\leq\Phi(x,y)\multimap\Phi(x',y')$.
fong-spivak-invitation_EQ0250	316	0.966 C -4	$\left\{(y,z)\mid\text{there exists }x\in R^m\text{ such that }y=S(g)(x)\text{ and }S(h)(x)=z\right\}$.	$\{(y,z)\mid\text{there exists }x\in R^m\text{ such that }y=S(g)(x)\text{ and }S(h)(x)=z\}.$	$\{(y,z)\mid\text{there exists }x\in R^m\text{ such that }y=S(g)(x)\text{ and }S(h)(x)=z\}.$
fong-spivak-invitation_EQ0168	261	0.967 N +1	$\forall (t:T).p(s,t)$.	$\forall (t:T).p(s,t).$	$\forall (t:T).p(s,t).$
fong-spivak-invitation_EQ0007	040	0.967 C +6	$f(a)\vee f(b)\supseteq f(a\vee b)$.	$f(a)\vee f(b)\not\supseteq f(a\vee b).$	$f(a)\vee f(b)\not\supseteq f(a\vee b).$
fong-spivak-invitation_EQ0101	158	0.969 N -1	$(\mathcal{X}\times\mathcal{Y})((x,y),(x',y')):=\mathcal{X}(x,x')\otimes\mathcal{Y}(y,y')$.	$(\mathcal{X}\times\mathcal{Y})((x,y),(x',y')):=\mathcal{X}(x,x')\otimes\mathcal{Y}(y,y')$.	$(\mathcal{X}\times\mathcal{Y})((x,y),(x',y')):=\mathcal{X}(x,x')\otimes\mathcal{Y}(y,y')$.
fong-spivak-invitation_EQ0094 (4.11)	145	0.970 N +0	$\mathcal{P}(p,G(q))\cong Q(F(p),q)$.	$\mathcal{P}(p,G(q))\cong Q(F(p),q).$	$\mathcal{P}(p,G(q))\cong Q(F(p),q).$
fong-spivak-invitation_EQ0213	299	0.971 N -1	$\lim_{\mathcal{J}}D:=\{(d_1,d_2)\mid d_1\in X\text{ and }d_2\in Y\},$	$\lim_{\mathcal{J}}D:=\{(d_1,d_2)\mid d_1\in X\text{ and }d_2\in Y\},$	$\lim_{\mathcal{J}}D:=\{(d_1,d_2)\mid d_1\in X\text{ and }d_2\in Y\},$
Conf. MathPix confidence: 0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_E00259	328	■0.972 ● S +0	$p \ ; \ h_{\{3\}}=r \ ; \ g \ ; \ h_{\{3\}}=r \ ; \ h_{\{2\}} \ ; \ g^{\{\backslash\prime\prime\}}=q \ ; \ f^{\{\backslash\prime\prime\}} \ ; \ g^{\{\backslash\prime\prime\}},$	$p;h_3=r;g;h_3=r;h_2;g'=q;f';g',$	$p\circ h_3=r\circ g\circ h_3=r\circ h_2\circ g'=q\circ f'\circ g',$
fong-spivak-invitation_E00152 (7.5)	241	■0.972 ● K +0	$\backslash\Omega_{\text{\text{Set}}\}\text{:}=\backslash\mathbb{B}=\backslash\{\text{\text{true}}\},\ \backslash\text{\text{true}}\ \backslash\{\text{\text{false}}\}\backslash\} \ .$	$\Omega_{\text{Set}} \text{:}=\mathbb{B}=\{\text{true},\text{false}\}.$	$\Omega_{\text{Set}} \text{:}=\mathbb{B}=\{\text{true},\text{false}\}.$
fong-spivak-invitation_E00068 (3.13)	117	■0.979 ● K +0	$f(p)\ \backslash\leq q\ \backslash\text{\text{iff}}\ \backslash\{p\ \backslash\leq g(q)\} \ .$	$f(p)\leq q\ \text{iff}\ p\leq g(q).$	$f(p)\leq q\ \text{iff}\ p\leq g(q).$
fong-spivak-invitation_E00111	176	■0.979 ● W +2	$\backslash\begin{array}{r} D \ x+3 \ D^{\{2\}} \ y-2 \ z=0 \ \backslash\backslash \ D^{\{2\}} \ y+5 \ D \ z=0 \ \backslash\end{array}$	$\begin{array}{l} Dx+3D^2y-2z=0 \\ D^2y+5Dz=0 \end{array}$	$\begin{array}{l} Dx+3D^2y-2z=0, \\ D^2y+5Dz=0, \end{array}$
fong-spivak-invitation_E00104 (5.2)	163	■0.979 ● N +0	$i \ \backslash\longmapsto \backslash\begin{cases} f(i) \ \& \ \backslash\text{\text{if}}\ \backslash\{1\ \backslash\leq i \ \backslash\leq m, \ \backslash\backslash \ m^{\{\backslash\prime\prime\}}+g(i) \ \& \ \backslash\text{\text{if}}\ \backslash\{m+1\ \backslash\leq i \ \backslash\leq m+n\} \ .\end{cases}$	$i\longmapsto\begin{cases} f(i) & \text{if } 1\leq i\leq m, \\ m'+g(i) & \text{if } m+1\leq i\leq m+n. \end{cases}$	$i\longmapsto\begin{cases} f(i) & \text{if } 1\leq i\leq m, \\ m'+g(i) & \text{if } m+1\leq i\leq m+n. \end{cases}$
fong-spivak-invitation_E00016	047	■0.980 ● N -1	$f_{\{!\}}\backslash\left(A^{\{\backslash\prime\prime\}}\backslash\right)\text{:}=\backslash\left\{b \ \backslash\in B \ \backslash\mid \backslash\text{\text{there exists}}\ \backslash\{a \ \backslash\in A^{\{\backslash\prime\prime\}}\} \ \backslash\text{\text{such that}}\ \backslash\{f(a)=b\}\backslash\right\} \ .$	$f_{!}(A'):=\{b\in B \mid \text{there exists } a\in A' \text{ such that } f(a)=b\}.$	$f_{!}(A'):=\{b\in B \mid \text{there exists } a\in A' \text{ such that } f(a)=b\}.$
fong-spivak-invitation_E00076	122	■0.982 ● N +0	$\backslash\begin{array}{c c} 1 \ \& \ \backslash\backslash \backslash\hline \backslash\text{\text{Bob-Grace-Pat-Emmy}} \ \& \ \backslash\backslash \backslash\text{\text{Sue-Doug}} \ \& \ \backslash\end{array}$	$\begin{array}{c c} 1 & \\ \hline \text{Bob-Grace-Pat-Emmy} & \\ \text{Sue-Doug} & \end{array}$	$\begin{array}{c c} \mathbf{1} & \\ \hline \text{Bob-Grace-Pat-Emmy} & \\ \text{Sue-Doug} & \end{array}$
fong-spivak-invitation_E00034	067	■0.983 ● K +0	$\backslash\text{\text{Cost}}\text{:}=(\backslash[0,\ \backslash\infty],\ \backslash\geq,\ 0,+),$	$\text{Cost} \text{:}=(\backslash[0,\infty],\geq,0,+),$	$\text{Cost} \text{:}=(\backslash[0,\infty],\geq,0,+),$
fong-spivak-invitation_E00131	206	■0.985 ● C +28	$\backslash\operatorname{colim}_{\{\mathcal{J}\}} D\text{:}=\backslash\{(v,\ d) \ \backslash\mid v \ \backslash\in V \ \backslash\text{\text{and}}\ \backslash\{d \ \backslash\in D(v)\}\backslash\} \ / \ \backslash\sim$	$\operatorname{colim}_{\mathcal{J}} D:=\{(v,d) \mid v\in V \text{ and } d\in D(v)\}/\sim$	$\operatorname{colim}_{\mathcal{J}} D:=\{(v,d) \mid v\in V \text{ and } d\in D(v)\}/\sim$
fong-spivak-invitation_E00247	316	■0.986 ● N -2	$\backslash\left\{(x,\ y) \ \backslash\mid \backslash\text{\text{there exists}}\ \backslash\{z \ \backslash\in R^{\{n\}} \ \backslash\text{\text{such that}}\ \backslash\{S(g)(x)=z \ \backslash\text{\text{and}}\ \backslash\{z=S(h)(y) \ \backslash\right\} \ .$	$\{(x,y) \mid \text{there exists } z\in R^n \text{ such that } S(g)(x)=z \text{ and } z=S(h)(y)\}.$	$\{(x,y) \mid \text{there exists } z\in R^n \text{ such that } S(g)(x)=z \text{ and } z=S(h)(y)\}.$
fong-spivak-invitation_E00017	047	■0.988 ● K +0	$f_{\{*\}}\backslash\left(A^{\{\backslash\prime\prime\}}\backslash\right)\text{:}=\backslash\left\{b \ \backslash\in B \ \backslash\mid \backslash\text{\text{for all}}\ \backslash\{a \ \backslash\text{\text{such that}}\ \backslash\{f(a)=b, \ \backslash\text{\text{we have}}\ \backslash\{a \ \backslash\in A^{\{\backslash\prime\prime\}}\}\backslash\right\} \ .$	$f_*(A'):=\{b\in B \mid \text{for all } a \text{ such that } f(a)=b, \text{ we have } a\in A'\}.$	$f_*(A'):=\{b\in B \mid \text{for all } a \text{ such that } f(a)=b, \text{ we have } a\in A'\}.$
fong-spivak-invitation_E00069	118	■0.988 ● C +8	$\backslash\alpha_{c,\ d}\text{:} \backslash\mathcal{C}(c,\ R(d)) \ \backslash\stackrel{\backslash\cong}{\longrightarrow} \backslash\mathcal{D}(L(c),\ d)$	$\alpha_{c,d}:\mathcal{C}(c,R(d))\stackrel{\cong}{\rightarrow}\mathcal{D}(L(c),d)$	$\alpha_{c,d}:\mathcal{C}(c,R(d))\stackrel{\cong}{\rightarrow}\mathcal{D}(L(c),d)$
fong-spivak-invitation_E00228	307	■0.992 ● K +0	$\backslash\sigma_{m,\ n}(i)\text{:}=\backslash\begin{cases} i+n \ \& \ \backslash\text{\text{if}}\ \backslash\{i \ \backslash\leq m, \ \backslash\backslash \ i-m \ \& \ \backslash\text{\text{if}}\ \backslash\{m+1\ \backslash\leq i\} \ .\end{cases}$	$\sigma_{m,n}(i):=\begin{cases} i+n & \text{if } i\leq m, \\ i-m & \text{if } m+1\leq i. \end{cases}$	$\sigma_{m,n}(i):=\begin{cases} i+n & \text{if } i\leq m, \\ i-m & \text{if } m+1\leq i. \end{cases}$
fong-spivak-invitation_E00126 (and)	191	■0.992 ● K +0	$\backslash\{(\{0,(x,\ x)\} \ \backslash\mid x \ \backslash\in R\} \ \backslash\subseteq R^{\{0\}} \ \backslash\times R^{\{2\}} \ \backslash\quad \backslash\text{\text{and}} \ \backslash\quad \backslash\{((x,\ x),\ 0) \ \backslash\mid x \ \backslash\in R\} \ \backslash\subseteq R^2 \times R^0 \ .$	$\{(0,(x,x)) \mid x\in R\}\subseteq R^0\times R^2 \quad \text{and} \quad \{((x,x),0) \mid x\in R\}\subseteq R^2\times R^0.$	$\{(0,(x,x)) \mid x\in R\}\subseteq R^0\times R^2 \quad \text{and} \quad \{((x,x),0) \mid x\in R\}\subseteq R^2\times R^0.$
fong-spivak-invitation_E00161 (7.13)	255	■0.992 ● K +0	$\backslash\Omega(U)\text{:}=\backslash\left\{U^{\{\backslash\prime\prime\}} \ \backslash\in \backslash\mathbf{0\ p} \ \backslash\mid U^{\{\backslash\prime\prime\}} \ \backslash\subseteq U^{\{\backslash\prime\prime\}} \ \backslash\right\} \ .$	$\Omega(U):=\{U'\in\mathbf{Op} \mid U'\subseteq U\}.$	$\Omega(U):=\{U'\in\mathbf{Op} \mid U'\subseteq U\}.$
fong-spivak-invitation_E00042	079	■0.993 ● K +0	$\backslash\mathcal{X}\backslash\left(x_{\{1\}},\ x_{\{2\}}\backslash\right) \ \backslash\leq y \backslash\left(F \backslash\left(x_{\{1\}}\backslash\right),\ F\backslash\left(x_{\{2\}}\backslash\right)\backslash\right),$	$\mathcal{X}(x_1,x_2)\leq y(F(x_1),F(x_2)),$	$\mathcal{X}(x_1,x_2)\leq y(F(x_1),F(x_2)),$
fong-spivak-invitation_E00164 (7.15)	257	■0.994 ● K +0	$(U \ \backslash\wedge V)\text{:}:=U \ \backslash\cap V \ \backslash\quad \backslash\text{\text{and}} \ \backslash\quad \backslash\quad (U \ \backslash\vee V)\text{:}:=U \ \backslash\cup V \ .$	$(U\wedge V):=U\cap V \quad \text{and} \quad (U\vee V):=U\cup V.$	$(U\wedge V):=U\cap V \quad \text{and} \quad (U\vee V):=U\cup V.$
fong-spivak-invitation_E00171	265	■0.994 ● N +0	$\backslash\mathbf{I}\backslash\backslash\mathbf{R}\text{:}=\backslash\{[d,\ u] \ \backslash\subseteq \backslash\mathbb{R} \ \backslash\mid d \ \backslash\leq u\} \ .$	$\mathbb{IR}:=\{[d,u]\subseteq\mathbb{R} \mid d\leq u\}.$	$\mathbb{IR}:=\{[d,u]\subseteq\mathbb{R} \mid d\leq u\}.$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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fong-spivak-invitation-EQ0192	289	■ 0.994 ● N +1	$\begin{aligned} & I_X * M(x, y) \&=\bigvee_{x^{\prime}\in X} I_X(x, x^{\prime}) \otimes M(x^{\prime}, y) \\ &= (I_X(x, x) \otimes M(x, y)) \vee \left(\bigvee_{x^{\prime}\in X, x^{\prime}\neq x} I_X(x, x^{\prime}) \otimes M(x^{\prime}, y) \right) \\ &= (I \otimes M(x, y)) \vee \left(\bigvee_{x^{\prime}\in X, x^{\prime}\neq x} 0 \otimes M(x^{\prime}, y) \right) \\ &= M(x, y) \vee 0 = M(x, y). \end{aligned}$	$I_X * M(x, y) = \bigvee_{x' \in X} I_X(x, x') \otimes M(x', y)$ $= (I_X(x, x) \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} I_X(x, x') \otimes M(x', y) \right)$ $= (I \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} 0 \otimes M(x', y) \right)$ $= M(x, y) \vee 0 = M(x, y)$	$I_X * M(x, y) = \bigvee_{x' \in X} I_X(x, x') \otimes M(x', y)$ $= (I_X(x, x) \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} I_X(x, x') \otimes M(x', y) \right)$ $= (I \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} 0 \otimes M(x', y) \right)$ $= M(x, y) \vee 0 = M(x, y).$
fong-spivak-invitation-EQ0113 (5.7)	178	■ 0.995 ● N +0	$M \ ; \ N(a, \ c):=\sum_{b \ \mathrm{in} \ \underline{n}} M(a, \ b) \ \times \ N(b, \ c) \ .$	$M; N(a, c) := \sum_{b \in \underline{n}} M(a, b) \times N(b, c).$	$M \circ N(a, c) := \sum_{b \in \underline{n}} M(a, b) \times N(b, c).$
fong-spivak-invitation-EQ0056 (3.6)	096	■ 0.995 ● N +2	$\mathbf{3} := \text{Free} \mid \bullet \xrightarrow{f_1} \bullet \xrightarrow{f_2} \bullet$	$\mathbf{3} := \text{Free} \mid \bullet \xrightarrow{f_1} \bullet \xrightarrow{f_2} \bullet$	$\mathbf{3} := \text{Free} \mid \bullet \xrightarrow{f_1} \bullet \xrightarrow{f_2} \bullet$
fong-spivak-invitation-EQ0195	290	■ 0.996 ● K +0	$t(p) := \begin{cases} v & \text{if } n = 0, \\ t(a_n) & \text{if } n \geq 1. \end{cases}$	$t(p) := \begin{cases} v & \text{if } n = 0, \\ t(a_n) & \text{if } n \geq 1. \end{cases}$	$t(p) := \begin{cases} v & \text{if } n = 0, \\ t(a_n) & \text{if } n \geq 1. \end{cases}$
fong-spivak-invitation-EQ0002 (1.5)	023	■ 0.996 ● C +3	$A = \bigcup_{p \in P} A_p \quad \text{and} \quad \text{if } p \neq q \text{ then } A_p \cap A_q = \emptyset.$	$A = \bigcup_{p \in P} A_p \quad \text{and} \quad \text{if } p \neq q \text{ then } A_p \cap A_q = \emptyset.$	$A = \bigcup_{p \in P} A_p \quad \text{and} \quad \text{if } p \neq q \text{ then } A_p \cap A_q = \emptyset.$
fong-spivak-invitation-EQ0189	288	■ 0.996 ● K +0	$A \cap B \subseteq (\bar{B} \cup C) \cap B = (\bar{B} \cap B) \cup (C \cap B) = C \cap B \subseteq C.$	$A \cap B \subseteq (\bar{B} \cup C) \cap B = (\bar{B} \cap B) \cup (C \cap B) = C \cap B \subseteq C.$	$A \cap B \subseteq (\bar{B} \cup C) \cap B = (\bar{B} \cap B) \cup (C \cap B) = C \cap B \subseteq C.$
fong-spivak-invitation-EQ0136	217	■ 0.996 ● C +15	$\varphi_{c_1, c_2} : F(c_1) \otimes_{\mathcal{D}} F(c_2) \rightarrow F(c_1 \otimes_{\mathcal{C}} c_2),$	$\varphi_{c_1, c_2} : F(c_1) \otimes_{\mathcal{D}} F(c_2) \rightarrow F(c_1 \otimes_{\mathcal{C}} c_2),$	$\varphi_{c_1, c_2} : F(c_1) \otimes_{\mathcal{D}} F(c_2) \rightarrow F(c_1 \otimes_{\mathcal{C}} c_2),$
fong-spivak-invitation-EQ0009	042	■ 0.997 ● N +0	$\lceil x/3 \rceil \leq y \text{ if and only if } x \leq 3y.$	$\lceil x/3 \rceil \leq y \text{ if and only if } x \leq 3y.$	$\lceil x/3 \rceil \leq y \text{ if and only if } x \leq 3y.$
fong-spivak-invitation-EQ0053	087	■ 0.997 ● C +7	$\begin{pmatrix} \text{false} & \text{false} \\ \text{false} & \text{true} \\ \text{true} & \text{true} \end{pmatrix} * \begin{pmatrix} \text{true} & \text{true} & \text{false} \\ \text{true} & \text{false} & \text{true} \end{pmatrix} = \begin{pmatrix} \text{false} & \text{false} & \text{false} \\ \text{true} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \end{pmatrix}$	$\begin{pmatrix} \text{false} & \text{false} \\ \text{false} & \text{true} \\ \text{true} & \text{true} \end{pmatrix} * \begin{pmatrix} \text{true} & \text{true} & \text{false} \\ \text{true} & \text{false} & \text{true} \end{pmatrix} = \begin{pmatrix} \text{false} & \text{false} & \text{false} \\ \text{true} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \end{pmatrix}$	$\begin{pmatrix} \text{false} & \text{false} \\ \text{false} & \text{true} \\ \text{true} & \text{true} \end{pmatrix} * \begin{pmatrix} \text{true} & \text{true} & \text{false} \\ \text{true} & \text{false} & \text{true} \end{pmatrix} = \begin{pmatrix} \text{false} & \text{false} & \text{false} \\ \text{true} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \end{pmatrix}$
fong-spivak-invitation-EQ0231	310	■ 0.997 ● N +0	$[], \quad [a], \quad [b], \quad [a, a], \quad [a, b], \quad \dots, \quad [b, a, b, b, a, b, a, a, a, a], \quad \dots$	$[], \quad [a], \quad [b], \quad [a, a], \quad [a, b], \quad \dots, \quad [b, a, b, b, a, b, a, a, a, a], \quad \dots$	$[], \quad [a], \quad [b], \quad [a, a], \quad [a, b], \quad \dots, \quad [b, a, b, b, a, b, a, a, a, a], \quad \dots$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_EQ0049	086	0.997 N +1	$d(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v), \quad \quad X(U, V) := \bigwedge_{u \in U} \bigvee_{v \in V} X(u, v)$	$d(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v), \quad X(U, V) := \bigwedge_{u \in U} \bigvee_{v \in V} X(u, v)$	$d(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v), \quad \quad \mathcal{X}(U, V) := \bigwedge_{u \in U} \bigvee_{v \in V} \mathcal{X}(u, v).$
fong-spivak-invitation_EQ0215	300	0.997 N +0	$\left(\mathrm{X}^{\mathrm{op}} \times \mathrm{Y}\right)\left((x, y),\left(x^{\prime}, y^{\prime}\right)\right) \leq \mathcal{V}\left(\Phi(x, y), \Phi\left(x^{\prime}, y^{\prime}\right)\right) .$	$\left(\mathcal{X}^{\mathrm{op}} \times \mathrm{Y}\right)\left((x, y),\left(x^{\prime}, y^{\prime}\right)\right) \leq \mathcal{V}\left(\Phi(x, y), \Phi\left(x^{\prime}, y^{\prime}\right)\right) .$	$\left(\mathcal{X}^{\mathrm{op}} \times \mathrm{Y}\right)\left((x, y),\left(x^{\prime}, y^{\prime}\right)\right) \leq \mathcal{V}\left(\Phi(x, y), \Phi\left(x^{\prime}, y^{\prime}\right)\right) .$
fong-spivak-invitation_EQ0108	169	0.998 N +0	$a \subset \bullet \supset b$	$a \subset \bullet \supset b$	$a \subset \bullet \supset b$
fong-spivak-invitation_EQ0166 (7.17)	260	0.998 K +0	$\left(\left \Omega^S\right ,\leq^S\right) .$	$\left(\left \Omega^S\right ,\leq^S\right) .$	$\left(\left \Omega^S\right ,\leq^S\right) .$
fong-spivak-invitation_EQ0011	045	0.998 C +14	$g(\bigwedge A) \cong \bigwedge g(A) \; .$	$g(\bigwedge A) \cong \bigwedge g(A).$	$g\left(\bigwedge A\right) \cong \bigwedge g(A).$
fong-spivak-invitation_EQ0106	166	0.998 K +0	$\mathrm{iota}^{\mathrm{\prime}}(\mathrm{x})=\begin{cases}\mathrm{iota}(\mathrm{x})\,\&\,\mathrm{if}\,\,\mathrm{x}\in I,\\ \mathrm{iota}^{\mathrm{\prime}}(\mathrm{x})\,\,\,\,\,\mathrm{if}\,\,\mathrm{x}\in \underline{n},\\ \mathrm{iota}^{\mathrm{\prime}}(\mathrm{x})\,\,\,\,\,\mathrm{if}\,\,\mathrm{x}\in O^{\mathrm{\prime}}.\end{cases}$	$\iota''(x)=\begin{cases}\iota(x) & \text{if } \iota(x)\in I, \\ \iota'(\iota(x)) & \text{if } \iota(x)\in \underline{n}, \\ \iota'(x) & \text{if } x\in O'.\end{cases}$	$\iota''(x)=\begin{cases}\iota(x) & \text{if } \iota(x)\in I, \\ \iota'(\iota(x)) & \text{if } \iota(x)\in \underline{n}, \\ \iota'(x) & \text{if } x\in O'.\end{cases}$
fong-spivak-invitation_EQ0008 (1.6)	041	0.999 C +3	$f(p) \leq q \quad \text{if and only if} \quad p \leq g(q) \; .$	$f(p) \leq q \quad \text{if and only if} \quad p \leq g(q).$	$f(p) \leq q \quad \text{if and only if} \quad p \leq g(q).$
fong-spivak-invitation_EQ0212	298	0.999 W +0	$\left(c_1,d_1\right) \xrightarrow{\left(f_1,g_1\right)} \left(c_2,d_2\right) \xrightarrow{\left(f_2,g_2\right)} \left(c_3,d_3\right) \xrightarrow{\left(f_3,g_3\right)} \left(c_4,d_4\right) .$	$\left(c_1,d_1\right) \xrightarrow{\left(f_1,g_1\right)} \left(c_2,d_2\right) \xrightarrow{\left(f_2,g_2\right)} \left(c_3,d_3\right) \xrightarrow{\left(f_3,g_3\right)} \left(c_4,d_4\right) .$	$\left(c_1,d_1\right) \xrightarrow{\left(f_1,g_1\right)} \left(c_2,d_2\right) \xrightarrow{\left(f_2,g_2\right)} \left(c_3,d_3\right) \xrightarrow{\left(f_3,g_3\right)} \left(c_4,d_4\right) .$
fong-spivak-invitation_EQ0232	310	0.999 N +0	$G:=\left\{\rho_{m,n}\colon m\rightarrow n\mid m,n\in\mathbb{N}\right\},\quad s\left(\rho_{m,n}\right):=m,\quad t\left(\rho_{m,n}\right):=n;$	$G:=\{\rho_{m,n}:m\rightarrow n\mid m,n\in\mathbb{N}\},\quad s(\rho_{m,n}):=m,\quad t(\rho_{m,n}):=n;$	$G:=\{\rho_{m,n}:m\rightarrow n\mid m,n\in\mathbb{N}\},\quad s(\rho_{m,n}):=m,\quad t(\rho_{m,n}):=n;$
fong-spivak-invitation_EQ0253	318	0.999 W -2	$\begin{aligned} f(m) *_R f(n) &= (\underbrace{1_R+\cdots+1_R}_{m\text{ summands}} + \underbrace{\cdots+\cdots+\cdots+1_R}_{n\text{ summands}}) *_R (\underbrace{1_R+\cdots+1_R}_{m\text{ summands}} + \underbrace{\cdots+\cdots+\cdots+1_R}_{n\text{ summands}}) \\ &= \underbrace{1_R *_R (1_R+\cdots+1_R)}_{n\text{ summands}} + \cdots + \underbrace{1_R *_R (1_R+\cdots+1_R)}_{n\text{ summands}} \\ &= \underbrace{1_R+\cdots+1_R}_{mn\text{ summands}} = f(m *_R n). \end{aligned}$	$f(m) *_R f(n) = (\underbrace{1_R+\cdots+1_R}_{m\text{ summands}} + \underbrace{\cdots+\cdots+\cdots+1_R}_{n\text{ summands}}) *_R (\underbrace{1_R+\cdots+1_R}_{m\text{ summands}} + \underbrace{\cdots+\cdots+\cdots+1_R}_{n\text{ summands}}) \\ = \underbrace{1_R *_R (1_R+\cdots+1_R)}_{n\text{ summands}} + \cdots + \underbrace{1_R *_R (1_R+\cdots+1_R)}_{n\text{ summands}} \\ = \underbrace{1_R+\cdots+1_R}_{mn\text{ summands}} = f(m *_R n).$	$f(m) *_R f(n) = (\underbrace{1_R+\cdots+1_R}_{m\text{ summands}} + \underbrace{\cdots+\cdots+\cdots+1_R}_{n\text{ summands}}) *_R (\underbrace{1_R+\cdots+1_R}_{m\text{ summands}} + \underbrace{\cdots+\cdots+\cdots+1_R}_{n\text{ summands}}) \\ = \underbrace{1_R *_R (1_R+\cdots+1_R)}_{n\text{ summands}} + \cdots + \underbrace{1_R *_R (1_R+\cdots+1_R)}_{n\text{ summands}} \\ = \underbrace{1_R+\cdots+1_R}_{mn\text{ summands}} = f(m *_R n).$
fong-spivak-invitation_EQ0146	230	0.999 C +17	$(f\circ ig):=(\mathrm{id}\otimes,\ldots,\otimes\mathrm{id}\otimes g\otimes\mathrm{id}\otimes,\ldots,\otimes\mathrm{id});f,$	$(f\circ ig):=(\mathrm{id}\otimes,\ldots,\otimes\mathrm{id}\otimes g\otimes\mathrm{id}\otimes,\ldots,\otimes\mathrm{id});f,$	$(f\circ_i g):=(\mathrm{id}\otimes,\ldots,\otimes\mathrm{id}\otimes g\otimes\mathrm{id}\otimes,\ldots,\otimes\mathrm{id})\circ f,$
Conf. MathPix confidence: 0.997 0.5-0.9 <0.5 no value					
Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

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fong-spivak-invitation_E00015	047	■0.999 ●K +0	$f\left(g\left(q_{\{0\}}\right)\right)=\bigwedge\left\{q \in Q \mid g\left(q_{\{0\}}\right) \leq g(q)\right\} \leq \bigwedge\left\{q_{\{0\}}\right\}=q_{\{0\}},$	$f\left(g\left(q_0\right)\right)=\bigwedge\left\{q \in Q \mid g\left(q_0\right) \leq g(q)\right\} \leq \bigwedge\left\{q_0\right\}=q_0,$	$f\left(g\left(q_0\right)\right)=\bigwedge\left\{q \in Q \mid g\left(q_0\right) \leq g(q)\right\} \leq \bigwedge\left\{q_0\right\}=q_0,$
fong-spivak-invitation_E00081 (4.3)	134	■0.999 ●K +0	$b \wedge c \leq d \quad \text { iff } \quad b \leq(c \Rightarrow d) \quad .$	$b \wedge c \leq d \quad \text { iff } \quad b \leq(c \Rightarrow d) .$	$b \wedge c \leq d \quad \text { iff } \quad b \leq(c \Rightarrow d) .$
fong-spivak-invitation_E00091 (4.10)	142	■0.999 ●S +0	$\begin{aligned} & \mathcal{P}\left(p, p_{\{1\}}\right) \otimes \Phi\left(p_{\{1\}}, q\right) \leq \mathcal{P}\left(p, p_{\{1\}}\right) \otimes \Phi\left(p_{\{1\}}, q\right) \otimes I \\ & \leq \mathcal{P}\left(p, p_{\{1\}}\right) \otimes \Phi\left(p_{\{1\}}, q\right) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q) . \end{aligned}$	$\mathcal{P}\left(p, p_{\{1\}}\right) \otimes \Phi\left(p_{\{1\}}, q\right)=\mathcal{P}\left(p, p_{\{1\}}\right) \otimes \Phi\left(p_{\{1\}}, q\right) \otimes I \\ \leq \mathcal{P}\left(p, p_{\{1\}}\right) \otimes \Phi\left(p_{\{1\}}, q\right) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q) .$	$\mathcal{P}\left(p, p_{\{1\}}\right) \otimes \Phi\left(p_{\{1\}}, q\right)=\mathcal{P}\left(p, p_{\{1\}}\right) \otimes \Phi\left(p_{\{1\}}, q\right) \otimes I \\ \leq \mathcal{P}\left(p, p_{\{1\}}\right) \otimes \Phi\left(p_{\{1\}}, q\right) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q) .$
fong-spivak-invitation_E00157	248	■0.999 ●N +1	$\mathbf{p}_{\{1\}}:=\{\varnothing,\{1\}, X\} \quad \text { and } \quad \mathbf{p}_{\{2\}}:=\{\varnothing,\{2\}, X\} \quad .$	$\mathbf{Op}_1:=\{\varnothing,\{1\}, X\} \quad \text { and } \quad \mathbf{Op}_2:=\{\varnothing,\{2\}, X\} .$	$\mathbf{Op}_1:=\{\varnothing,\{1\}, X\} \quad \text { and } \quad \mathbf{Op}_2:=\{\varnothing,\{2\}, X\} \quad .$
fong-spivak-invitation_E00183	276	■0.999 ●W +0	$f^*(P)=\left\{\left\{x \in \mathbb{Z} \mid x \leq 0\right\},\left\{x \in \mathbb{Z} \mid x \geq 1\right\}\right\} \quad \text { and } \quad f^*(Q)=\left\{\{0\},\left\{x \in \mathbb{Z} \mid x \neq 0\right\}\right\} .$	$f^*(P)=\left\{\left\{x \in \mathbb{Z} \mid x \leq 0\right\},\left\{x \in \mathbb{Z} \mid x \geq 1\right\}\right\} \quad \text { and } \quad f^*(Q)=\left\{\{0\},\left\{x \in \mathbb{Z} \mid x \neq 0\right\}\right\} .$	$f^*(P)=\left\{\left\{x \in \mathbb{Z} \mid x \leq 0\right\},\left\{x \in \mathbb{Z} \mid x \geq 1\right\}\right\} \quad \text { and } \quad f^*(Q)=\left\{\{0\},\left\{x \in \mathbb{Z} \mid x \neq 0\right\}\right\} .$
fong-spivak-invitation_E00245	316	■0.999 ●N -1	$B+C:=\left\{(w, y, x, z) \in R^{m+p} \times R^{n+q} \mid(w, x) \in B \text { and }(y, z) \in C\right\} \quad .$	$B+C:=\left\{(w, y, x, z) \in R^{m+p} \times R^{n+q} \mid(w, x) \in B \text { and }(y, z) \in C\right\} \quad .$	$B+C:=\left\{(w, y, x, z) \in R^{m+p} \times R^{n+q} \mid(w, x) \in B \text { and }(y, z) \in C\right\} .$
fong-spivak-invitation_E00039	078	■0.999 ●W -2	$\begin{array}{l} I_{\{W\}} \leq f\left(I_{\{V\}}\right) \\ \& \left(f \text { is monoidal monotone }\right) \\ \& \leq f\left(\mathcal{C}(c, c)\right) \\ \& =\mathcal{C}_f(c, c) \end{array}$	$\begin{array}{l} I_W \leq f\left(I_V\right) \\ \leq f\left(\mathcal{C}(c, c)\right) \\ =\mathcal{C}_f(c, c) \end{array}$	$\begin{array}{l} I_W \leq f\left(I_V\right) \\ \leq f\left(\mathcal{C}(c, c)\right) \\ =\mathcal{C}_f(c, c) \end{array}$
fong-spivak-invitation_E00142 (6.16)	221	■0.999 ●W +0	$\begin{aligned} & \psi_{V, V^{\prime}}:\operatorname{Circ}(V) \times \operatorname{Circ}\left(V^{\prime}\right) \longrightarrow \operatorname{Circ}\left(V+V^{\prime}\right) ; \\ & \left((V, A, s, t, \ell),\left(V^{\prime}, A^{\prime}, s^{\prime}, t^{\prime}, \ell^{\prime}\right)\right) \longmapsto\left(V+V^{\prime}, A+A^{\prime}, s+s^{\prime}, t+t^{\prime},\left[\ell, \ell^{\prime}\right]\right) . \end{aligned}$	$\begin{aligned} & \psi_{V, V^{\prime}}:\operatorname{Circ}(V) \times \operatorname{Circ}\left(V^{\prime}\right) \longrightarrow \operatorname{Circ}\left(V+V^{\prime}\right) ; \\ & \left((V, A, s, t, \ell),\left(V^{\prime}, A^{\prime}, s^{\prime}, t^{\prime}, \ell^{\prime}\right)\right) \longmapsto\left(V+V^{\prime}, A+A^{\prime}, s+s^{\prime}, t+t^{\prime},\left[\ell, \ell^{\prime}\right]\right) . \end{aligned}$	$\begin{aligned} & \psi_{V, V^{\prime}}:\operatorname{Circ}(V) \times \operatorname{Circ}\left(V^{\prime}\right) \longrightarrow \operatorname{Circ}\left(V+V^{\prime}\right) ; \\ & \left((V, A, s, t, \ell),\left(V^{\prime}, A^{\prime}, s^{\prime}, t^{\prime}, \ell^{\prime}\right)\right) \longmapsto\left(V+V^{\prime}, A+A^{\prime}, s+s^{\prime}, t+t^{\prime},\left[\ell, \ell^{\prime}\right]\right) . \end{aligned}$
fong-spivak-invitation_E00052 (2.26)	087	■1.000 ●N -1	$(M * N)(x, z):=\bigvee_{y \in Y} M(x, y) \otimes N(y, z) \quad .$	$(M * N)(x, z):=\bigvee_{y \in Y} M(x, y) \otimes N(y, z) .$	$(M * N)(x, z):=\bigvee_{y \in Y} M(x, y) \otimes N(y, z) .$
fong-spivak-invitation_E00054	087	■1.000 ●W +2	$I_{\{X\}}(x, y):=\begin{cases} I & \text { if } x=y \\ 0 & \text { if } x \neq y \end{cases}$	$I_X(x, y):=\begin{cases} I & \text { if } x=y \\ 0 & \text { if } x \neq y \end{cases}$	$I_X(x, y):=\begin{cases} I & \text { if } x=y, \\ 0 & \text { if } x \neq y . \end{cases}$
fong-spivak-invitation_E00063	113	■1.000 ●C -3	$\alpha_{\text {Vertex }}: G(\text { Vertex }) \rightarrow H(\text { Vertex }) \quad \text { and } \quad \alpha_{\text {Arrow }}: G(\text { Arrow }) \rightarrow H(\text { Arrow }) ,$	$\alpha_{\text {Vertex }}: G(\text { Vertex }) \rightarrow H(\text { Vertex }) \quad \text { and } \quad \alpha_{\text {Arrow }}: G(\text { Arrow }) \rightarrow H(\text { Arrow }) ,$	$\alpha_{\text {Vertex }}: G(\text { Vertex }) \rightarrow H(\text { Vertex }) \quad \text { and } \quad \alpha_{\text {Arrow }}: G(\text { Arrow }) \rightarrow H(\text { Arrow }) ,$
fong-spivak-invitation_E00067 (3.12)	116	■1.000 ●N +0	$\text {Gr} \xrightarrow{F} \text {DDS} \xrightarrow{I} \text {Set} .$	$\text {Gr} \xrightarrow{F} \text {DDS} \xrightarrow{I} \text {Set} .$	$\text {Gr} \xrightarrow{F} \text {DDS} \xrightarrow{I} \text {Set} .$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value					
Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

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fong-spivak-invitation-EQ0181	273	■1.000 ●K +0	$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4.$	$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4.$	$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4.$
fong-spivak-invitation-EQ0220	301	■1.000 ●N +1	$\left(\begin{array}{ccc} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{array}\right).$	$\left(\begin{array}{ccc} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{array}\right).$	$= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix}.$
fong-spivak-invitation-EQ0258	324	■1.000 ●W +0	$\begin{aligned} &\varphi_{S^{\prime}, T^{\prime}}\left(\varphi_{S, T}(A, B)\right) \\ &= \left\{\left(f(a), g(b)\right) \mid a \in A, b \in B\right\} \\ &= \operatorname{im}_{f \times g}(A \times B) \\ &= \operatorname{im}_{f \times g}\left(\varphi_{S, T}(A, B)\right) . \end{aligned}$	$\begin{aligned} \varphi_{S^{\prime}, T^{\prime}}\left(\left(\operatorname{im}_f \times \operatorname{im}_g\right)(A \times B)\right) &= \varphi_{S^{\prime}, T^{\prime}}\left(\left\{f(a) \mid a \in A\right\} \times\left\{g(b) \mid b \in B\right\}\right) \\ &= \left\{\left(f(a), g(b)\right) \mid a \in A, b \in B\right\} \\ &= \operatorname{im}_{f \times g}(A \times B) \\ &= \operatorname{im}_{f \times g}\left(\varphi_{S, T}(A, B)\right) . \end{aligned}$	$\begin{aligned} \varphi_{S^{\prime}, T^{\prime}}\left(\left(\operatorname{im}_f \times \operatorname{im}_g\right)(A \times B)\right) &= \varphi_{S^{\prime}, T^{\prime}}\left(\left\{f(a) \mid a \in A\right\} \times\left\{g(b) \mid b \in B\right\}\right) \\ &= \left\{\left(f(a), g(b)\right) \mid a \in A, b \in B\right\} \\ &= \operatorname{im}_{f \times g}(A \times B) \\ &= \operatorname{im}_{f \times g}\left(\varphi_{S, T}(A, B)\right) . \end{aligned}$
fong-spivak-invitation-EQ0001 (1.4)	020	■1.000 ●K +0	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$
fong-spivak-invitation-EQ0004	026	■1.000 ●K +0	$f(11)=a, \quad f(12)=a, \quad f(13)=b, \quad f(21)=c, \quad f(22)=d, \quad f(23)=d.$	$f(11)=a, \quad f(12)=a, \quad f(13)=b, \quad f(21)=c, \quad f(22)=d, \quad f(23)=d.$	$f(11)=a, \quad f(12)=a, \quad f(13)=b, \quad f(21)=c, \quad f(22)=d, \quad f(23)=d.$
fong-spivak-invitation-EQ0006	037	■1.000 ●K +0	$\left\{\left.\frac{1}{n+1}\right n \in \mathbb{N}\right\}=\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\} \subseteq \mathbb{R}$	$\left\{\left.\frac{1}{n+1}\right n \in \mathbb{N}\right\}=\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\} \subseteq \mathbb{R}$	$\left\{\left.\frac{1}{n+1}\right n \in \mathbb{N}\right\}=\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\} \subseteq \mathbb{R}$
fong-spivak-invitation-EQ0012	045	■1.000 ●C +7	$f\left(\bigvee A\right) \cong \bigvee f(A)$	$f\left(\bigvee A\right) \cong \bigvee f(A)$	$f\left(\bigvee A\right) \cong \bigvee f(A).$
fong-spivak-invitation-EQ0013 (1.8)	046	■1.000 ●K +0	$f(p):=\bigwedge\{q \in Q \mid p \leq g(q)\} ;$	$f(p):=\bigwedge\{q \in Q \mid p \leq g(q)\};$	$f(p):=\bigwedge\{q \in Q \mid p \leq g(q)\};$
fong-spivak-invitation-EQ0014	047	■1.000 ●N +0	$p_{\{0\}} \leq \bigwedge\left\{g(q) \in P \mid p_{\{0\}} \leq g(q)\right\} \cong g\left(\bigwedge\{q \in Q \mid p_{\{0\}} \leq g(q)\}\right)=g\left(f\left(p_{\{0\}}\right)\right),$	$p_0 \leq \bigwedge\{g(q) \in P \mid p_0 \leq g(q)\} \cong g\left(\bigwedge\{q \in Q \mid p_0 \leq g(q)\}\right)=g\left(f\left(p_0\right)\right),$	$p_0 \leq \bigwedge\{g(q) \in P \mid p_0 \leq g(q)\} \cong g\left(\bigwedge\{q \in Q \mid p_0 \leq g(q)\}\right)=g\left(f\left(p_0\right)\right),$
fong-spivak-invitation-EQ0019 (2.2)	055	■1.000 ●N +0	$m * e=m, \quad e * m=m, \quad(m * n) * p=m *(n * p)$	$m * e=m, \quad e * m=m, \quad(m * n) * p=m *(n * p)$	$m * e=m, \quad e * m=m, \quad(m * n) * p=m *(n * p)$
fong-spivak-invitation-EQ0026 (2.8)	060	■1.000 ●N +0	$t \leq v+w, \quad w+u \leq x+z, \quad v+x \leq y,$	$t \leq v+w, \quad w+u \leq x+z, \quad v+x \leq y,$	$t \leq v+w, \quad w+u \leq x+z, \quad v+x \leq y,$
fong-spivak-invitation-EQ0027 (2.9)	060	■1.000 ●N +0	$t+u \leq y+z.$	$t+u \leq y+z.$	$t+u \leq y+z.$
fong-spivak-invitation-EQ0028 (2.10)	060	■1.000 ●N +0	$t+u \leq v+w+u \leq v+x+z \leq y+z.$	$t+u \leq v+w+u \leq v+x+z \leq y+z.$	$t+u \leq v+w+u \leq v+x+z \leq y+z.$
fong-spivak-invitation-EQ0029	061	■1.000 ●K +0	$\mathrm{H}_2 \mathrm{O}, \quad \mathrm{NaCl}, \quad 2 \mathrm{NaOH}, \quad \mathrm{CH}_4+3 \mathrm{O}_2$	$\mathrm{H}_2 \mathrm{O}, \quad \mathrm{NaCl}, \quad 2 \mathrm{NaOH}, \quad \mathrm{CH}_4+3 \mathrm{O}_2$	$\mathrm{H}_2 \mathrm{O}, \quad \mathrm{NaCl}, \quad 2 \mathrm{NaOH}, \quad \mathrm{CH}_4+3 \mathrm{O}_2$
fong-spivak-invitation-EQ0030	061	■1.000 ●N +0	$2 \mathrm{H}_2 \mathrm{O}+2 \mathrm{Na} \rightarrow 2 \mathrm{NaOH}+\mathrm{H}_2.$	$2 \mathrm{H}_2 \mathrm{O}+2 \mathrm{Na} \rightarrow 2 \mathrm{NaOH}+\mathrm{H}_2.$	$2 \mathrm{H}_2 \mathrm{O}+2 \mathrm{Na} \rightarrow 2 \mathrm{NaOH}+\mathrm{H}_2.$
fong-spivak-invitation-EQ0031 (2.11)	062	■1.000 ●N +0	$y+k \rightarrow y^{\prime}+k^{\prime}, \quad x+y^{\prime} \rightarrow z^{\prime}, \quad z^{\prime}+k^{\prime} \rightarrow z+k.$	$y+k \rightarrow y^{\prime}+k^{\prime}, \quad x+y^{\prime} \rightarrow z^{\prime}, \quad z^{\prime}+k^{\prime} \rightarrow z+k.$	$y+k \rightarrow y^{\prime}+k^{\prime}, \quad x+y^{\prime} \rightarrow z^{\prime}, \quad z^{\prime}+k^{\prime} \rightarrow z+k.$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_E00032 (2.12)	062	■1.000 K +0	x+y+k \rightarrow x+y^{\prime}+k^{\prime} \rightarrow z^{ \prime }+k^{\prime} \rightarrow z+k .	$x + y + k \rightarrow x + y' + k' \rightarrow z' + k' \rightarrow z + k.$	$x + y + k \rightarrow x + y' + k' \rightarrow z' + k' \rightarrow z + k.$
fong-spivak-invitation_E00033 (2.14)	063	■1.000 K +0	\mathrm{H}_{_2}\mathrm{O}+\mathrm{NaCl}\rightarrow\mathrm{H}_{_2}\mathrm{O}	H ₂ O + NaCl → ? H ₂ O	H ₂ O + NaCl →? H ₂ O
fong-spivak-invitation_E00035	069	■1.000 W +1	d(x):=\left\{\begin{array}{ll} \text { false } & \& \text { if } x>0, \\ \text { true } & \& \text { if } x=0, \end{array}\right.\quad u(x):=\begin{cases} \text { false } & \& \text { if } x=\infty, \\ \text { true } & \& \text { if } x<\infty . \end{cases}	$d(x) := \begin{cases} \text{false} & \text{if } x > 0, \\ \text{true} & \text{if } x = 0, \end{cases} \quad u(x) := \begin{cases} \text{false} & \text{if } x = \infty, \\ \text{true} & \text{if } x < \infty. \end{cases}$	$d(x) := \begin{cases} \text{false} & \text{if } x > 0, \\ \text{true} & \text{if } x = 0, \end{cases} \quad u(x) := \begin{cases} \text{false} & \text{if } x = \infty, \\ \text{true} & \text{if } x < \infty. \end{cases}$
fong-spivak-invitation_E00036	073	■1.000 N +0	d_{L}(U,V):=\sup _{u \in U} \inf _{v \in V} d(u,v),	$d_L(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$	$d_L(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$
fong-spivak-invitation_E00038 (2.20)	076	■1.000 N +0	M_{Y}:=\begin{array}{c ccc} & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & \infty \\ z & \infty & 4 & 0 \end{array}	$M_Y := \begin{array}{c ccc} \nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & \infty \\ z & \infty & 4 & 0 \end{array}$	$M_Y := \begin{array}{c ccc} \nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & \infty \\ z & \infty & 4 & 0 \end{array}$
fong-spivak-invitation_E00041 (2.21)	078	■1.000 N +0	f(x):= \begin{cases} \text { true } & \& \text { if } x=0, \\ \text { false } & \& \text { if } x>0 . \end{cases}	$f(x) := \begin{cases} \text{true} & \text{if } x = 0, \\ \text{false} & \text{if } x > 0. \end{cases}$	$f(x) := \begin{cases} \text{true} & \text{if } x = 0, \\ \text{false} & \text{if } x > 0. \end{cases}$
fong-spivak-invitation_E00044 (2.23)	082	■1.000 C +8	(a \otimes v) \leq w \quad \& \quad \text { iff } \quad a \leq(v \multimap w)	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w)$	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w)$
fong-spivak-invitation_E00045	083	■1.000 K +0	a+x \geq y \quad \& \quad \text { iff } \quad a \geq y-x \quad \& \quad \text { iff } \quad \max (0, a) \geq \max (0, y-x) \quad \& \quad \text { iff } \quad a \geq(x \multimap y)	$a + x \geq y \quad \text{iff} \quad a \geq y - x \quad \text{iff} \quad \max(0, a) \geq \max(0, y - x) \quad \text{iff} \quad a \geq (x \multimap y)$	$a + x \geq y \quad \text{iff} \quad a \geq y - x \quad \text{iff} \quad \max(0, a) \geq \max(0, y - x) \quad \text{iff} \quad a \geq (x \multimap y)$
fong-spivak-invitation_E00050	086	■1.000 N +0	v \multimap w:=\bigvee_{\{a \in V \mid a \otimes v \leq w\}} a .	$v \multimap w := \bigvee_{\{a \in V a \otimes v \leq w\}} a.$	$v \multimap w := \bigvee_{\{a \in V a \otimes v \leq w\}} a.$
fong-spivak-invitation_E00051 (2.25)	087	■1.000 N +0	(M * N)(i, k)=\sum_{j} M(i, j) * N(j, k) .	$(M * N)(i, k) = \sum_j M(i, j) * N(j, k).$	$(M * N)(i, k) = \sum_j M(i, j) * N(j, k).$
fong-spivak-invitation_E00058	107	■1.000 S +0	f^{\prime} ; h^{\prime}=F(f) ; F(h)=F(f ; h)=F(g ; i)=F(g) ; F(i)=g^{\prime} ; i^{\prime} .	$f'; h' = F(f); F(h) = F(f; h) = F(g; i) = F(g); F(i) = g'; i'.$	$f' \circledast h' = F(f) \circledast F(h) = F(f \circledast h) = F(g \circledast i) = F(g) \circledast F(i) = g' \circledast i'.$
fong-spivak-invitation_E00060 (3.9)	110	■1.000 C +8	F(f) ; \alpha_d=\alpha_c ; G(f) .	$F(f); \alpha_d = \alpha_c; G(f).$	$F(f) \circledast \alpha_d = \alpha_c \circledast G(f).$
fong-spivak-invitation_E00078	124	■1.000 N +2	X \times Y:=\{(x, y) \mid x \in X, y \in Y\},	$X \times Y := \{(x, y) \mid x \in X, y \in Y\},$	$X \times Y := \{(x, y) \mid x \in X, y \in Y\},$
fong-spivak-invitation_E00084 (4.5)	137	■1.000 N +2	\Phi(B, x)=11, \quad \Phi(A, z)=20, \quad \Phi(C, y)=17 .	$\Phi(B, x) = 11, \quad \Phi(A, z) = 20, \quad \Phi(C, y) = 17.$	$\Phi(B, x) = 11, \quad \Phi(A, z) = 20, \quad \Phi(C, y) = 17.$
fong-spivak-invitation_E00086 (4.7)	140	■1.000 N +1	(\Phi ; \Psi)(p, r):=\bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r)	$(\Phi; \Psi)(p, r) := \bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r)$	$(\Phi \circledast \Psi)(p, r) := \bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r).$
fong-spivak-invitation_E00087	141	■1.000 C +13	(\Phi ; \Psi)(p, r)=\bigvee_{q \in Q}(\Phi(p, q) \otimes \Psi(q, r)) .	$(\Phi; \Psi)(p, r) = \bigvee_{q \in Q} (\Phi(p, q) \otimes \Psi(q, r)).$	$(\Phi \circledast \Psi)(p, r) = \bigvee_{q \in Q} (\Phi(p, q) \otimes \Psi(q, r)).$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

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fong-spivak-invitation_EQ0098	150	■ 1.000 ● N +1	$\alpha_{s,t,u}:\{(s,(t,u))\mid s\in S,t\in T,u\in U\}\xrightarrow{\cong}\{((s,t),u)\mid s\in S,t\in T,u\in U\}.$	$\alpha_{s,t,u}:\{(s,(t,u))\mid s\in S,t\in T,u\in U\}\xrightarrow{\cong}\{((s,t),u)\mid s\in S,t\in T,u\in U\}.$	$\alpha_{s,t,u}:\{(s,(t,u))\mid s\in S,t\in T,u\in U\}\xrightarrow{\cong}\{((s,t),u)\mid s\in S,t\in T,u\in U\}.$
fong-spivak-invitation_EQ0102	158	■ 1.000 ● N -1	$(\Phi\times\Psi)\left(\left(x_1,y_1\right),\left(x_2,y_2\right)\right):=\Phi\left(x_1,x_2\right)\otimes\Psi\left(y_1,y_2\right).$	$(\Phi\times\Psi)\left(\left(x_1,y_1\right),\left(x_2,y_2\right)\right):=\Phi\left(x_1,x_2\right)\otimes\Psi\left(y_1,y_2\right).$	$(\Phi\times\Psi)\left(\left(x_1,y_1\right),\left(x_2,y_2\right)\right):=\Phi\left(x_1,x_2\right)\otimes\Psi\left(y_1,y_2\right).$
fong-spivak-invitation_EQ0105	164	■ 1.000 ● N -1	$R\text{ ; }S:=\{(i,k)\in\underline{m}\times\underline{p}\mid\exists(j\in\underline{n}).(i,j)\in R\text{ and }(j,k)\in S\}.$	$R;S:=\{(i,k)\in\underline{m}\times\underline{p}\mid\exists(j\in\underline{n}).(i,j)\in R\text{ and }(j,k)\in S\}.$	$R\circ S:=\{(i,k)\in\underline{m}\times\underline{p}\mid\exists(j\in\underline{n}).(i,j)\in R\text{ and }(j,k)\in S\}.$
fong-spivak-invitation_EQ0109 (5.5)	170	■ 1.000 ● N +0	$G:=\left\{\rho_{m,n}\mid m,n\in\mathbb{N}\right\},\quad\quad s\left(\rho_{m,n}\right):=m,\quad\quad t\left(\rho_{m,n}\right):=n,$	$G:=\left\{\rho_{m,n}\mid m,n\in\mathbb{N}\right\},\quad\quad s\left(\rho_{m,n}\right):=m,\quad\quad t\left(\rho_{m,n}\right):=n,$	$G:=\left\{\rho_{m,n}\mid m,n\in\mathbb{N}\right\},\quad\quad s\left(\rho_{m,n}\right):=m,\quad\quad t\left(\rho_{m,n}\right):=n,$
fong-spivak-invitation_EQ0115	178	■ 1.000 ● K +0	$A=\left(\begin{array}{lll}3&3&1\\2&0&4\end{array}\right),\quad\quad B=\left(\begin{array}{llll}2&5&6&1\end{array}\right).$	$A=\left(\begin{array}{lll}3&3&1\\2&0&4\end{array}\right),\quad\quad B=\left(\begin{array}{llll}2&5&6&1\end{array}\right).$	$A=\left(\begin{array}{lll}3&3&1\\2&0&4\end{array}\right),\quad\quad B=\left(\begin{array}{llll}2&5&6&1\end{array}\right).$
fong-spivak-invitation_EQ0117	181	■ 1.000 ● K +0	$\sum_{j\in\underline{n}}\alpha(i,j)*\beta(j,k).$	$\sum_{j\in\underline{n}}\alpha(i,j)*\beta(j,k).$	$\sum_{j\in\underline{n}}\alpha(i,j)*\beta(j,k).$
fong-spivak-invitation_EQ0120 (5.11)	188	■ 1.000 ● K +0	$\mathrm{B}(g)=\{(x,S(g)(x))\mid x\in R^m\}\subseteq R^m\times R^n.$	$\mathrm{B}(g)=\{(x,S(g)(x))\mid x\in R^m\}\subseteq R^m\times R^n.$	$\mathrm{B}(g)=\{(x,S(g)(x))\mid x\in R^m\}\subseteq R^m\times R^n.$
fong-spivak-invitation_EQ0121 (5.12)	188	■ 1.000 ● K +0	$\mathrm{B}\left(g^{\mathrm{op}}\right):=\left\{\left(S(g)(x),x\right)\mid x\in R^m\right\}\subseteq R^n\times R^m.$	$\mathrm{B}\left(g^{\mathrm{op}}\right):=\left\{\left(S(g)(x),x\right)\mid x\in R^m\right\}\subseteq R^n\times R^m.$	$\mathrm{B}\left(g^{\mathrm{op}}\right):=\left\{\left(S(g)(x),x\right)\mid x\in R^m\right\}\subseteq R^n\times R^m.$
fong-spivak-invitation_EQ0125	190	■ 1.000 ● N +0	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right);h\right)=\left\{\left(S(g)(x),S(h)(x)\right)\mid x\in R^m\right\}.$	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right);h\right)=\left\{\left(S(g)(x),S(h)(x)\right)\mid x\in R^m\right\}.$	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right)\circ h\right)=\left\{\left(S(g)(x),S(h)(x)\right)\mid x\in R^m\right\}.$
fong-spivak-invitation_EQ0127	192	■ 1.000 ● K +0	$v=\int\frac{1}{m}F(t)dt+u(t)+p\int au(t)+bv(t)dt.$	$v=\int\frac{1}{m}F(t)dt+u(t)+p\int au(t)+bv(t)dt.$	$v=\int\frac{1}{m}F(t)dt+u(t)+p\int au(t)+bv(t)dt.$
fong-spivak-invitation_EQ0130	204	■ 1.000 ● N +0	$y(0)=y(g(0))=x(f(0))=x(0).$	$y(0)=y(g(0))=x(f(0))=x(0).$	$y(0)=y(g(0))=x(f(0))=x(0).$
fong-spivak-invitation_EQ0133	207	■ 1.000 ● K +0	$\{(v,d)\mid d\in D(v)\}.$	$\{(v,d)\mid d\in D(v)\}.$	$\{(v,d)\mid d\in D(v)\}.$
fong-spivak-invitation_EQ0137 (6.15)	219	■ 1.000 ● K +0	$F\left(\left[\iota_N,\iota_P\right]\right)\left(\varphi_{N,P}(s,t)\right).$	$F\left(\left[\iota_N,\iota_P\right]\right)\left(\varphi_{N,P}(s,t)\right).$	$F\left(\left[\iota_N,\iota_P\right]\right)\left(\varphi_{N,P}(s,t)\right).$
fong-spivak-invitation_EQ0140	221	■ 1.000 ● N +0	$\mathrm{Circ}(V):=\{(V,A,s,t,\ell)\mid\text{where }s,t:A\rightarrow V,\ell:E\rightarrow C\}.$	$\mathrm{Circ}(V):=\{(V,A,s,t,\ell)\mid\text{ where }s,t:A\rightarrow V,\ell:E\rightarrow C\}.$	$\mathrm{Circ}(V):=\{(V,A,s,t,\ell)\mid\text{ where }s,t:A\rightarrow V,\ell:E\rightarrow C\}.$
fong-spivak-invitation_EQ0147	230	■ 1.000 ● C +6	$F:\mathcal{O}(t_1,\ldots,t_n;t)\rightarrow\mathcal{P}(f(t_1),\ldots,f(t_n);f(t))$	$F:\mathcal{O}(t_1,\ldots,t_n;t)\rightarrow\mathcal{P}(f(t_1),\ldots,f(t_n);f(t))$	$F:\mathcal{O}(t_1,\ldots,t_n;t)\rightarrow\mathcal{P}(f(t_1),\ldots,f(t_n);f(t))$
fong-spivak-invitation_EQ0154 (7.6)	241	■ 1.000 ● N -1	$X=\{y\in Y\mid p(y)=\text{true}\}.$	$X=\{y\in Y\mid p(y)=\text{true}\}.$	$X=\{y\in Y\mid p(y)=\text{true}\}.$
fong-spivak-invitation_EQ0158	249	■ 1.000 ● N +1	$s_i _{U_i\cap U_j}=s_j _{U_i\cap U_j}.$	$s_i _{U_i\cap U_j}=s_j _{U_i\cap U_j}.$	$s_i _{U_i\cap U_j}=s_j _{U_i\cap U_j}.$
fong-spivak-invitation_EQ0162 (7.14)	255	■ 1.000 ● N +0	$\begin{aligned}\Omega(U)&\rightarrow\Omega(V),\\U^{\prime}&\mapsto U^{\prime}\cap V.\end{aligned}$	$\Omega(U)\rightarrow\Omega(V),\\U^{\prime}\mapsto U^{\prime}\cap V.$	$\Omega(U)\rightarrow\Omega(V),\\U^{\prime}\mapsto U^{\prime}\cap V.$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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fong-spivak-invitation_EQ0163	256	■ 1.000 ● K +0	$V \cap U_{\{j\}} = \left(\bigcup_{i \in I} V_{\{i\}} \right) \cap U_{\{j\}} = \bigcup_{i \in I} \left(V_{\{i\}} \cap U_{\{j\}} \right) = \bigcup_{i \in I} \left(V_{\{i\}} \cap U_{\{i\}} \right) = \bigcup_{i \in I} V_{\{i\}} = V_{\{j\}}.$	$V \cap U_j = \left(\bigcup_{i \in I} V_i \right) \cap U_j = \bigcup_{i \in I} (V_i \cap U_j) = \bigcup_{i \in I} (V_j \cap U_i) = \left(\bigcup_{i \in I} U_i \right) \cap V_j = V_j.$	$V \cap U_j = \left(\bigcup_{i \in I} V_i \right) \cap U_j = \bigcup_{i \in I} (V_i \cap U_j) = \bigcup_{i \in I} (V_j \cap U_i) = \left(\bigcup_{i \in I} U_i \right) \cap V_j = V_j.$
fong-spivak-invitation_EQ0169	261	■ 1.000 ● W +1	$p(n, z) = \begin{cases} \text{true} & \text{if } n \leq z , \\ \text{false} & \text{if } n > z . \end{cases}$	$p(n, z) = \begin{cases} \text{true} & \text{if } n \leq z , \\ \text{false} & \text{if } n > z . \end{cases}$	$p(n, z) = \begin{cases} \text{true} & \text{if } n \leq z , \\ \text{false} & \text{if } n > z . \end{cases}$
fong-spivak-invitation_EQ0173	266	■ 1.000 ● N +0	$S \left(o_{[a, b]} \right) \cong \lim_{\epsilon > 0} S \left(o_{[a - \epsilon, b + \epsilon]} \right)$	$S \left(o_{[a, b]} \right) \cong \lim_{\epsilon > 0} S \left(o_{[a - \epsilon, b + \epsilon]} \right)$	$S(o_{[a, b]}) \cong \lim_{\epsilon > 0} S(o_{[a - \epsilon, b + \epsilon]}).$
fong-spivak-invitation_EQ0175	271	■ 1.000 ● K +0	$\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}.$	$\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}.$	$\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}.$
fong-spivak-invitation_EQ0176	271	■ 1.000 ● K +0	$(h, 1), (h, 2), (h, 3), (1, 1), (1, 2), (1, 3).$	$(h, 1), (h, 2), (h, 3), (1, 1), (1, 2), (1, 3).$	$(h, 1), (h, 2), (h, 3), (1, 1), (1, 2), (1, 3).$
fong-spivak-invitation_EQ0177	271	■ 1.000 ● K +0	$(h, 1), (1, 1), (1, 2), (2, 2), (3, 2).$	$(h, 1), (1, 1), (1, 2), (2, 2), (3, 2).$	$(h, 1), (1, 1), (1, 2), (2, 2), (3, 2).$
fong-spivak-invitation_EQ0178	271	■ 1.000 ● K +0	$h, 1, 2, 3.$	$h, 1, 2, 3.$	$h, 1, 2, 3.$
fong-spivak-invitation_EQ0179	272	■ 1.000 ● S +1	$\begin{array}{l} (11, 11), (11, 12), (12, 11), (12, 12), (13, 13), \\ (21, 21), (22, 22), (22, 23), (23, 22), (23, 23). \end{array}$	$(11, 11), (11, 12), (12, 11), (12, 12), (13, 13), \\ (21, 21), (22, 22), (22, 23), (23, 22), (23, 23).$	$(11, 11), (11, 12), (12, 11), (12, 12), (13, 13), \\ (21, 21), (22, 22), (22, 23), (23, 22), (23, 23).$
fong-spivak-invitation_EQ0182	273	■ 1.000 ● K +0	$(\bullet)(\circ)(*), (\bullet\circ)(*), (\bullet*)(\circ), (\bullet)(\circ*), (\bullet\circ*).$	$(\bullet)(\circ)(*), (\bullet\circ)(*), (\bullet*)(\circ), (\bullet)(\circ*), (\bullet\circ*).$	$(\bullet)(\circ)(*), (\bullet\circ)(*), (\bullet*)(\circ), (\bullet)(\circ*), (\bullet\circ*).$
fong-spivak-invitation_EQ0184	280	■ 1.000 ● N +0	$g(q) \leq g^{\prime}(f(g(q))) \leq g^{\prime}(q).$	$g(q) \leq g^{\prime}(f(g(q))) \leq g^{\prime}(q).$	$g(q) \leq g^{\prime}(f(g(q))) \leq g^{\prime}(q).$
fong-spivak-invitation_EQ0188	288	■ 1.000 ● K +0	$A = (A \cap \bar{B}) \cup (A \cap B) \subseteq \bar{B} \cup C.$	$A = (A \cap \bar{B}) \cup (A \cap B) \subseteq \bar{B} \cup C.$	$A = (A \cap \bar{B}) \cup (A \cap B) \subseteq \bar{B} \cup C.$
fong-spivak-invitation_EQ0190	288	■ 1.000 ● W +0	$\begin{array}{l} \left(\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right), \left(\begin{array}{cc} \text{true} & \text{false} \\ \text{false} & \text{true} \end{array} \right), \text{ and } \left(\begin{array}{cc} 0 & \infty \\ \infty & 0 \end{array} \right). \end{array}$	$\left(\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right), \left(\begin{array}{cc} \text{true} & \text{false} \\ \text{false} & \text{true} \end{array} \right), \text{ and } \left(\begin{array}{cc} 0 & \infty \\ \infty & 0 \end{array} \right).$	$\left(\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right), \left(\begin{array}{cc} \text{true} & \text{false} \\ \text{false} & \text{true} \end{array} \right), \text{ and } \left(\begin{array}{cc} 0 & \infty \\ \infty & 0 \end{array} \right).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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fong-spivak-invitation_ EQ0194	289	■1.000 ● N +1	$\begin{gathered} M_{\{X\}}=\left(\begin{array}{cccc} 0 & \infty & 3 & \infty \\ \infty & 3 & 0 & \infty \\ \infty & \infty & 6 & 0 \end{array}\right), \\ M_{\{X\}^2}=\left(\begin{array}{cccc} 0 & 6 & 3 & \infty \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ \infty & 9 & 6 & 0 \end{array}\right), \\ M_{\{X\}^3}=\left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array}\right). \end{gathered}$	$M_X = \begin{pmatrix} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & 5 \\ \infty & 3 & 0 & \infty \\ \infty & \infty & 6 & 0 \end{pmatrix}, \quad M_X^2 = \begin{pmatrix} 0 & 6 & 3 & \infty \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ \infty & 9 & 6 & 0 \end{pmatrix},$ $M_X^3 = M_X^4 = \begin{pmatrix} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{pmatrix}.$	$M_X = \begin{pmatrix} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & 5 \\ \infty & 3 & 0 & \infty \\ \infty & \infty & 6 & 0 \end{pmatrix}, \quad M_X^2 = \begin{pmatrix} 0 & 6 & 3 & \infty \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ \infty & 9 & 6 & 0 \end{pmatrix},$ $M_X^3 = M_X^4 = \begin{pmatrix} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{pmatrix}.$																				
fong-spivak-invitation_ EQ0196	290	■1.000 ● N +0	$(p \ ; \ q) := \left(v, a_{\{1\}}, \ \ldots, a_{\{m\}}, b_{\{1\}}, \ \ldots, b_{\{n\}} \right) \ .$	$(p;q) := (v, a_1, \dots, a_m, b_1, \dots, b_n) \,.$	$(p \circ q) := (v, a_1, \dots, a_m, b_1, \dots, b_n) \,.$																				
fong-spivak-invitation_ EQ0197	290	■1.000 ● K +0	$\left(v, a_{\{1\}}, \ \ldots, a_{\{m\}}, b_{\{1\}}, \ \ldots, b_{\{n\}}, c_{\{1\}}, \ \ldots, c_{\{o\}} \right) \ .$	$(v, a_1, \dots, a_m, b_1, \dots, b_n, c_1, \dots, c_o) \,.$	$(v, a_1, \dots, a_m, b_1, \dots, b_n, c_1, \dots, c_o) \,.$																				
fong-spivak-invitation_ EQ0203	292	■1.000 ● S +0	$\begin{array}{lll} (1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (2,3) \\ (3,1) & (3,2) & (3,3) \end{array}$	$\begin{array}{lll} (1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (2,3) \\ (3,1) & (3,2) & (3,3) \end{array}$	$\begin{array}{lll} (1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (2,3) \\ (3,1) & (3,2) & (3,3) \end{array}$																				
fong-spivak-invitation_ EQ0204	292	■1.000 ● N +0	$f^{-1}\{1\}=b, \quad \text{\rm \quad} f^{-1}\{2\}=a, \quad \text{\rm \quad} f^{-1}\{3\}=c \ .$	$f^{-1}(1) = b, \quad f^{-1}(2) = a, \quad f^{-1}(3) = c.$	$f^{-1}(1) = b, \quad f^{-1}(2) = a, \quad f^{-1}(3) = c.$																				
fong-spivak-invitation_ EQ0205	293	■1.000 ● W +0	$\begin{gathered} A^{\prime} \mapsto A, \quad f^{\prime} \mapsto f, \quad g^{\prime} \mapsto g, \quad f^{\prime}; h^{\prime} \mapsto f; h, \quad g^{\prime}; i^{\prime} \mapsto f; h, \\ B^{\prime} \mapsto B, \quad h^{\prime} \mapsto h, \quad C^{\prime} \mapsto C, \quad i^{\prime} \mapsto i, \quad D^{\prime} \mapsto D. \end{gathered}$	$\begin{gathered} A^{\prime} \mapsto A, \quad f^{\prime} \mapsto f, \quad g^{\prime} \mapsto g, \quad f^{\prime}; h^{\prime} \mapsto f; h, \quad g^{\prime}; i^{\prime} \mapsto f; h, \\ B^{\prime} \mapsto B, \quad h^{\prime} \mapsto h, \quad C^{\prime} \mapsto C, \quad i^{\prime} \mapsto i, \quad D^{\prime} \mapsto D. \end{gathered}$	$\begin{gathered} A^{\prime} \mapsto A, \quad f^{\prime} \mapsto f, \quad g^{\prime} \mapsto g, \quad f^{\prime} \circ h^{\prime} \mapsto f \circ h, \quad g^{\prime} \circ i^{\prime} \mapsto f \circ h, \\ B^{\prime} \mapsto B, \quad h^{\prime} \mapsto h, \quad C^{\prime} \mapsto C, \quad i^{\prime} \mapsto i, \quad D^{\prime} \mapsto D. \end{gathered}$																				
fong-spivak-invitation_ EQ0208	293	■1.000 ● S +0	$((F \ ; \ G) \ ; \ H)(c) = H((F \ ; \ G)(c)) = H(G(F(c))) = (G \ ; \ H)(F(c)) = (F \ ; \ (G \ ; \ H))(c) \ .$	$((F;G);H)(c) = H((F;G)(c)) = H(G(F(c))) = (G;H)(F(c)) = (F;(G;H))(c).$	$((F \circ G) \circ H)(c) = H((F \circ G)(c)) = H(G(F(c))) = (G \circ H)(F(c)) = (F \circ (G \circ H))(c).$																				
fong-spivak-invitation_ EQ0211	297	■1.000 ● S +0	$\left(f_{\{1\}} \ ; \ f_{\{2\}} \right)^{\{B\}}(g) = g \ ; \ \left(f_{\{1\}} \ ; \ f_{\{2\}} \right) = \left(g \ ; \ f_{\{1\}} \right) \ ; \ f_{\{2\}} = \left(f_{\{1\}}^{\{B\}} \ ; \ f_{\{2\}}^{\{B\}} \right)(g)$	$(f_1;f_2)^B(g) = g; (f_1;f_2) = (g;f_1); f_2 = (f_1^B;f_2^B)(g)$	$(f_1 \circ f_2)^B(g) = g \circ (f_1 \circ f_2) = (g \circ f_1) \circ f_2 = (f_1^B \circ f_2^B)(g)$																				
fong-spivak-invitation_ EQ0218	300	■1.000 ● N +0	$\begin{array}{c ccc} \Phi & x & y & z \\ \hline A & 17 & 20 & 20 \\ B & 11 & 14 & 14 \\ C & 14 & 17 & 17 \\ D & 12 & 9 & 15 \end{array}$	$\begin{array}{c ccc} \Phi & x & y & z \\ \hline A & 17 & 20 & 20 \\ B & 11 & 14 & 14 \\ C & 14 & 17 & 17 \\ D & 12 & 9 & 15 \end{array}$	<table><tr><td>Φ</td><td>x</td><td>y</td><td>z</td></tr><tr><td>A</td><td>17</td><td>20</td><td>20</td></tr><tr><td>B</td><td>11</td><td>14</td><td>14</td></tr><tr><td>C</td><td>14</td><td>17</td><td>17</td></tr><tr><td>D</td><td>12</td><td>9</td><td>15</td></tr></table>	Φ	x	y	z	A	17	20	20	B	11	14	14	C	14	17	17	D	12	9	15
Φ	x	y	z																						
A	17	20	20																						
B	11	14	14																						
C	14	17	17																						
D	12	9	15																						
fong-spivak-invitation_ EQ0229	307	■1.000 ● K +0	$(f+g)(i) := \begin{cases} f(i) & \text{\rm \quad if } i \leq m, \\ g(i-m)+m^{\prime} & \text{\rm \quad if } m+1 \leq i. \end{cases}$	$(f+g)(i) := \begin{cases} f(i) & \text{\rm \quad if } i \leq m, \\ g(i-m)+m' & \text{\rm \quad if } m+1 \leq i. \end{cases}$	$(f+g)(i) := \begin{cases} f(i) & \text{\rm \quad if } i \leq m, \\ g(i-m)+m' & \text{\rm \quad if } m+1 \leq i. \end{cases}$																				
fong-spivak-invitation_ EQ0230	310	■1.000 ● K +0	$a^{\{0\}}, a^{\{1\}}, a^{\{2\}}, a^{\{3\}}, \ \ldots, a^{\{2019\}}, \ \ldots$	$a^0, a^1, a^2, a^3, \dots, a^{2019}, \dots$	$a^0, a^1, a^2, a^3, \dots, a^{2019}, \dots$																				
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured																									

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation-EQ0233	311	■1.000 ● K +0	$\left(\begin{array}{llll} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$.	$\left(\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$.	$\left(\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$.
fong-spivak-invitation-EQ0234	311	■1.000 ● K +0	$A * B=\left(\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right) * \left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right) \neq \left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right) * \left(\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right) = B * A.$	$A * B = \left(\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array}\right) * \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right) \neq \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right) * \left(\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array}\right) = B * A.$	$A * B = \left(\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array}\right) * \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right) \neq \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right) * \left(\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array}\right) = B * A.$
fong-spivak-invitation-EQ0236	312	■1.000 ● K +0	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$
fong-spivak-invitation-EQ0237	312	■1.000 ● K +0	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$
fong-spivak-invitation-EQ0238	313	■1.000 ● K +0	$g_{\{1\}}:=c_{\{n\}}+\cdots+c_{\{n\}}: m \rightarrow(m \times n),$	$g_1:=c_n+\cdots+c_n: m \rightarrow(m \times n),$	$g_1:=c_n+\cdots+c_n: m \rightarrow(m \times n),$
fong-spivak-invitation-EQ0240	313	■1.000 ● N +0	$\begin{aligned} g_{\{2\}}:= & s_{\{M(1,1)\}}+\cdots+s_{\{M(1, n)\}} \\ & \wedge +s_{\{M(2,1)\}}+\cdots+s_{\{M(2, n)\}} \\ & \wedge +\cdots \\ & \wedge +s_{\{M(m, 1)\}}+\cdots+s_{\{M(m, n)\}}:(m \times n) \\ & \rightarrow(m \times n), \end{aligned}$	$g_2:=s_{M(1,1)}+\cdots+s_{M(1,n)}+s_{M(2,1)}+\cdots+s_{M(2,n)}+\cdots+s_{M(m,1)}+\cdots+s_{M(m,n)}:(m \times n) \rightarrow(m \times n),$	$g_2:=s_{M(1,1)}+\cdots+s_{M(1,n)}+s_{M(2,1)}+\cdots+s_{M(2,n)}+\cdots+s_{M(m,1)}+\cdots+s_{M(m,n)}:(m \times n) \rightarrow(m \times n),$
fong-spivak-invitation-EQ0241	313	■1.000 ● K +0	$g_{\{4\}}:=a_{\{m\}}+\cdots+a_{\{m\}}:(m \times n) \rightarrow n,$	$g_4:=a_m+\cdots+a_m:(m \times n) \rightarrow n,$	$g_4:=a_m+\cdots+a_m:(m \times n) \rightarrow n,$
fong-spivak-invitation-EQ0243	315	■1.000 ● K +0	$\mu:=\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{array}\right)$	$\mu:=\left(\begin{array}{cc} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{array}\right)$	$\mu:=\left(\begin{array}{cc} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{array}\right)$
fong-spivak-invitation-EQ0244	315	■1.000 ● N +0	$U(\mu)\left(\left(a_1, \ldots, a_m\right),\left(b_1, \ldots, b_m\right)\right):=\left(a_1+b_1, \ldots, a_m+b_m\right) .$	$U(\mu)\left(\left(a_1, \ldots, a_m\right),\left(b_1, \ldots, b_m\right)\right):=\left(a_1+b_1, \ldots, a_m+b_m\right) .$	$U(\mu)\left(\left(a_1, \ldots, a_m\right),\left(b_1, \ldots, b_m\right)\right):=\left(a_1+b_1, \ldots, a_m+b_m\right) .$
fong-spivak-invitation-EQ0246	316	■1.000 ● W -1	$\begin{aligned} \mathrm{B}(g) &=\left\{(x, z) \in R^m \times R^n \mid S(g)(x)=z\right\}, \\ \mathrm{B}\left(h^{\mathrm{op}}\right) &=\left\{(z, y) \in R^n \times R^\ell \mid z=S(h)(y)\right\}, \end{aligned}$	$\mathrm{B}(g)=\{(x, z) \in R^m \times R^n \mid S(g)(x)=z\},$ $\mathrm{B}\left(h^{\mathrm{op}}\right)=\{(z, y) \in R^n \times R^\ell \mid z=S(h)(y)\},$	$\mathrm{B}(g)=\{(x, z) \in R^m \times R^n \mid S(g)(x)=z\},$ $\mathrm{B}\left(h^{\mathrm{op}}\right)=\{(z, y) \in R^n \times R^\ell \mid z=S(h)(y)\},$
fong-spivak-invitation-EQ0249	316	■1.000 ● W -1	$\begin{aligned} \mathrm{B}\left(g^{\mathrm{op}}\right) &=\left\{(y, x) \in R^n \times R^m \mid y=S(g)(x)\right\}, \\ \mathrm{B}(h) &=\left\{(x, z) \in R^m \times R^p \mid S(h)(x)=z\right\}, \end{aligned}$	$\mathrm{B}\left(g^{\mathrm{op}}\right)=\{(y, x) \in R^n \times R^m \mid y=S(g)(x)\},$ $\mathrm{B}(h)=\{(x, z) \in R^m \times R^p \mid S(h)(x)=z\},$	$\mathrm{B}\left(g^{\mathrm{op}}\right)=\{(y, x) \in R^n \times R^m \mid y=S(g)(x)\},$ $\mathrm{B}(h)=\{(x, z) \in R^m \times R^p \mid S(h)(x)=z\},$
fong-spivak-invitation-EQ0251	316	■1.000 ● N +1	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right); h\right)=\left\{(S(g)(x), S(h)(x)) \mid x \in R^m\right\} .$	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right); h\right)=\{(S(g)(x), S(h)(x)) \mid x \in R^m\} .$	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right) ; h\right)=\{(S(g)(x), S(h)(x)) \mid x \in R^m\} .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																		
fong-spivak-invitation_EQ0252	318	■1.000 ● C -12	f(m)=f(\underbrace{1+1+\cdots+1}_{m \text{ summands}})=\underbrace{f(1)+f(1)+\cdots+f(1)}_{m \text{ summands}}=\underbrace{1_{\text{R}}+1_{\text{R}}+\cdots+1_{\text{R}}}_{m \text{ summands}}.	$f(m) = f(\underbrace{1+1+\cdots+1}_{m \text{ summands}}) = \underbrace{f(1)+f(1)+\cdots+f(1)}_{m \text{ summands}} = \underbrace{1_R+1_R+\cdots+1_R}_{m \text{ summands}}.$	$f(m) = f(\underbrace{1+1+\cdots+1}_{m \text{ summands}}) = \underbrace{f(1)+f(1)+\cdots+f(1)}_{m \text{ summands}} = \underbrace{1_R+1_R+\cdots+1_R}_{m \text{ summands}}.$																																		
fong-spivak-invitation_EQ0254	318	■1.000 ● N +0	\begin{aligned} \{[f, g]: A \sqcup B \} &\& \longrightarrow T \\ \text{apple } 1 &\longmapsto a \end{aligned}	$[f, g]: A \sqcup B \longrightarrow T$ apple 1 $\mapsto a$	$[f, g]: A \sqcup B \longrightarrow T$ apple1 $\mapsto a$																																		
fong-spivak-invitation_EQ0261	338	■1.000 ● K +0	U=\bigcup_{i \in I}\left\{x \in \mathbb{R} \mid a_{\{i\}<x<b_{\{i\}}\right\}	$U = \bigcup_{i \in I} \{x \in \mathbb{R} \mid a_i < x < b_i\}$	$U = \bigcup_{i \in I} \{x \in \mathbb{R} \mid a_i < x < b_i\}$																																		
fong-spivak-invitation_EQ0262	338	■1.000 ● K +0	U=\bigcup_{j \in J}\left\{x \in \mathbb{R} \mid a_{\{j\}<x<b_{\{j\}}\right\}=\bigcup_{j \in J}\left(o_{\left[\left[a_{\{j\}}, b_{\{j\}}\right]\right]} \cap \mathbb{R}\right)=\left(\bigcup_{j \in J} o_{\left[\left[a_{\{j\}}, b_{\{j\}}\right]\right]} \cap \mathbb{R}\right)	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$																																		
fong-spivak-invitation_EQ0064	114	— ● C +31	—	—	$G :=$ $\begin{array}{c} 1 \\ \bullet \end{array} \xrightarrow{a} \begin{array}{c} 2 \\ \bullet \end{array} \xrightarrow{b} \begin{array}{c} 3 \\ \bullet \end{array}$ $H :=$ $\begin{array}{c} 4 \\ \bullet \end{array} \xrightarrow[c]{d} \begin{array}{c} 5 \\ \bullet \end{array} \curvearrowright e$																																		
fong-spivak-invitation_EQ0065	114	— ● C +77	—	—	<table><tr><td rowspan="3">G :=</td><td></td><td>Arrow</td><td>source</td><td>target</td><td>Vertex</td></tr><tr><td></td><td>a</td><td>1</td><td>2</td><td>1</td></tr><tr><td></td><td>b</td><td>2</td><td>3</td><td>2 3</td></tr><tr><td rowspan="4">H :=</td><td></td><td>Arrow</td><td>source</td><td>target</td><td>Vertex</td></tr><tr><td>c</td><td>4</td><td>5</td><td>4</td><td rowspan="3"></td></tr><tr><td>d</td><td>4</td><td>5</td><td>5</td></tr><tr><td>e</td><td>5</td><td>5</td></tr></table>	G :=		Arrow	source	target	Vertex		a	1	2	1		b	2	3	2 3	H :=		Arrow	source	target	Vertex	c	4	5	4		d	4	5	5	e	5	5
G :=		Arrow	source	target	Vertex																																		
		a	1	2	1																																		
		b	2	3	2 3																																		
H :=		Arrow	source	target	Vertex																																		
	c	4	5	4																																			
	d	4	5	5																																			
	e	5	5																																				
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured																																							

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																				
fong-spivak-invitation_Table 2	028	—	(no LaTeX source; 702 × 280 px region)	—	<table><tr><th>Arrow a</th><th>source $s(a) \in V$</th><th>target $t(a) \in V$</th></tr><tr><td>a</td><td>1</td><td>?</td></tr><tr><td>b</td><td>1</td><td>3</td></tr><tr><td>c</td><td>?</td><td>?</td></tr><tr><td>d</td><td>?</td><td>?</td></tr><tr><td>e</td><td>?</td><td>?</td></tr></table>	Arrow a	source $s(a) \in V$	target $t(a) \in V$	a	1	?	b	1	3	c	?	?	d	?	?	e	?	?																		
Arrow a	source $s(a) \in V$	target $t(a) \in V$																																							
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d	?	?																																							
e	?	?																																							
fong-spivak-invitation_Table 3	065	—	(no LaTeX source; 834 × 144 px region)	—	<table><tr><td>$\wedge$</td><td>false</td><td>true</td><td>$*$</td><td>0</td><td>1</td></tr><tr><td>false</td><td>false</td><td>false</td><td></td><td>0</td><td>0</td></tr><tr><td>true</td><td>false</td><td>true</td><td></td><td>1</td><td>0</td></tr></table>	$\wedge$	false	true	$*$	0	1	false	false	false		0	0	true	false	true		1	0																		
$\wedge$	false	true	$*$	0	1																																				
false	false	false		0	0																																				
true	false	true		1	0																																				
fong-spivak-invitation_Table 6	066	—	(no LaTeX source; 406 × 144 px region)	—	<table><tr><td>$\vee$</td><td>false</td><td>true</td></tr><tr><td>false</td><td>false</td><td>true</td></tr><tr><td>true</td><td>true</td><td>true</td></tr></table>	$\vee$	false	true	false	false	true	true	true	true																											
$\vee$	false	true																																							
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fong-spivak-invitation_Table 6	066	—	(no LaTeX source; 208 × 146 px region)	—	<table><tr><td>max</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	max	0	1	0	0	1	1	1	1																											
max	0	1																																							
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fong-spivak-invitation_Table 6	066	—	(no LaTeX source; 458 × 186 px region)	—	<table><tr><td>min</td><td>no</td><td>maybe</td><td>yes</td></tr><tr><td>no</td><td>?</td><td>?</td><td>?</td></tr><tr><td>maybe</td><td>?</td><td>?</td><td>?</td></tr><tr><td>yes</td><td>?</td><td>?</td><td>?</td></tr></table>	min	no	maybe	yes	no	?	?	?	maybe	?	?	?	yes	?	?	?																				
min	no	maybe	yes																																						
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maybe	?	?	?																																						
yes	?	?	?																																						
fong-spivak-invitation_Table 6	071	—	(no LaTeX source; 778 × 278 px region)	—	<table><tr><td>$\cdot \leq \cdot$</td><td>p</td><td>q</td><td>r</td><td>s</td><td>t</td></tr><tr><td>p</td><td>true</td><td>true</td><td>true</td><td>true</td><td>true</td></tr><tr><td>q</td><td>false</td><td>true</td><td>false</td><td>true</td><td>true</td></tr><tr><td>r</td><td>false</td><td>false</td><td>true</td><td>true</td><td>true</td></tr><tr><td>s</td><td>false</td><td>false</td><td>false</td><td>true</td><td>true</td></tr><tr><td>t</td><td>false</td><td>false</td><td>false</td><td>false</td><td>true</td></tr></table>	$\cdot \leq \cdot$	p	q	r	s	t	p	true	true	true	true	true	q	false	true	false	true	true	r	false	false	true	true	true	s	false	false	false	true	true	t	false	false	false	false	true
$\cdot \leq \cdot$	p	q	r	s	t																																				
p	true	true	true	true	true																																				
q	false	true	false	true	true																																				
r	false	false	true	true	true																																				
s	false	false	false	true	true																																				
t	false	false	false	false	true																																				
fong-spivak-invitation_Table 6	075	—	(no LaTeX source; 284 × 190 px region)	—	<table><tr><td>$d(\nearrow)$</td><td>x</td><td>y</td><td>z</td></tr><tr><td>x</td><td>0</td><td>4</td><td>3</td></tr><tr><td>y</td><td>3</td><td>0</td><td>6</td></tr><tr><td>z</td><td>7</td><td>4</td><td>0</td></tr></table>	$d(\nearrow)$	x	y	z	x	0	4	3	y	3	0	6	z	7	4	0																				
$d(\nearrow)$	x	y	z																																						
x	0	4	3																																						
y	3	0	6																																						
z	7	4	0																																						
fong-spivak-invitation_Table 6	075	—	(no LaTeX source; 368 × 234 px region)	—	<table><tr><td>$d(\nearrow)$</td><td>A</td><td>B</td><td>C</td><td>D</td></tr><tr><td>A</td><td>0</td><td>?</td><td>?</td><td>?</td></tr><tr><td>B</td><td>2</td><td>?</td><td>5</td><td>?</td></tr><tr><td>C</td><td>?</td><td>?</td><td>?</td><td>?</td></tr><tr><td>D</td><td>?</td><td>?</td><td>?</td><td>?</td></tr></table>	$d(\nearrow)$	A	B	C	D	A	0	?	?	?	B	2	?	5	?	C	?	?	?	?	D	?	?	?	?											
$d(\nearrow)$	A	B	C	D																																					
A	0	?	?	?																																					
B	2	?	5	?																																					
C	?	?	?	?																																					
D	?	?	?	?																																					
fong-spivak-invitation_Table 6	076	—	(no LaTeX source; 432 × 232 px region)	—	<table><tr><td>$M_X =$</td><td>$\nearrow$</td><td>A</td><td>B</td><td>C</td><td>D</td></tr><tr><td>A</td><td>0</td><td>?</td><td>?</td><td>?</td><td></td></tr><tr><td>B</td><td>2</td><td>0</td><td>∞</td><td>?</td><td></td></tr><tr><td>C</td><td>?</td><td>?</td><td>?</td><td>?</td><td></td></tr><tr><td>D</td><td>?</td><td>?</td><td>?</td><td>?</td><td></td></tr></table>	$M_X =$	$\nearrow$	A	B	C	D	A	0	?	?	?		B	2	0	∞	?		C	?	?	?	?		D	?	?	?	?							
$M_X =$	$\nearrow$	A	B	C	D																																				
A	0	?	?	?																																					
B	2	0	∞	?																																					
C	?	?	?	?																																					
D	?	?	?	?																																					
fong-spivak-invitation_Table 6	081	—	(no LaTeX source; 272 × 190 px region)	—	<table><tr><td>$\mathcal{X}$</td><td>A</td><td>B</td><td>C</td></tr><tr><td>A</td><td>0</td><td>2</td><td>5</td></tr><tr><td>B</td><td>∞</td><td>0</td><td>3</td></tr><tr><td>C</td><td>∞</td><td>∞</td><td>0</td></tr></table>	$\mathcal{X}$	A	B	C	A	0	2	5	B	∞	0	3	C	∞	∞	0																				
$\mathcal{X}$	A	B	C																																						
A	0	2	5																																						
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C	∞	∞	0																																						
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value																																									
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured																																									

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																	
fong-spivak-invitation_Table 0	081	—	(no LaTeX source; 178 × 148 px region)	—	<table><tr><td>y</td><td>p</td><td>q</td></tr><tr><td>p</td><td>0</td><td>5</td></tr><tr><td>q</td><td>8</td><td>0</td></tr></table>	y	p	q	p	0	5	q	8	0																																								
y	p	q																																																				
p	0	5																																																				
q	8	0																																																				
fong-spivak-invitation_Table 0	081	—	(no LaTeX source; 890 × 336 px region)	—	<table><tr><td>$\mathcal{X} \times \mathcal{Y}$</td><td>$(A, p)$</td><td>$(B, p)$</td><td>$(C, p)$</td><td>$(A, q)$</td><td>$(B, q)$</td><td>$(C, q)$</td></tr><tr><td>$(A, p)$</td><td>0</td><td>2</td><td>5</td><td>5</td><td>7</td><td>10</td></tr><tr><td>(B, p)</td><td>∞</td><td>0</td><td>3</td><td>∞</td><td>5</td><td>8</td></tr><tr><td>(C, p)</td><td>∞</td><td>∞</td><td>0</td><td>∞</td><td>∞</td><td>5</td></tr><tr><td>(A, q)</td><td>8</td><td>10</td><td>13</td><td>0</td><td>2</td><td>5</td></tr><tr><td>(B, q)</td><td>∞</td><td>8</td><td>11</td><td>∞</td><td>0</td><td>3</td></tr><tr><td>(C, q)</td><td>∞</td><td>∞</td><td>8</td><td>∞</td><td>∞</td><td>0</td></tr></table>	$\mathcal{X} \times \mathcal{Y}$	(A, p)	(B, p)	(C, p)	(A, q)	(B, q)	(C, q)	(A, p)	0	2	5	5	7	10	(B, p)	∞	0	3	∞	5	8	(C, p)	∞	∞	0	∞	∞	5	(A, q)	8	10	13	0	2	5	(B, q)	∞	8	11	∞	0	3	(C, q)	∞	∞	8	∞	∞	0
$\mathcal{X} \times \mathcal{Y}$	(A, p)	(B, p)	(C, p)	(A, q)	(B, q)	(C, q)																																																
(A, p)	0	2	5	5	7	10																																																
(B, p)	∞	0	3	∞	5	8																																																
(C, p)	∞	∞	0	∞	∞	5																																																
(A, q)	8	10	13	0	2	5																																																
(B, q)	∞	8	11	∞	0	3																																																
(C, q)	∞	∞	8	∞	∞	0																																																
fong-spivak-invitation_Table 0	088	—	(no LaTeX source; 300 × 224 px region)	—	<table><tr><td>M_Y</td><td>x</td><td>y</td><td>z</td></tr><tr><td>x</td><td>0</td><td>4</td><td>3</td></tr><tr><td>y</td><td>3</td><td>0</td><td>∞</td></tr><tr><td>z</td><td>∞</td><td>4</td><td>0</td></tr></table>	M_Y	x	y	z	x	0	4	3	y	3	0	∞	z	∞	4	0																																	
M_Y	x	y	z																																																			
x	0	4	3																																																			
y	3	0	∞																																																			
z	∞	4	0																																																			
fong-spivak-invitation_Table 0	088	—	(no LaTeX source; 256 × 224 px region)	—	<table><tr><td>d_Y</td><td>x</td><td>y</td><td>z</td></tr><tr><td>x</td><td>0</td><td>4</td><td>3</td></tr><tr><td>y</td><td>3</td><td>0</td><td>6</td></tr><tr><td>z</td><td>7</td><td>4</td><td>0</td></tr></table>	d_Y	x	y	z	x	0	4	3	y	3	0	6	z	7	4	0																																	
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x	0	4	3																																																			
y	3	0	6																																																			
z	7	4	0																																																			
fong-spivak-invitation_Table 0	088	—	(no LaTeX source; 334 × 190 px region)	—	$M_Y^2 = \begin{array}{c ccc} \nearrow & x & y & z \\ x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$																																																	
fong-spivak-invitation_Table 0	088	—	(no LaTeX source; 334 × 190 px region)	—	$M_Y^3 = \begin{array}{c ccc} \nearrow & x & y & z \\ x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$																																																	
fong-spivak-invitation_Table 0	091	—	(no LaTeX source; 640 × 188 px region)	—	<table><tr><th>Employee</th><th>FName</th><th>WorksIn</th><th>Mngr</th></tr><tr><td>1</td><td>Alan</td><td>101</td><td>2</td></tr><tr><td>2</td><td>Ruth</td><td>101</td><td>2</td></tr><tr><td>3</td><td>Kris</td><td>102</td><td>3</td></tr></table>	Employee	FName	WorksIn	Mngr	1	Alan	101	2	2	Ruth	101	2	3	Kris	102	3																																	
Employee	FName	WorksIn	Mngr																																																			
1	Alan	101	2																																																			
2	Ruth	101	2																																																			
3	Kris	102	3																																																			
fong-spivak-invitation_Table 0	091	—	(no LaTeX source; 490 × 188 px region)	—	<table><tr><th>Department</th><th>DName</th><th>Secr</th></tr><tr><td>101</td><td>Sales</td><td>1</td></tr><tr><td>102</td><td>IT</td><td>3</td></tr></table>	Department	DName	Secr	101	Sales	1	102	IT	3																																								
Department	DName	Secr																																																				
101	Sales	1																																																				
102	IT	3																																																				
fong-spivak-invitation_Table 0	104	—	(no LaTeX source; 1038 × 432 px region)	—	<table><tr><th>Planet of Sol</th><th>Prime number</th><th>Flying pig</th></tr><tr><td>Mercury</td><td>2</td><td></td></tr><tr><td>Venus</td><td>3</td><td></td></tr><tr><td>Earth</td><td>5</td><td></td></tr><tr><td>Mars</td><td>7</td><td></td></tr><tr><td>Jupiter</td><td>11</td><td></td></tr><tr><td>Saturn</td><td>13</td><td></td></tr><tr><td>Uranus</td><td>17</td><td></td></tr><tr><td>Neptune</td><td>$\vdots$</td><td></td></tr></table>	Planet of Sol	Prime number	Flying pig	Mercury	2		Venus	3		Earth	5		Mars	7		Jupiter	11		Saturn	13		Uranus	17		Neptune	$\vdots$																							
Planet of Sol	Prime number	Flying pig																																																				
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Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured																																																						

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image				
fong-spivak-invitation_TAL_064	104	—	(no LaTeX source; 382 × 232 px region)	—	Beatle	Played			
					George	lead guitar			
					John	rhythm guitar			
					Paul	bass guitar			
					Ringo	drums			
fong-spivak-invitation_TAL_065	105	—	(no LaTeX source; 382 × 278 px region)	—	Beatle	Played			
					George	lead guitar			
					John	rhythm guitar			
					Paul	bass guitar			
					Ringo	drums			
fong-spivak-invitation_TAL_065	105	—	(no LaTeX source; 428 × 276 px region)	—	Rock-and-roll instrument				
					bass guitar				
					drums				
					keyboard				
					lead guitar				
					rhythm guitar				
fong-spivak-invitation_TAL_069	109	—	(no LaTeX source; 426 × 454 px region)	—	$N_{\geq 2}$	Smallest prime factor			
					2	2			
					3	3			
					4	2			
					$\vdots$	$\vdots$			
					49	7			
					50	2			
					51	3			
					$\vdots$	$\vdots$			
fong-spivak-invitation_TAL_123	113	—	(no LaTeX source; 806 × 280 px region)	—	Arrow	source	target	Vertex	
					a	1	2	1	
					b	1	3	2	
					c	1	3	3	
					d	2	2	4	
					e	2	3		
fong-spivak-invitation_TAL_125	115	—	(no LaTeX source; 216 × 368 px region)	—	State	next			
					1	4			
					2	4			
					3	5			
					4	5			
					5	5			
					6	7			
					7	6			
fong-spivak-invitation_TAL_126	116	—	(no LaTeX source; 804 × 368 px region)	—	Arrow	source	target	Vertex	
					1	1	4	1	
					2	2	4	2	
					3	3	5	3	
					4	4	5	4	
					5	5	5	5	
					6	6	7	6	
					7	7	6	7	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value									
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured									

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image			
fong-spivak-invitation_TAL_060	120	—	(no LaTeX source; 1158 × 278 px region)	—	Migration functor	pronounced	reminiscent of	database idea
					Δ	delta	duplicate or destroy	duplicate or destroy tables or columns
					Σ	sigma	sum	union (sum up) data
					Π	pi	product	pair ⁹ and query data
fong-spivak-invitation_TAL_060	121	—	(no LaTeX source; 922 × 324 px region)	—	Email	sent_by	received_by	Address
					Em_1	Bob	Grace	Bob
					Em_2	Grace	Pat	Doug
					Em_3	Bob	Emmy	Emmy
					Em_4	Sue	Doug	Grace
					Em_5	Doug	Sue	Pat
					Em_6	Bob	Bob	Sue
fong-spivak-invitation_TAL_060	133	—	(no LaTeX source; 536 × 140 px region)	—	Bool-category	preorder		
					Bool-functor	monotone map		
					Bool-profunctor	feasibility relation		
fong-spivak-invitation_TAL_060	134	—	(no LaTeX source; 426 × 234 px region)	—	c	d	$c \Rightarrow d$	
					true	true	true	
					true	false	false	
					false	true	true	
					false	false	true	
fong-spivak-invitation_TAL_060	136	—	(no LaTeX source; 588 × 288 px region)	—	Φ	a	b	c
					N	?	?	?
					E	true	?	?
					W	?	?	?
					S	?	?	?
fong-spivak-invitation_TAL_060	137	—	(no LaTeX source; 280 × 232 px region)	—	Φ	x	y	z
					A	?	?	20
					B	11	?	?
					C	?	17	?
					D	?	?	?
fong-spivak-invitation_TAL_060	137	—	(no LaTeX source; 362 × 234 px region)	—	M_X	A	B	C
					A	0	∞	3
					B	2	0	∞
					C	∞	3	0
					D	∞	∞	4
fong-spivak-invitation_TAL_060	137	—	(no LaTeX source; 276 × 232 px region)	—	M_Φ	x	y	z
					A	∞	∞	∞
					B	11	∞	∞
					C	∞	∞	∞
					D	∞	9	∞
fong-spivak-invitation_TAL_060	137	—	(no LaTeX source; 254 × 186 px region)	—	M_Y	x	y	z
					x	0	4	3
					y	3	0	∞
					z	∞	4	0

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																														
fong-spivak-invitation_TAL_450	140	—	(no LaTeX source; 698 × 236 px region)	—	<table><tr><th>Φ</th><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>N</td><td>true</td><td>false</td><td>true</td><td>false</td><td>false</td></tr><tr><td>E</td><td>true</td><td>false</td><td>true</td><td>false</td><td>true</td></tr><tr><td>W</td><td>true</td><td>true</td><td>true</td><td>true</td><td>false</td></tr><tr><td>S</td><td>true</td><td>true</td><td>true</td><td>true</td><td>true</td></tr></table>	Φ	a	b	c	d	e	N	true	false	true	false	false	E	true	false	true	false	true	W	true	true	true	true	false	S	true	true	true	true	true
Φ	a	b	c	d	e																														
N	true	false	true	false	false																														
E	true	false	true	false	true																														
W	true	true	true	true	false																														
S	true	true	true	true	true																														
fong-spivak-invitation_TAL_450	140	—	(no LaTeX source; 352 × 280 px region)	—	<table><tr><th>Ψ</th><th>x</th><th>y</th></tr><tr><td>a</td><td>false</td><td>true</td></tr><tr><td>b</td><td>true</td><td>true</td></tr><tr><td>c</td><td>false</td><td>true</td></tr><tr><td>d</td><td>true</td><td>true</td></tr><tr><td>e</td><td>false</td><td>false</td></tr></table>	Ψ	x	y	a	false	true	b	true	true	c	false	true	d	true	true	e	false	false												
Ψ	x	y																																	
a	false	true																																	
b	true	true																																	
c	false	true																																	
d	true	true																																	
e	false	false																																	
fong-spivak-invitation_TAL_450	140	—	(no LaTeX source; 384 × 232 px region)	—	<table><tr><th>$\Phi \circ \Psi$</th><th>x</th><th>y</th></tr><tr><td>N</td><td>false</td><td>true</td></tr><tr><td>E</td><td>false</td><td>true</td></tr><tr><td>W</td><td>true</td><td>true</td></tr><tr><td>S</td><td>true</td><td>true</td></tr></table>	$\Phi \circ \Psi$	x	y	N	false	true	E	false	true	W	true	true	S	true	true															
$\Phi \circ \Psi$	x	y																																	
N	false	true																																	
E	false	true																																	
W	true	true																																	
S	true	true																																	
fong-spivak-invitation_TAL_451	141	—	(no LaTeX source; 354 × 232 px region)	—	<table><tr><th>$\Phi \circ \Psi$</th><th>p</th><th>q</th><th>r</th><th>s</th></tr><tr><td>A</td><td>?</td><td>24</td><td>?</td><td>?</td></tr><tr><td>B</td><td>?</td><td>?</td><td>?</td><td>?</td></tr><tr><td>C</td><td>?</td><td>?</td><td>?</td><td>?</td></tr><tr><td>D</td><td>?</td><td>?</td><td>9</td><td>?</td></tr></table>	$\Phi \circ \Psi$	p	q	r	s	A	?	24	?	?	B	?	?	?	?	C	?	?	?	?	D	?	?	9	?					
$\Phi \circ \Psi$	p	q	r	s																															
A	?	24	?	?																															
B	?	?	?	?																															
C	?	?	?	?																															
D	?	?	9	?																															
fong-spivak-invitation_TAL_450	146	—	(no LaTeX source; 818 × 142 px region)	—	<table><tr><th>$\mathcal{X}$</th><th>A</th><th>B</th><th>Φ</th><th>x</th><th>y</th><th>$\mathcal{Y}$</th><th>x</th><th>y</th></tr><tr><td>A</td><td>0</td><td>2</td><td>A</td><td>5</td><td>8</td><td>x</td><td>0</td><td>3</td></tr><tr><td>B</td><td>∞</td><td>0</td><td>B</td><td>∞</td><td>∞</td><td>y</td><td>4</td><td>0</td></tr></table>	$\mathcal{X}$	A	B	Φ	x	y	$\mathcal{Y}$	x	y	A	0	2	A	5	8	x	0	3	B	∞	0	B	∞	∞	y	4	0			
$\mathcal{X}$	A	B	Φ	x	y	$\mathcal{Y}$	x	y																											
A	0	2	A	5	8	x	0	3																											
B	∞	0	B	∞	∞	y	4	0																											
fong-spivak-invitation_TAL_450	146	—	(no LaTeX source; 408 × 266 px region)	—	<table><tr><th>$\text{Col}(\Phi)$</th><th>A</th><th>B</th><th>x</th><th>y</th></tr><tr><td>A</td><td>0</td><td>2</td><td>5</td><td>8</td></tr><tr><td>B</td><td>∞</td><td>0</td><td>∞</td><td>∞</td></tr><tr><td>x</td><td>0</td><td>0</td><td>0</td><td>3</td></tr><tr><td>y</td><td>0</td><td>0</td><td>4</td><td>0</td></tr></table>	$\text{Col}(\Phi)$	A	B	x	y	A	0	2	5	8	B	∞	0	∞	∞	x	0	0	0	3	y	0	0	4	0					
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fong-spivak-invitation_TAL_450	179	—	(no LaTeX source; 604 × 144 px region)	—	<table><tr><th></th><th>a</th><th>b</th></tr><tr><td>x</td><td>15</td><td>3</td></tr><tr><td>y</td><td>0</td><td>21</td></tr></table> $\begin{pmatrix} 15 & 3 \\ 0 & 21 \end{pmatrix}$		a	b	x	15	3	y	0	21																					
	a	b																																	
x	15	3																																	
y	0	21																																	
fong-spivak-invitation_TAL_450	179	—	(no LaTeX source; 714 × 408 px region)	—	<table><tr><th>Generator</th><th>Icon</th><th>Matrix</th><th>Arity</th></tr><tr><td>amplify by $a \in R$</td><td></td><td>(a)</td><td>$1 \rightarrow 1$</td></tr><tr><td>add</td><td></td><td>$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$</td><td>$2 \rightarrow 1$</td></tr><tr><td>zero</td><td></td><td>$()$</td><td>$0 \rightarrow 1$</td></tr><tr><td>copy</td><td></td><td>$\begin{pmatrix} 1 & 1 \end{pmatrix}$</td><td>$1 \rightarrow 2$</td></tr><tr><td>discard</td><td></td><td>$()$</td><td>$1 \rightarrow 0$</td></tr></table>	Generator	Icon	Matrix	Arity	amplify by $a \in R$		(a)	$1 \rightarrow 1$	add		$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$2 \rightarrow 1$	zero		$()$	$0 \rightarrow 1$	copy		$\begin{pmatrix} 1 & 1 \end{pmatrix}$	$1 \rightarrow 2$	discard		$()$	$1 \rightarrow 0$						
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fong-spivak-invitation_TAL_450	242	—	(no LaTeX source; 674 × 236 px region)	—	<table><tr><th>P</th><th>Q</th><th>$\lceil (\text{true}, \text{true})^\neg(P, Q)$</th></tr><tr><td>true</td><td>true</td><td>true</td></tr><tr><td>true</td><td>false</td><td>false</td></tr><tr><td>false</td><td>true</td><td>false</td></tr><tr><td>false</td><td>false</td><td>false</td></tr></table>	P	Q	$\lceil (\text{true}, \text{true})^\neg(P, Q)$	true	true	true	true	false	false	false	true	false	false	false	false															
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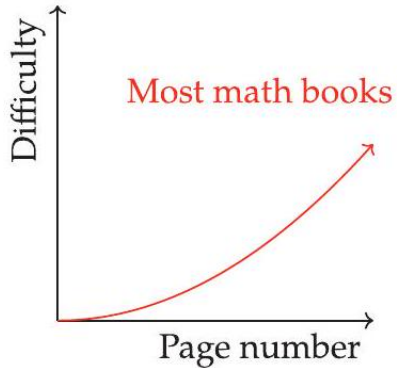
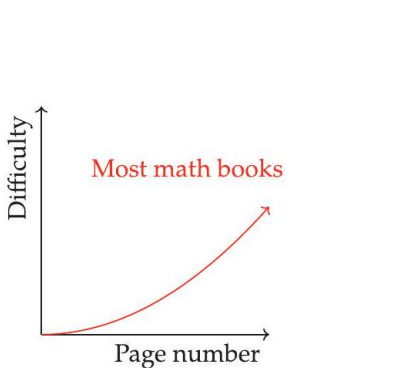
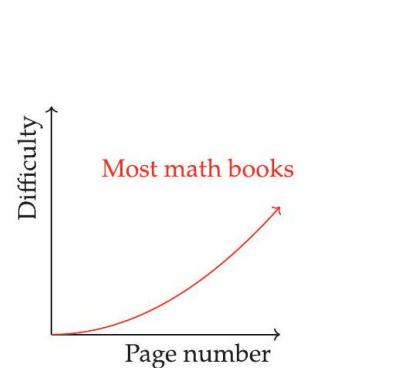
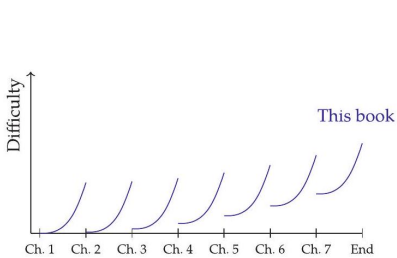
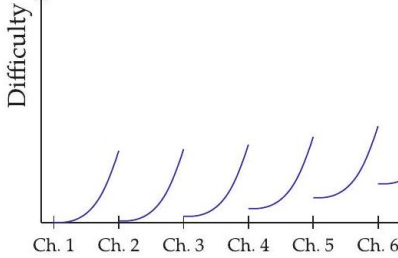
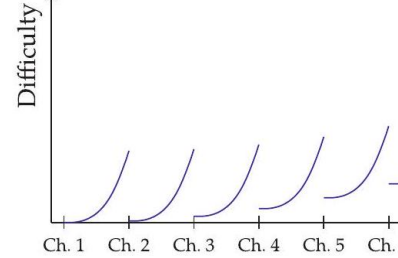
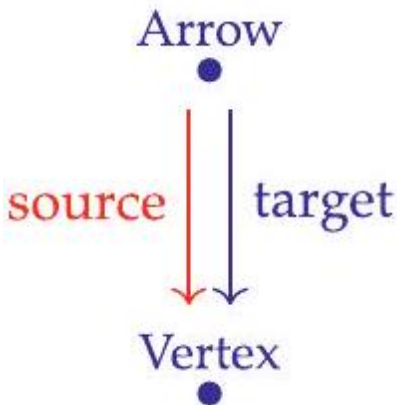
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																																																											
fong-spivak-invitation_T4_442	242	—	(no LaTeX source; 434 × 236 px region)	—	<table><tr><th>P</th><th>Q</th><th>$P \vee Q$</th></tr><tr><td>true</td><td>true</td><td>true</td></tr><tr><td>true</td><td>false</td><td>true</td></tr><tr><td>false</td><td>true</td><td>true</td></tr><tr><td>false</td><td>false</td><td>false</td></tr></table>	P	Q	$P \vee Q$	true	true	true	true	false	true	false	true	true	false	false	false																																																																												
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fong-spivak-invitation_T4_443	243	—	(no LaTeX source; 656 × 236 px region)	—	<table><tr><th>P</th><th>Q</th><th>$P \wedge Q$</th><th>$P = (P \wedge Q)$</th></tr><tr><td>true</td><td>true</td><td>?</td><td>?</td></tr><tr><td>true</td><td>false</td><td>?</td><td>?</td></tr><tr><td>false</td><td>true</td><td>?</td><td>?</td></tr><tr><td>false</td><td>false</td><td>?</td><td>?</td></tr></table>	P	Q	$P \wedge Q$	$P = (P \wedge Q)$	true	true	?	?	true	false	?	?	false	true	?	?	false	false	?	?																																																																							
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fong-spivak-invitation_T4_443	273	—	(no LaTeX source; 648 × 272 px region)	—	<table><tr><th>Arrow a</th><th>Source $s(a) \in V$</th><th>Target $t(a) \in V$</th></tr><tr><td>a</td><td>1</td><td>2</td></tr><tr><td>b</td><td>1</td><td>3</td></tr><tr><td>c</td><td>1</td><td>3</td></tr><tr><td>d</td><td>2</td><td>2</td></tr><tr><td>e</td><td>2</td><td>3</td></tr></table>	Arrow a	Source $s(a) \in V$	Target $t(a) \in V$	a	1	2	b	1	3	c	1	3	d	2	2	e	2	3																																																																									
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fong-spivak-invitation_T4_443	280	—	(no LaTeX source; 794 × 550 px region)	—	<table><tr><th>p</th><th>q</th><th>$f(p)$</th><th>$g(q)$</th><th>$f(p) \leq^? q$</th><th>$p \leq^? g(q)$</th><th>Same?</th></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>yes</td><td>yes</td><td>yes!</td></tr><tr><td>1</td><td>2</td><td>1</td><td>2</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>1</td><td>4</td><td>1</td><td>4</td><td>yes</td><td>yes</td><td>yes!</td></tr><tr><td>2</td><td>1</td><td>2</td><td>1</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>2</td><td>2</td><td>2</td><td>2</td><td>yes</td><td>yes</td><td>yes!</td></tr><tr><td>2</td><td>4</td><td>2</td><td>4</td><td>yes</td><td>yes</td><td>yes!</td></tr><tr><td>3.9</td><td>1</td><td>4</td><td>1</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>3.9</td><td>2</td><td>4</td><td>2</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>3.9</td><td>4</td><td>4</td><td>4</td><td>yes</td><td>yes</td><td>yes!</td></tr><tr><td>4</td><td>1</td><td>4</td><td>1</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>4</td><td>2</td><td>4</td><td>2</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>4</td><td>4</td><td>4</td><td>4</td><td>yes</td><td>yes</td><td>yes!</td></tr></table>	p	q	$f(p)$	$g(q)$	$f(p) \leq^? q$	$p \leq^? g(q)$	Same?	1	1	1	1	yes	yes	yes!	1	2	1	2	no	no	yes!	1	4	1	4	yes	yes	yes!	2	1	2	1	no	no	yes!	2	2	2	2	yes	yes	yes!	2	4	2	4	yes	yes	yes!	3.9	1	4	1	no	no	yes!	3.9	2	4	2	no	no	yes!	3.9	4	4	4	yes	yes	yes!	4	1	4	1	no	no	yes!	4	2	4	2	no	no	yes!	4	4	4	4	yes	yes	yes!
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fong-spivak-invitation_T4_443	283	—	(no LaTeX source; 464 × 174 px region)	—	<table><tr><th>min</th><th>no</th><th>maybe</th><th>yes</th></tr><tr><td>no</td><td>no</td><td>no</td><td>no</td></tr><tr><td>maybe</td><td>no</td><td>maybe</td><td>maybe</td></tr><tr><td>yes</td><td>no</td><td>maybe</td><td>yes</td></tr></table>	min	no	maybe	yes	no	no	no	no	maybe	no	maybe	maybe	yes	no	maybe	yes																																																																											
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fong-spivak-invitation_T4_443	285	—	(no LaTeX source; 362 × 218 px region)	—	<table><tr><th>$d(\nearrow)$</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>A</td><td>0</td><td>6</td><td>3</td><td>11</td></tr><tr><td>B</td><td>2</td><td>0</td><td>5</td><td>5</td></tr><tr><td>C</td><td>5</td><td>3</td><td>0</td><td>8</td></tr><tr><td>D</td><td>11</td><td>9</td><td>6</td><td>0</td></tr></table>	$d(\nearrow)$	A	B	C	D	A	0	6	3	11	B	2	0	5	5	C	5	3	0	8	D	11	9	6	0																																																																		
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fong-spivak-invitation_T4_443	285	—	(no LaTeX source; 432 × 214 px region)	—	<table><tr><th>$\nearrow$</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>A</td><td>0</td><td>∞</td><td>3</td><td>∞</td></tr><tr><td>B</td><td>2</td><td>0</td><td>∞</td><td>5</td></tr><tr><td>C</td><td>∞</td><td>3</td><td>0</td><td>∞</td></tr><tr><td>D</td><td>∞</td><td>∞</td><td>6</td><td>0</td></tr></table>	$\nearrow$	A	B	C	D	A	0	∞	3	∞	B	2	0	∞	5	C	∞	3	0	∞	D	∞	∞	6	0																																																																		
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fong-spivak-invitation_T4_443	285	—	(no LaTeX source; 680 × 214 px region)	—	<table><tr><th>$\mathcal{X}(\nearrow)$</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>A</td><td>M</td><td>{boat}</td><td>$\emptyset$</td><td>{boat}</td></tr><tr><td>B</td><td>$\emptyset$</td><td>M</td><td>$\emptyset$</td><td>{boat}</td></tr><tr><td>C</td><td>{foot, boat}</td><td>{boat}</td><td>M</td><td>M</td></tr><tr><td>D</td><td>{foot}</td><td>$\emptyset$</td><td>{foot, car}</td><td>M</td></tr></table>	$\mathcal{X}(\nearrow)$	A	B	C	D	A	M	{boat}	$\emptyset$	{boat}	B	$\emptyset$	M	$\emptyset$	{boat}	C	{foot, boat}	{boat}	M	M	D	{foot}	$\emptyset$	{foot, car}	M																																																																		
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fong-spivak-invitation_T4_443	286	—	(no LaTeX source; 326 × 176 px region)	—	<table><tr><th>$M(\nearrow)$</th><th>A</th><th>B</th><th>C</th></tr><tr><td>A</td><td>∞</td><td>10</td><td>10</td></tr><tr><td>B</td><td>6</td><td>∞</td><td>6</td></tr><tr><td>C</td><td>10</td><td>10</td><td>∞</td></tr></table>	$M(\nearrow)$	A	B	C	A	∞	10	10	B	6	∞	6	C	10	10	∞																																																																											
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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																	
fong-spivak-invitation_T4_288	288	—	(no LaTeX source; 374 × 134 px region)	—	<table><tr><td>$\Rightarrow$</td><td>false</td><td>true</td></tr><tr><td>false</td><td>true</td><td>true</td></tr><tr><td>true</td><td>false</td><td>true</td></tr></table>	$\Rightarrow$	false	true	false	true	true	true	false	true																																								
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fong-spivak-invitation_T4_290	290	—	(no LaTeX source; 694 × 300 px region)	—	<table><tr><td>$\nearrow$</td><td>v_1</td><td>f_1</td><td>$f_1 \circ f_2$</td><td>v_2</td><td>f_2</td><td>v_3</td></tr><tr><td>v_1</td><td>v_1</td><td>f_1</td><td>$f_1 \circ f_2$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td></tr><tr><td>f_1</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>f_1</td><td>$f_1 \circ f_2$</td><td>$\boxtimes$</td></tr><tr><td>$f_1 \circ f_2$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$f_1 \circ f_2$</td></tr><tr><td>v_2</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>v_2</td><td>f_2</td><td>$\boxtimes$</td></tr><tr><td>f_2</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>f_2</td></tr><tr><td>v_3</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>v_3</td></tr></table>	$\nearrow$	v_1	f_1	$f_1 \circ f_2$	v_2	f_2	v_3	v_1	v_1	f_1	$f_1 \circ f_2$	$\boxtimes$	$\boxtimes$	$\boxtimes$	f_1	$\boxtimes$	$\boxtimes$	$\boxtimes$	f_1	$f_1 \circ f_2$	$\boxtimes$	$f_1 \circ f_2$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$f_1 \circ f_2$	v_2	$\boxtimes$	$\boxtimes$	$\boxtimes$	v_2	f_2	$\boxtimes$	f_2	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	f_2	v_3	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	v_3
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fong-spivak-invitation_T4_295	295	—	(no LaTeX source; 532 × 256 px region)	—	<table><tr><th>Arrow</th><th>source</th><th>target</th></tr><tr><td>Mngr</td><td>Employee</td><td>Employee</td></tr><tr><td>WorksIn</td><td>Employee</td><td>Department</td></tr><tr><td>Secr</td><td>Department</td><td>Employee</td></tr><tr><td>FName</td><td>Employee</td><td>string</td></tr><tr><td>DName</td><td>Department</td><td>string</td></tr></table>	Arrow	source	target	Mngr	Employee	Employee	WorksIn	Employee	Department	Secr	Department	Employee	FName	Employee	string	DName	Department	string																															
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fong-spivak-invitation_T4_295	295	—	(no LaTeX source; 196 × 256 px region)	—	<table><tr><th>Vertex</th></tr><tr><td>Department</td></tr><tr><td>Employee</td></tr><tr><td>string</td></tr></table>	Vertex	Department	Employee	string																																													
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fong-spivak-invitation_T4_296	296	—	(no LaTeX source; 826 × 354 px region)	—	Arrow source target Vertex Gr G State next DDS State next Set <table><tr><th>State</th><th>next</th></tr><tr><td>1</td><td>4</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>5</td></tr><tr><td>4</td><td>5</td></tr><tr><td>5</td><td>5</td></tr><tr><td>6</td><td>7</td></tr><tr><td>7</td><td>6</td></tr></table>	State	next	1	4	2	4	3	5	4	5	5	5	6	7	7	6																																	
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fong-spivak-invitation_T4_296	296	—	(no LaTeX source; 330 × 316 px region)	—	<table><tr><th>Arrow</th><th>source</th><th>target</th></tr><tr><td>1</td><td>4</td><td>1</td></tr><tr><td>2</td><td>4</td><td>2</td></tr><tr><td>3</td><td>5</td><td>3</td></tr><tr><td>4</td><td>5</td><td>4</td></tr><tr><td>5</td><td>5</td><td>5</td></tr><tr><td>6</td><td>7</td><td>6</td></tr><tr><td>7</td><td>6</td><td>7</td></tr></table>	Arrow	source	target	1	4	1	2	4	2	3	5	3	4	5	4	5	5	5	6	7	6	7	6	7																									
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fong-spivak-invitation_T4_296	296	—	(no LaTeX source; 120 × 316 px region)	—	<table><tr><th>Vertex</th></tr><tr><td>1</td></tr><tr><td>2</td></tr><tr><td>3</td></tr><tr><td>4</td></tr><tr><td>5</td></tr><tr><td>6</td></tr><tr><td>7</td></tr></table>	Vertex	1	2	3	4	5	6	7																																									
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fong-spivak-invitation_T4_297	297	—	(no LaTeX source; 872 × 298 px region)	—	<table><tr><th>Email</th><th>sent_by</th><th>received_by</th><th>Address</th></tr><tr><td>Em_1</td><td>Bob</td><td>Grace</td><td>Bob</td></tr><tr><td>Em_2</td><td>Grace</td><td>Pat</td><td>Doug</td></tr><tr><td>Em_3</td><td>Bob</td><td>Emmy</td><td>Emmy</td></tr><tr><td>Em_4</td><td>Sue</td><td>Doug</td><td>Grace</td></tr><tr><td>Em_5</td><td>Doug</td><td>Sue</td><td>Pat</td></tr><tr><td>Em_6</td><td>Bob</td><td>Bob</td><td>Sue</td></tr></table>	Email	sent_by	received_by	Address	Em_1	Bob	Grace	Bob	Em_2	Grace	Pat	Doug	Em_3	Bob	Emmy	Emmy	Em_4	Sue	Doug	Grace	Em_5	Doug	Sue	Pat	Em_6	Bob	Bob	Sue																					
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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																
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fong-spivak-invitation_TAB_061	301	—	(no LaTeX source; 382 × 216 px region)	—	<table><tr><th>$\Phi \circ \Psi$</th><th>p</th><th>q</th><th>r</th><th>s</th></tr><tr><td>A</td><td>22</td><td>24</td><td>20</td><td>21</td></tr><tr><td>B</td><td>16</td><td>18</td><td>14</td><td>15</td></tr><tr><td>C</td><td>19</td><td>21</td><td>17</td><td>18</td></tr><tr><td>D</td><td>11</td><td>13</td><td>9</td><td>10</td></tr></table>	$\Phi \circ \Psi$	p	q	r	s	A	22	24	20	21	B	16	18	14	15	C	19	21	17	18	D	11	13	9	10							
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fong-spivak-invitation_TAB_060	325	—	(no LaTeX source; 522 × 174 px region)	—	<table><tr><th></th><th>r1</th><th>r2</th><th>r3</th><th>c1</th><th>i1</th></tr><tr><td>$s(-)$</td><td>dl</td><td>ul</td><td>ur</td><td>ul</td><td>dl</td></tr><tr><td>$t(-)$</td><td>ul</td><td>ur</td><td>dr</td><td>ur</td><td>dr</td></tr><tr><td>$\ell(-)$</td><td>1Ω</td><td>2Ω</td><td>1Ω</td><td>3F</td><td>1H</td></tr></table>		r1	r2	r3	c1	i1	$s(-)$	dl	ul	ur	ul	dl	$t(-)$	ul	ur	dr	ur	dr	$\ell(-)$	1 Ω	2 Ω	1 Ω	3 F	1 H								
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$t''(-)$	ul	m	dr	m	dr	r	m																														
$\ell''(-)$	1 Ω	2 Ω	1 Ω	3 F	1 H	5 Ω	8 Ω																														
fong-spivak-invitation_TAB_068	331	—	(no LaTeX source; 614 × 216 px region)	—	<table><tr><th>P</th><th>Q</th><th>$P \wedge Q$</th><th>$P = (P \wedge Q)$</th></tr><tr><td>true</td><td>true</td><td>true</td><td>true</td></tr><tr><td>true</td><td>false</td><td>false</td><td>false</td></tr><tr><td>false</td><td>true</td><td>false</td><td>true</td></tr><tr><td>false</td><td>false</td><td>false</td><td>true</td></tr></table>	P	Q	$P \wedge Q$	$P = (P \wedge Q)$	true	true	true	true	true	false	false	false	false	true	false	true	false	false	false	true												
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Image regions — rendered against scan

Two comparable cells per row. **Rendered**: the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan**: the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

Identifier	Page	Class	Source	Author source	Rendered	Scan
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contested: 4 rows

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source

source

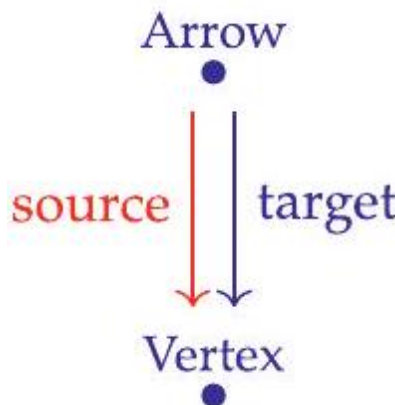
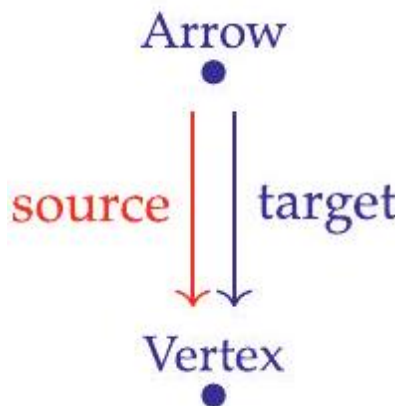
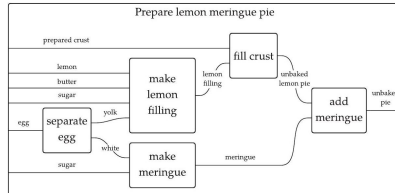
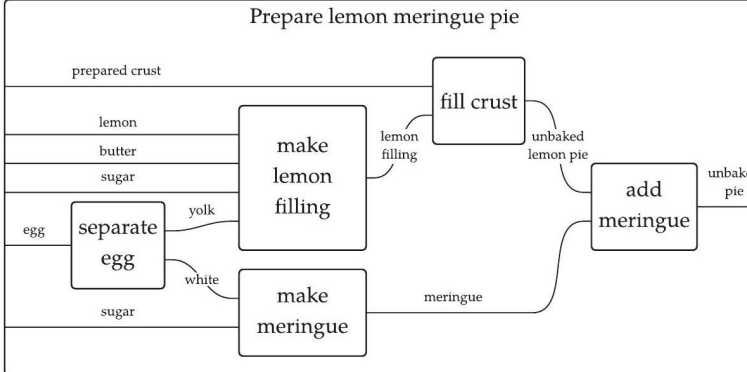
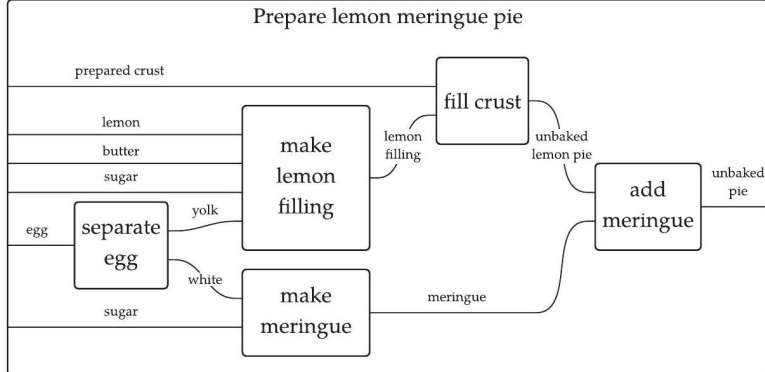
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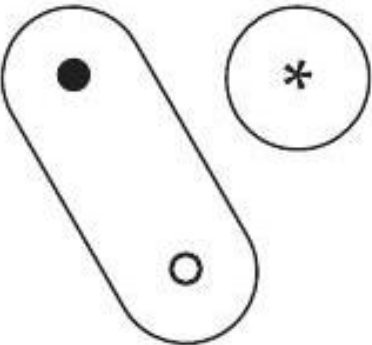
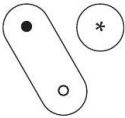
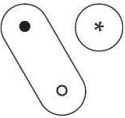
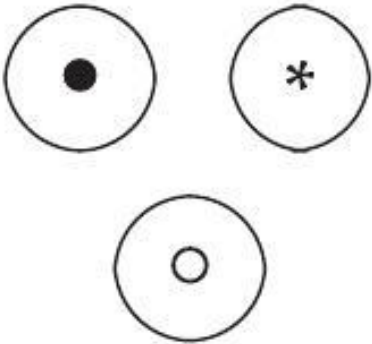
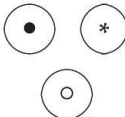
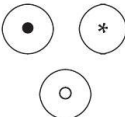
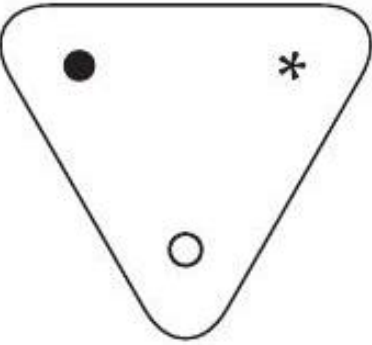
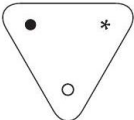
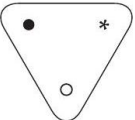
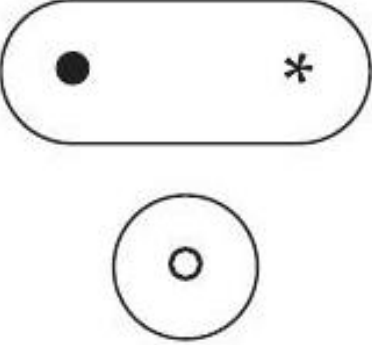
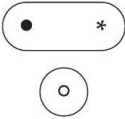
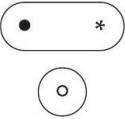
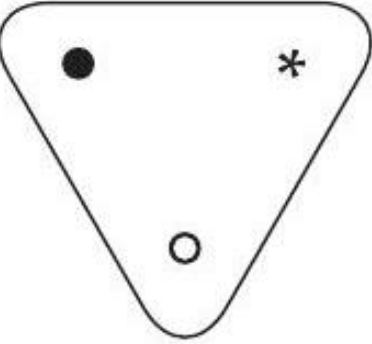
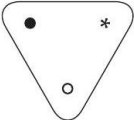
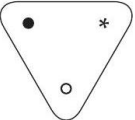
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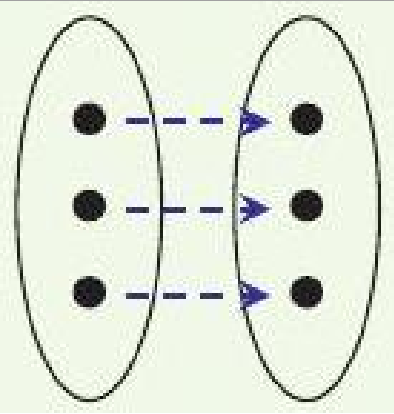
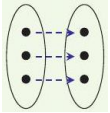
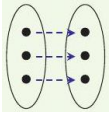
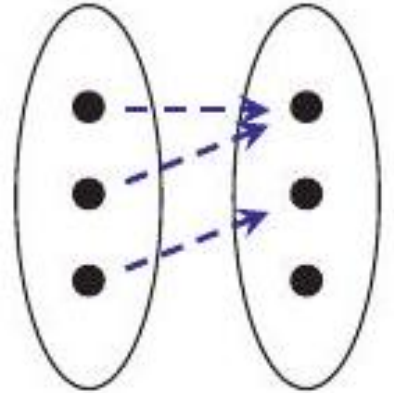
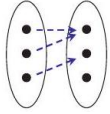
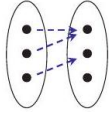
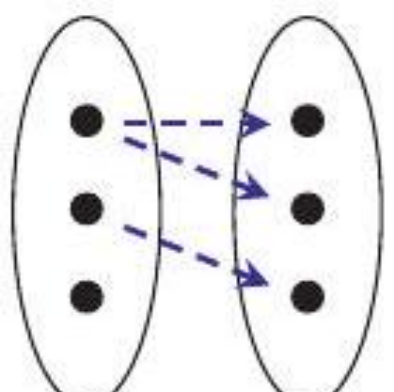
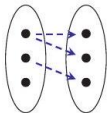
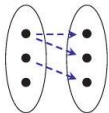
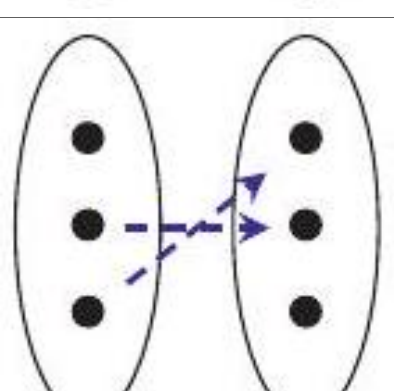
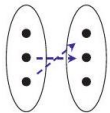
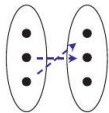
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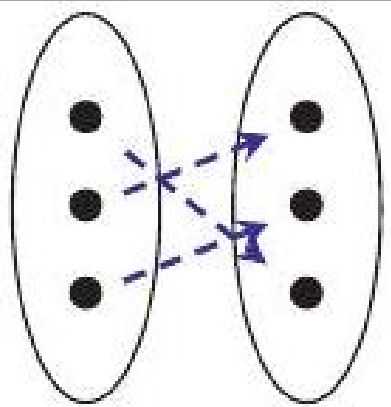
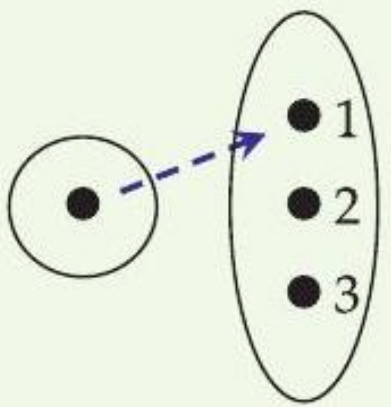
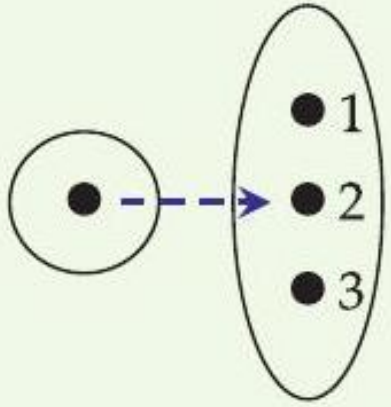
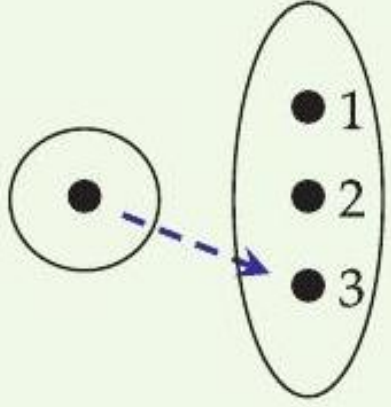
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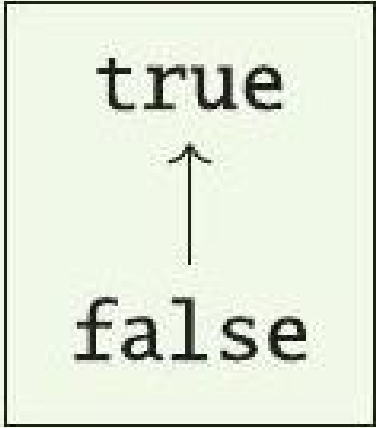
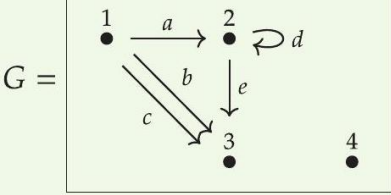
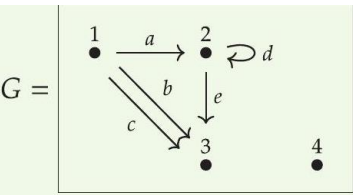
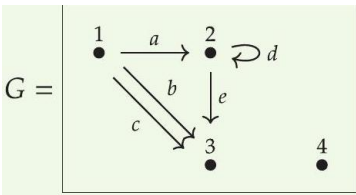
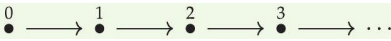
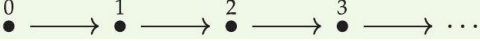
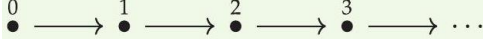
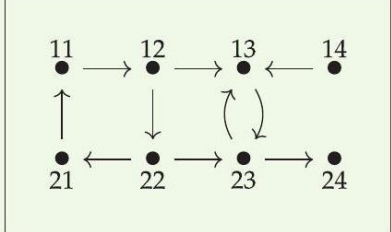
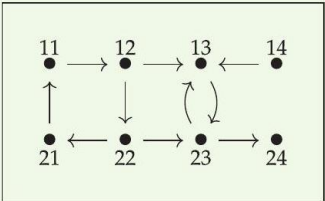
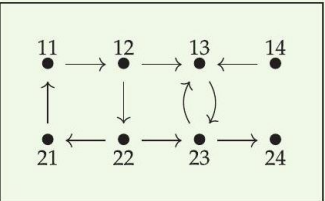
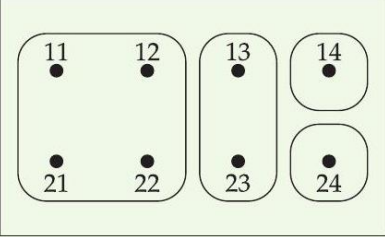
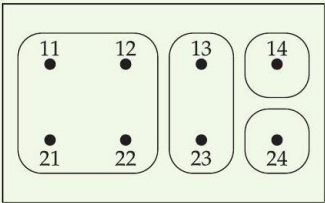
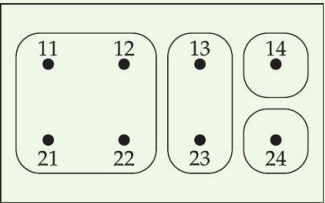
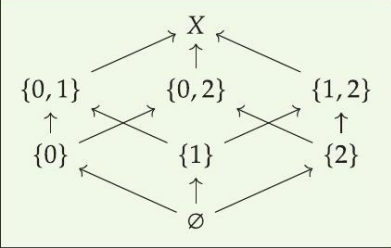
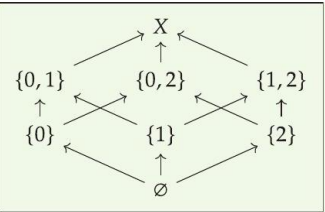
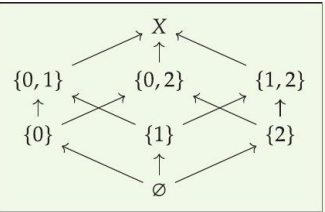
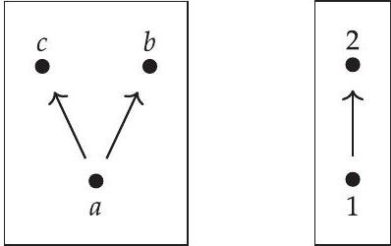
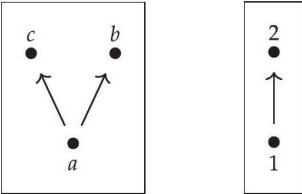
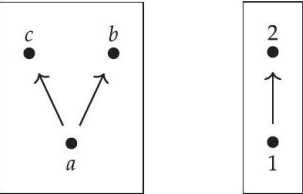
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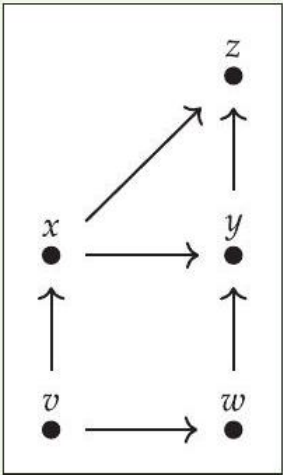
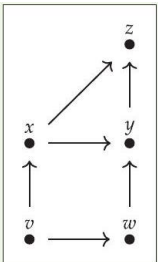
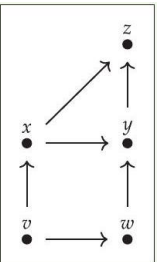
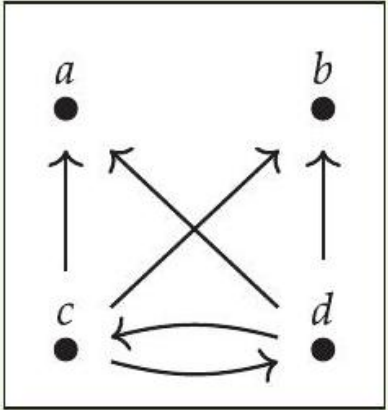
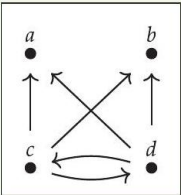
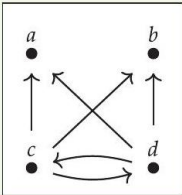
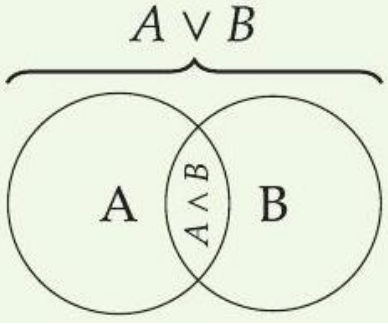
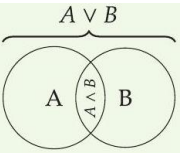
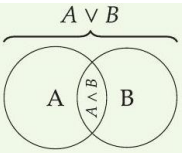
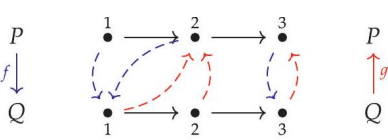
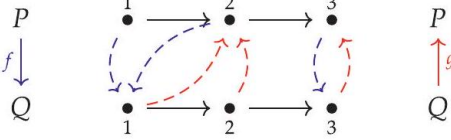
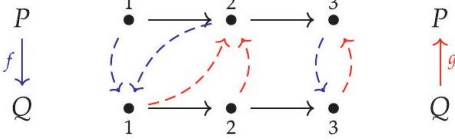
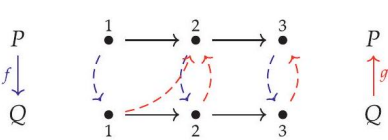
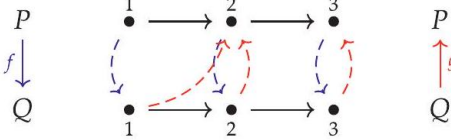
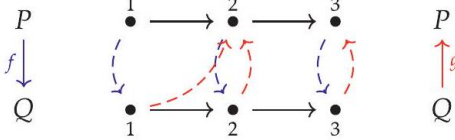
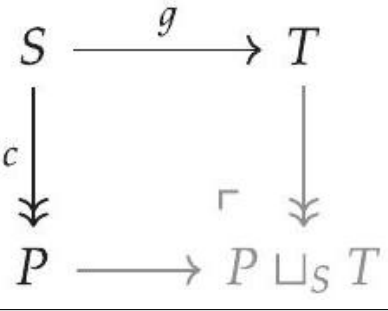
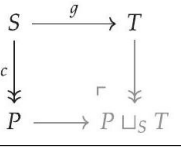
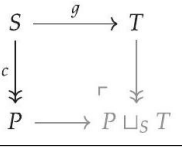
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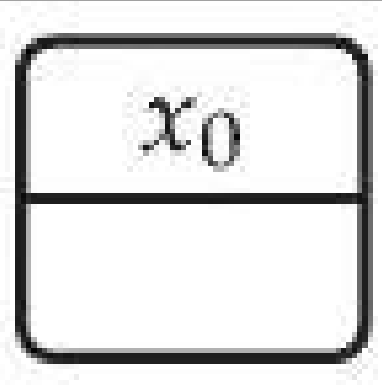
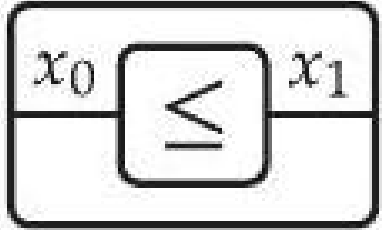
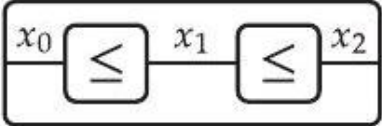
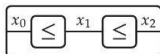
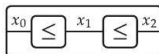
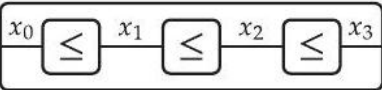
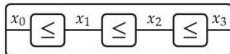
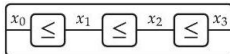
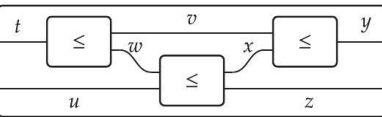
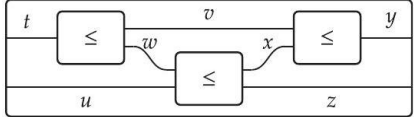
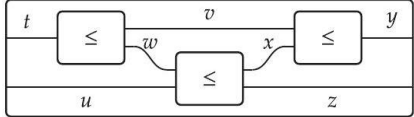
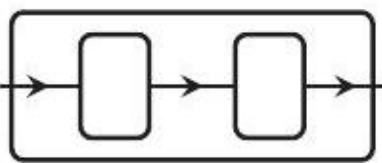
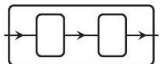
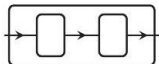
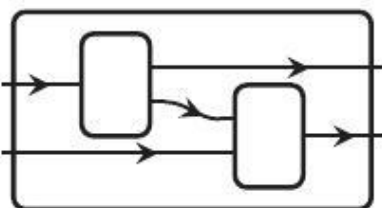
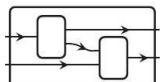
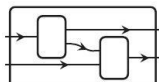
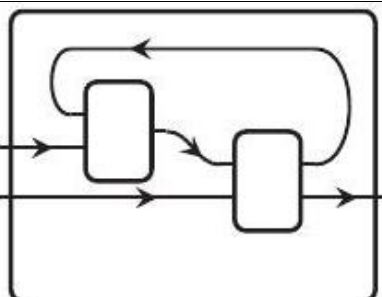
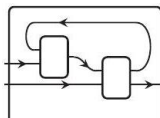
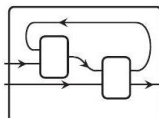
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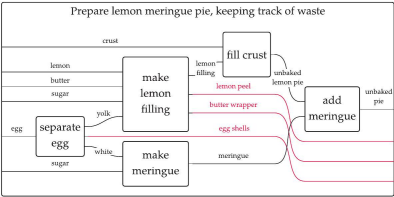
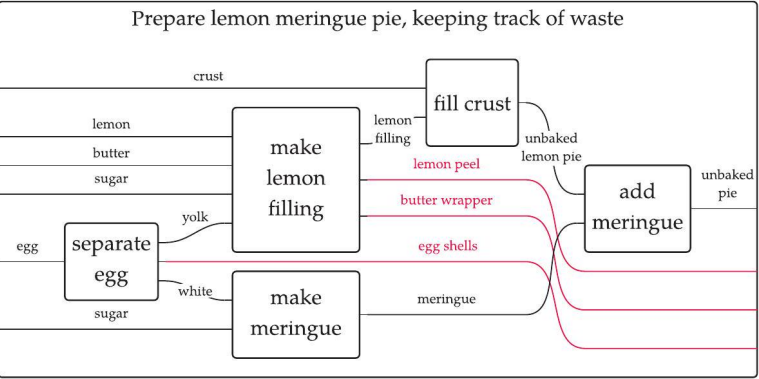
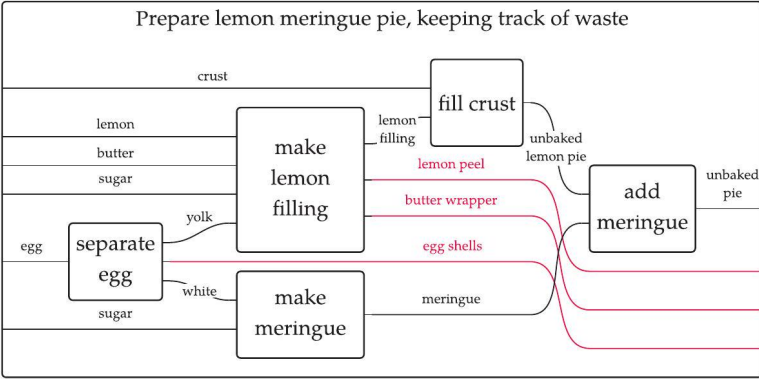
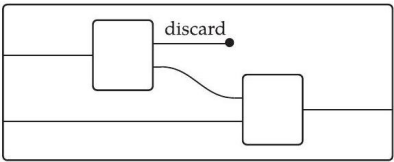
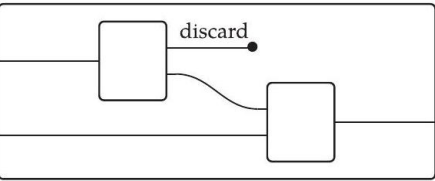
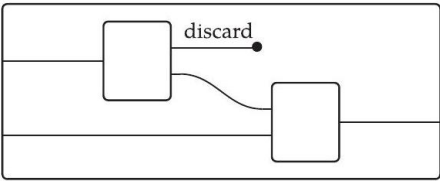
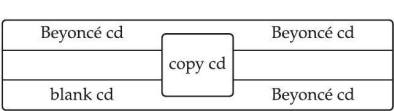
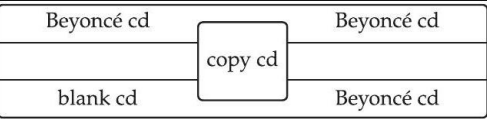
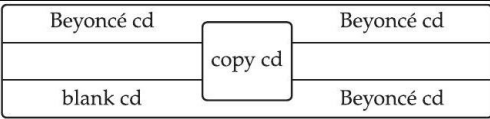
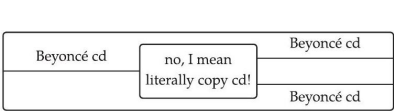
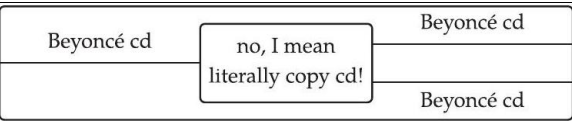
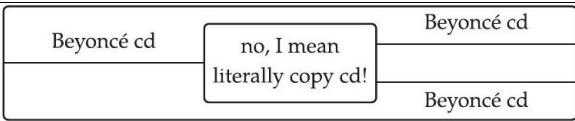
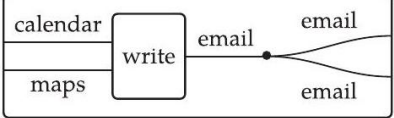
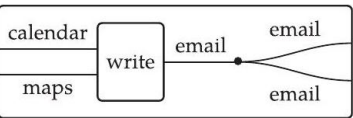
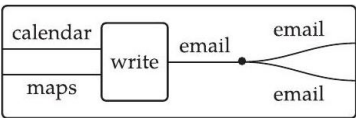
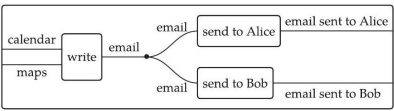
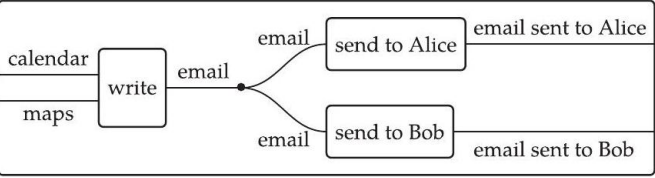
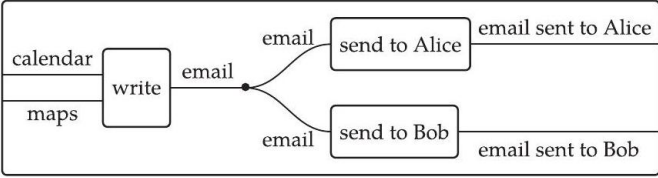
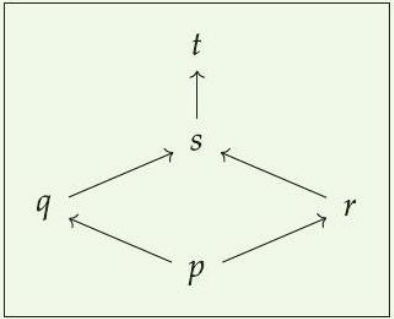
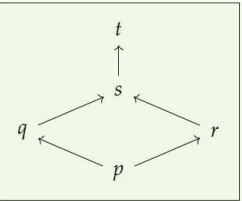
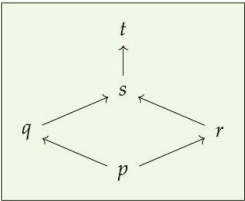
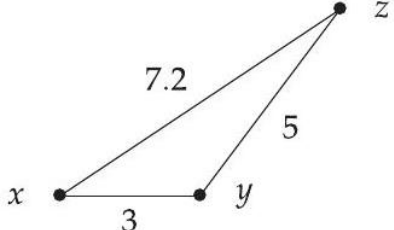
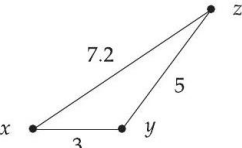
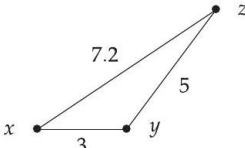
fong-spivak-invitation_D0_033.jpg	033	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-033_396_846_512_514.jpg			
fong-spivak-invitation_D0_033.jpg	033	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-033_124_780_1148_548.jpg			
fong-spivak-invitation_D0_033.jpg	033	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-033_566_1056_1651_412.jpg			
fong-spivak-invitation_D0_036.jpg	036	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-036_344_330_1188_410.jpg	$P :=$	$P :=$	$P :=$
fong-spivak-invitation_D0_036.jpg	036	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-036_344_286_1188_818.jpg	$Q :=$	$Q :=$	$Q :=$

fong-spivak-invitation_D0_036	036	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-036_344_282_1188_1184.jpg	$R :=$ 	$R :=$ 	$R :=$ 
fong-spivak-invitation_D0_039	039	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-039_256_242_302_818.jpg			
fong-spivak-invitation_D0_039	039	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-039_194_236_1657_820.jpg			
fong-spivak-invitation_D0_042	042	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-042_186_602_996_661.jpg			
fong-spivak-invitation_D0_042	042	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-042_190_604_1364_659.jpg			
fong-spivak-invitation_D0_043	043	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-043_174_228_824_580.jpg			

fong-spivak-invitation_D0_043	043	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-043_180_252_818_1046.jpg			
fong-spivak-invitation_D0_043	043	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-043_310_846_1853_514.jpg			
fong-spivak-invitation_D0_044	044	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-044_266_1178_578_350.jpg			
fong-spivak-invitation_D0_046	046	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-046_312_800_864_540.jpg			
fong-spivak-invitation_D0_050	050	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-050_60_234_474_822.jpg			
fong-spivak-invitation_D0_053	053	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-053_598_1207_270_336.jpg			
fong-spivak-invitation_D0_053	053	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-053_280_1188_1104_344.jpg			
fong-spivak-invitation_D0_056	056	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-056_84_134_1553_874.jpg			

fong-spivak-invitation_D10_056.jpg	056	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-056_72_74_1801_368.jpg			
fong-spivak-invitation_D10_056.jpg	056	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-056_72_118_1801_530.jpg			
fong-spivak-invitation_D10_056.jpg	056	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-056_74_212_1801_736.jpg			
fong-spivak-invitation_D10_056.jpg	056	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-056_74_304_1801_1036.jpg			
fong-spivak-invitation_D10_056.jpg	056	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-056_158_538_2041_672.jpg			
fong-spivak-invitation_D10_057.jpg	057	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-057_90_206_644_336.jpg			
fong-spivak-invitation_D10_057.jpg	057	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-057_116_208_632_640.jpg			
fong-spivak-invitation_D10_057.jpg	057	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-057_168_214_608_964.jpg			

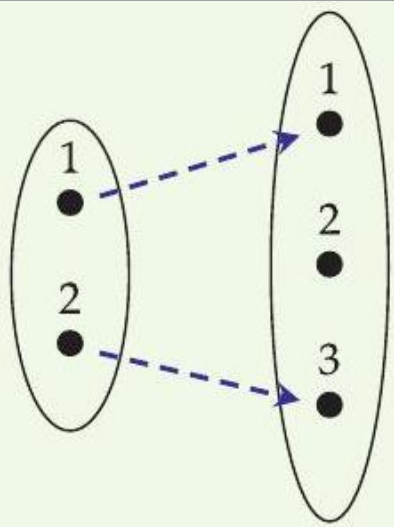
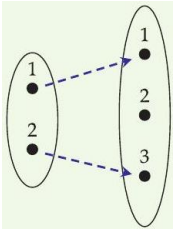
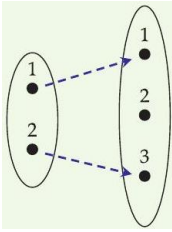
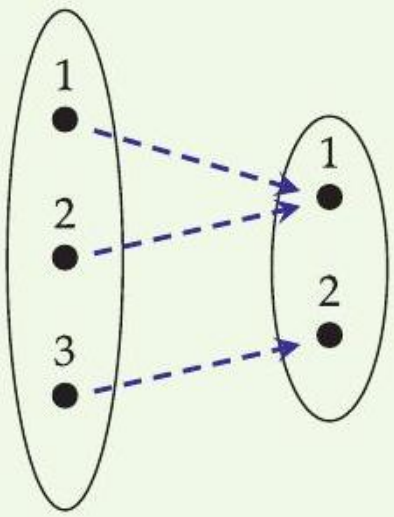
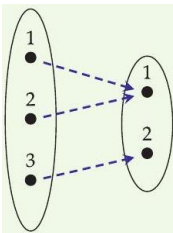
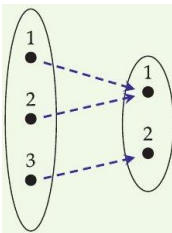
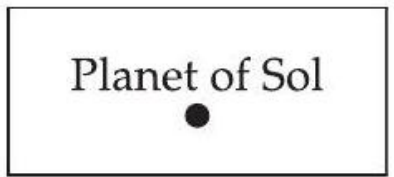
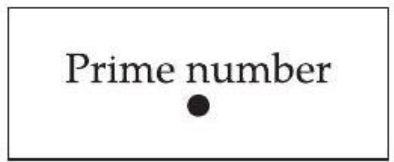
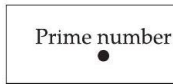
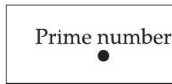
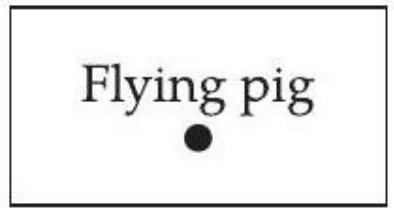
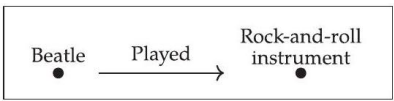
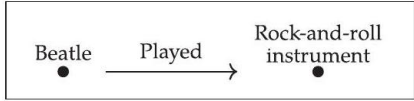
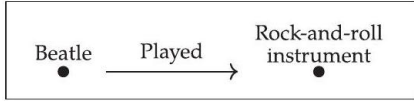
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fong-spivak-invitation_D0_009	058	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-058_116_442_1080_722.jpg			
fong-spivak-invitation_D0_010	058	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-058_36_246_1527_816.jpg			
fong-spivak-invitation_D0_011	058	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-058_76_922_1743_478.jpg			
fong-spivak-invitation_D0_012	058	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-058_66_548_1955_668.jpg			
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fong-spivak-invitation_D0_014	059	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-059_86_304_618_784.jpg			
fong-spivak-invitation_D0_015	059	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-059_64_164_1527_688.jpg			
fong-spivak-invitation_D0_016	059	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-059_64_166_1527_1024.jpg			
fong-spivak-invitation_D0_017	059	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-059_74_352_2005_762.jpg			
fong-spivak-invitation_D0_018	060	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-060_220_744_260_568.jpg			
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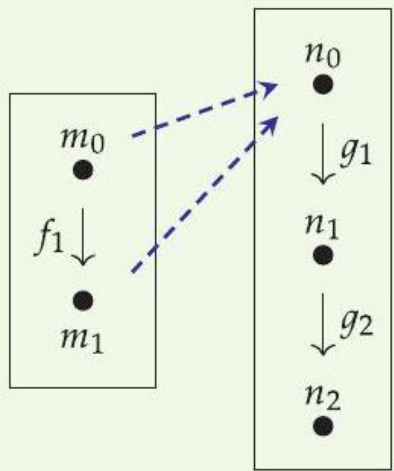
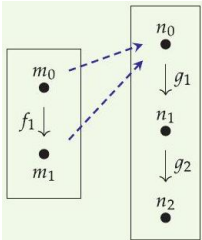
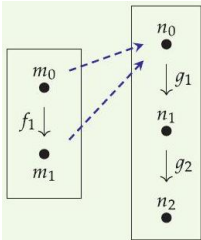
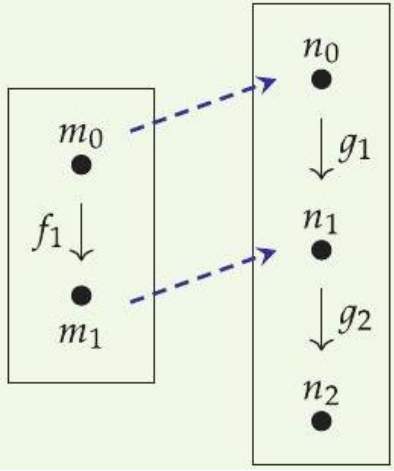
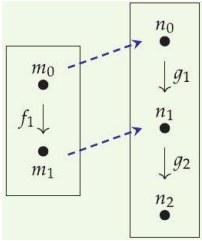
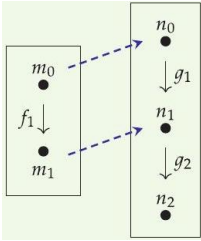
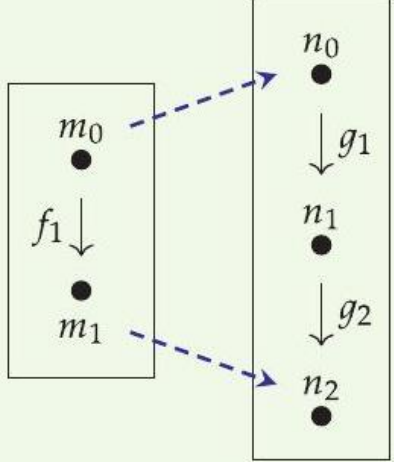
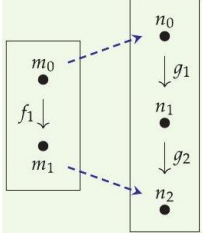
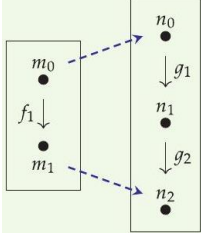
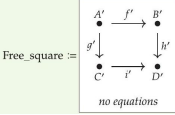
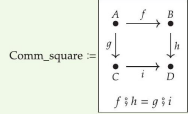
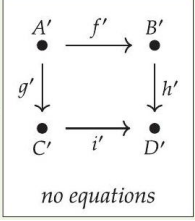
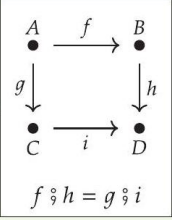
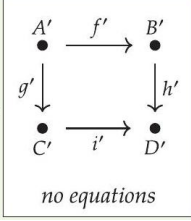
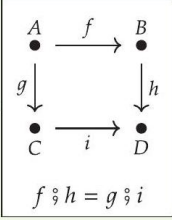
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fong-spivak-invitation_D0_064	064	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-064_154_644_648_618.jpg			
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fong-spivak-invitation_D0_070	070	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-070_262_322_1673_798.jpg			
fong-spivak-invitation_D0_073	073	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-073_203_327_382_775.jpg			

fong-spivak-invitation_D0_075	075	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-075_330_384_602_398.jpg	$X :=$	$X :=$	$X :=$
fong-spivak-invitation_D0_075	075	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-075_330_344_604_1052.jpg	$=: Y$	$=: Y$	$=: Y$
fong-spivak-invitation_D0_080	080	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-080_116_444_2017_540.jpg	$\mathcal{X} :=$	$\mathcal{X} :=$	$\mathcal{X} :=$
fong-spivak-invitation_D0_080	080	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-080_288_234_1939_1108.jpg	$=: \mathcal{Y}$	$=: \mathcal{Y}$	$=: \mathcal{Y}$
fong-spivak-invitation_D0_081	081	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-081_328_652_536_610.jpg	$\mathcal{X} \times \mathcal{Y} =$	$\mathcal{X} \times \mathcal{Y} =$	$\mathcal{X} \times \mathcal{Y} =$
fong-spivak-invitation_D0_088	088	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-088_278_346_670_370.jpg	$Y :=$	$Y :=$	$Y :=$

fong-spivak-invitation_D10_092	092	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-092_294_1042_814_386.jpg			
fong-spivak-invitation_D10_093	093	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-093_396_1032_258_388.jpg			
fong-spivak-invitation_D10_093	093	— (dup)	no image for this region (500 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
fong-spivak-invitation_D10_094	094	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-094_488_1194_620_342.jpg			
fong-spivak-invitation_D10_097	097	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-097_176_106_962_884.jpg			
fong-spivak-invitation_D10_098	098	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-098_312_984_550_446.jpg			

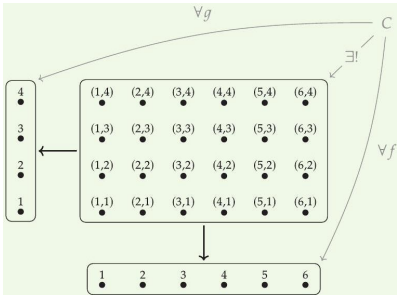
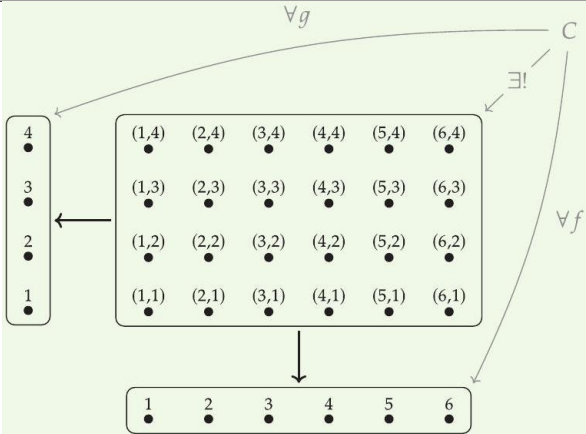
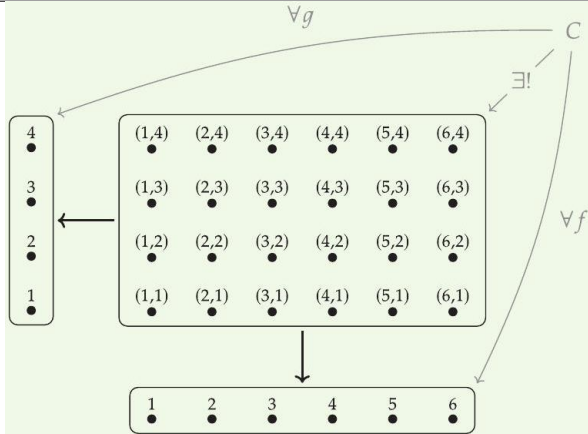
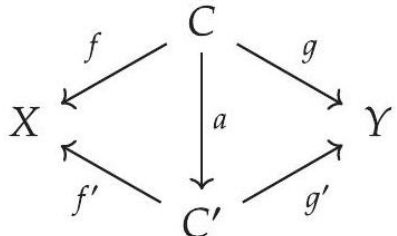
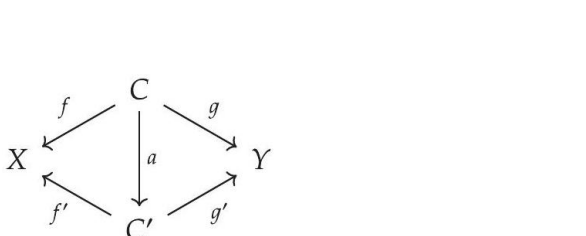
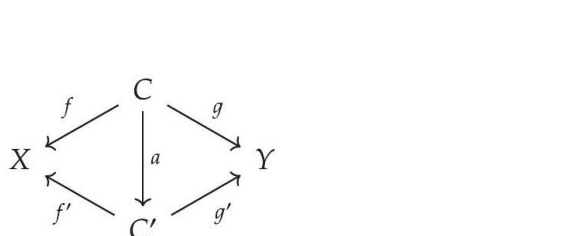
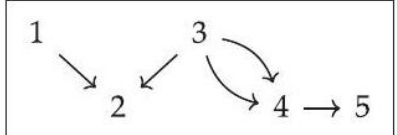
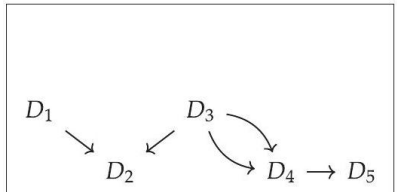
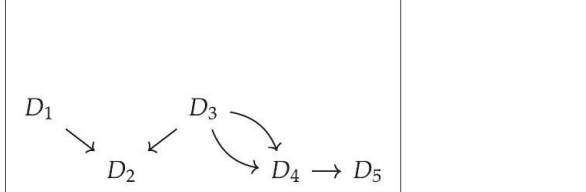
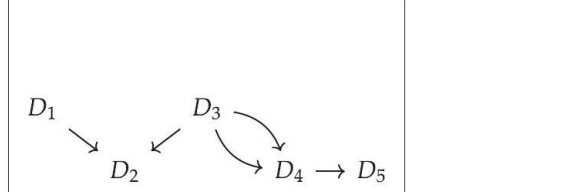
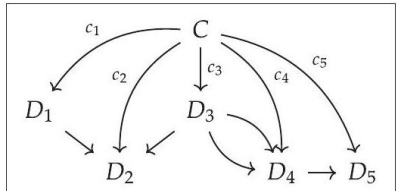
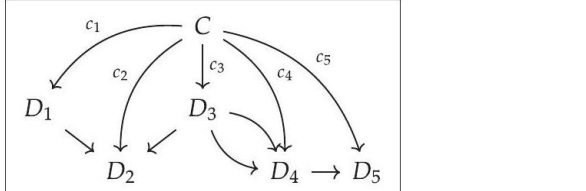
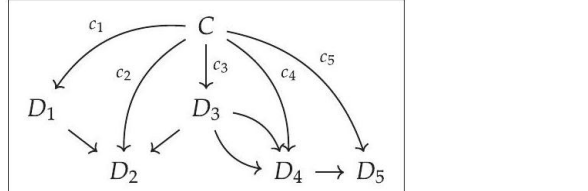
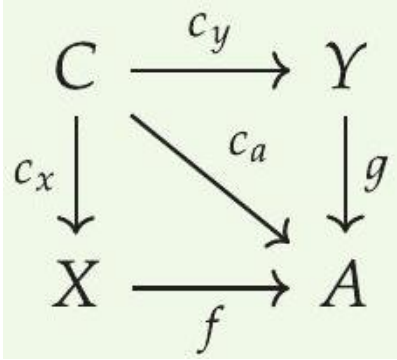
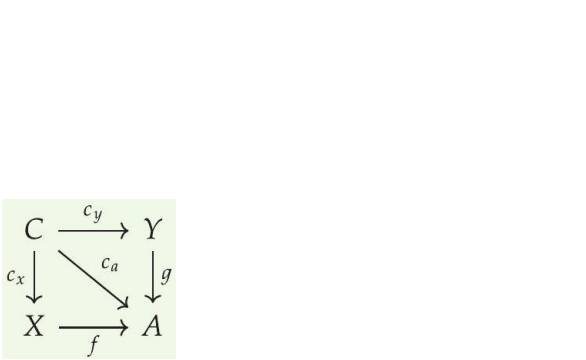
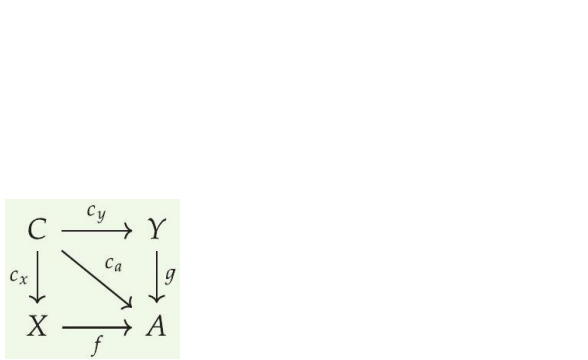
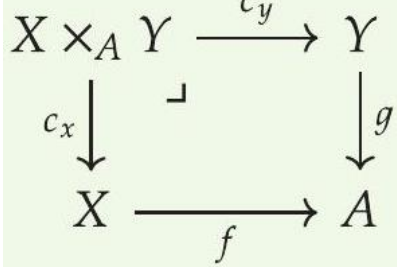
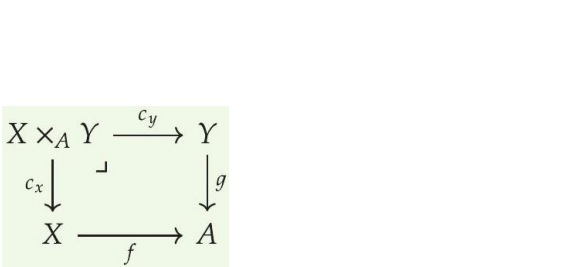
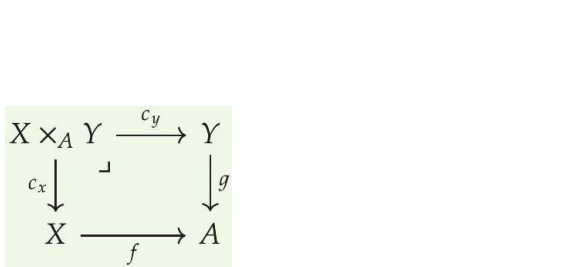
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fong-spivak-invitation_D10_100	100	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-100_238_322_314_364.jpg			
fong-spivak-invitation_D10_100	100	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-100_282_320_270_780.jpg			
fong-spivak-invitation_D10_100	100	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-100_282_320_270_1196.jpg			
fong-spivak-invitation_D10_100	100	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-100_272_1202_1018_346.jpg			

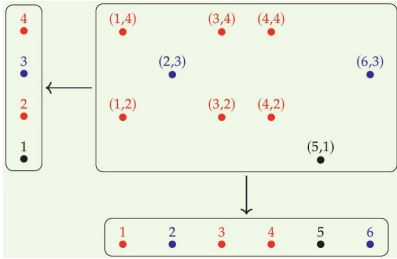
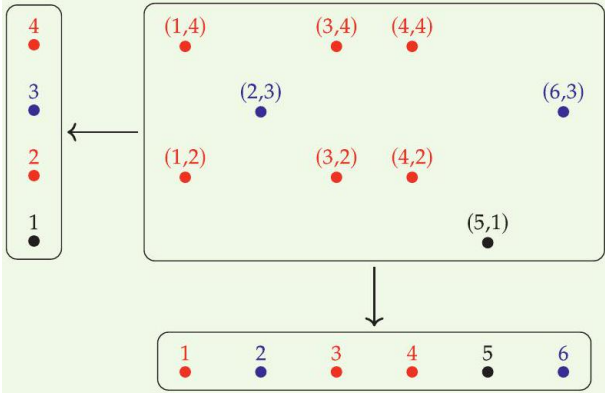
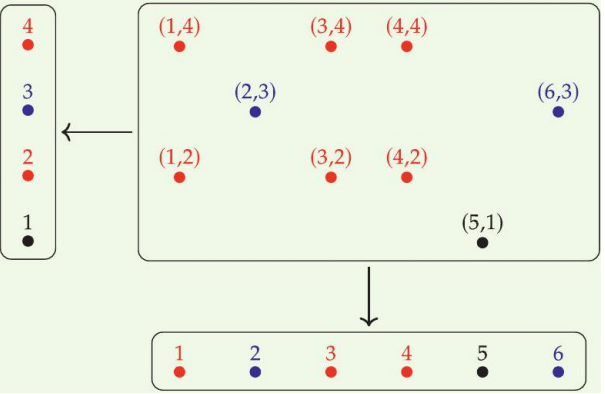
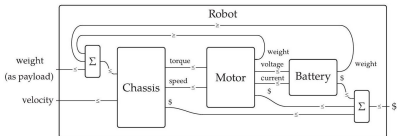
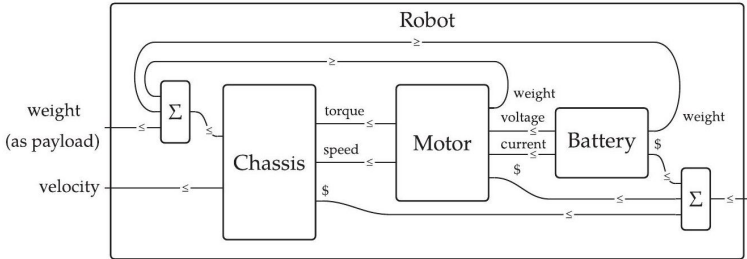
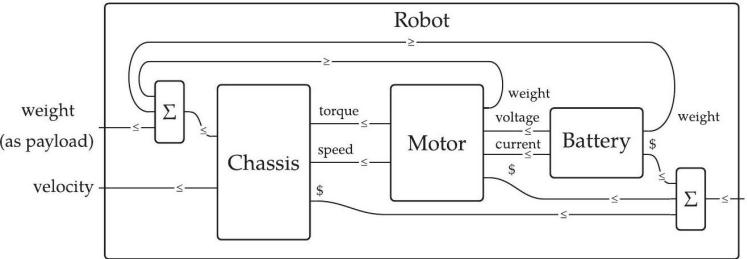
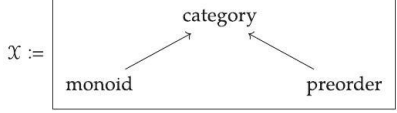
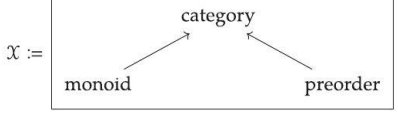
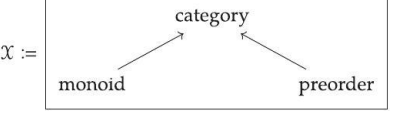
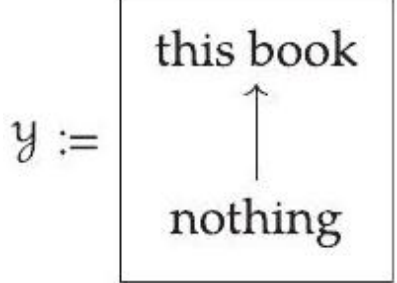
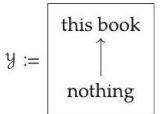
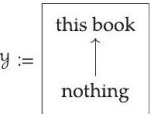
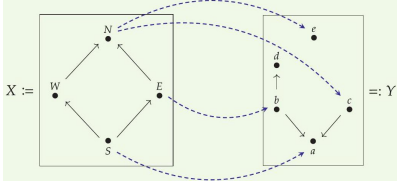
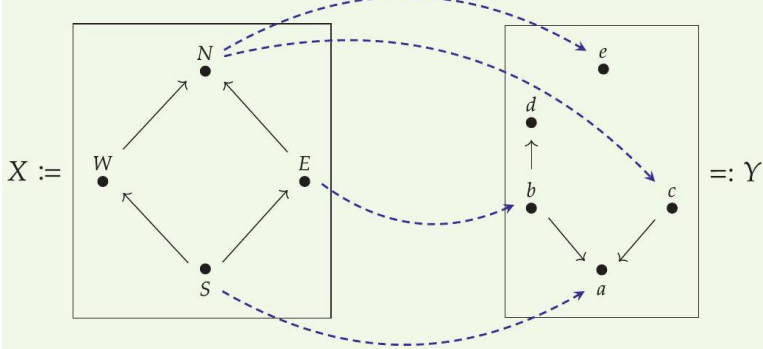
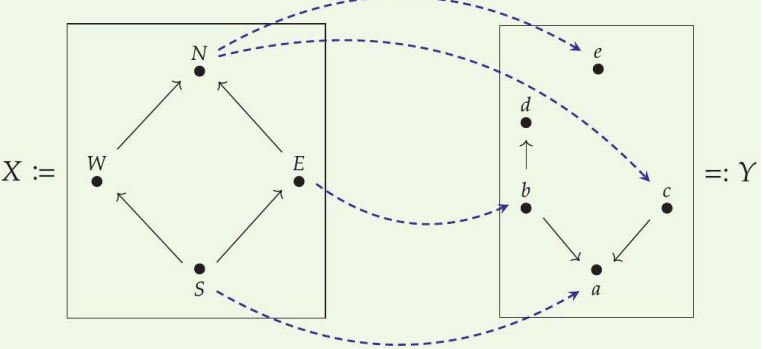
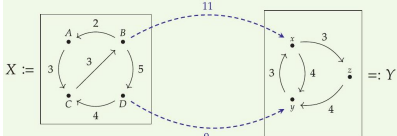
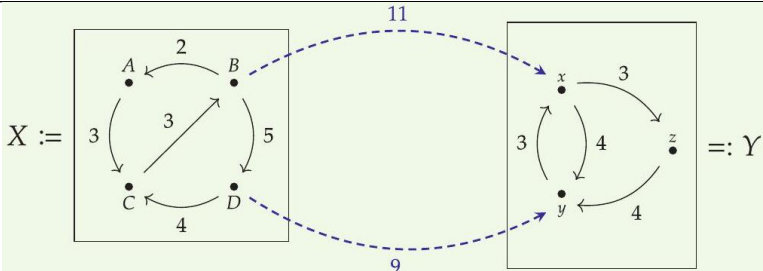
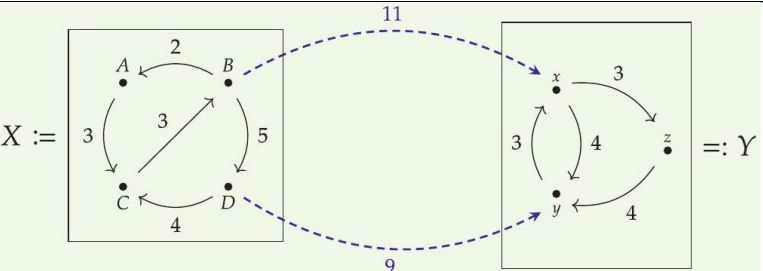
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fong-spivak-invitation_D1_004	104	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-104_110_238_1509_342.jpg			
fong-spivak-invitation_D1_004	104	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-104_110_262_1509_824.jpg			
fong-spivak-invitation_D1_004	104	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-104_114_210_1505_1331.jpg			
fong-spivak-invitation_D1_005	105	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-105_134_540_626_668.jpg			

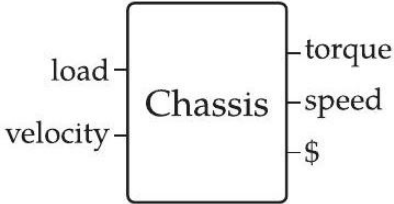
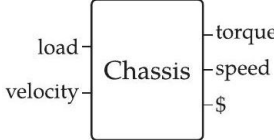
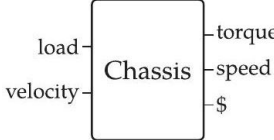
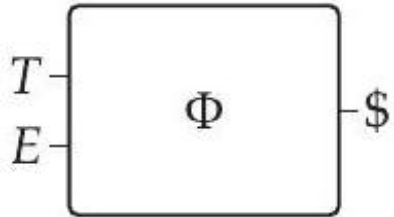
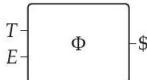
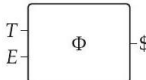
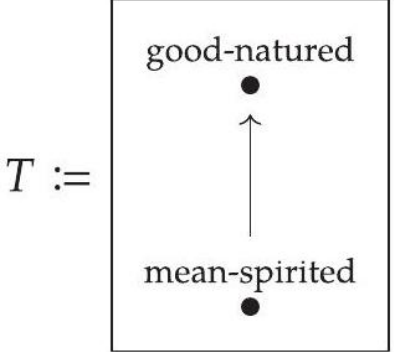
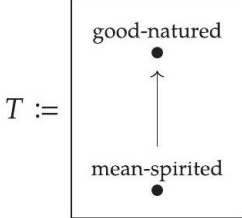
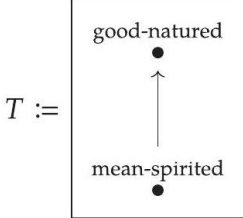
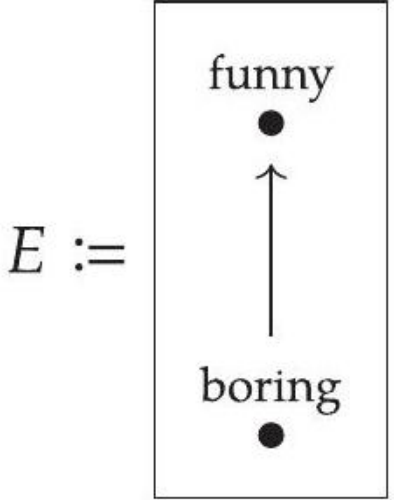
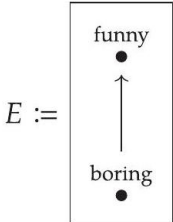
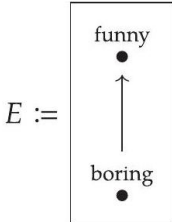
fong-spivak-invitation_D11040	105	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-105_312_260_1917_364.jpg			
fong-spivak-invitation_D11040	105	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-105_310_260_1921_808.jpg			
fong-spivak-invitation_D11040	105	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-105_308_260_1921_1251.jpg			
fong-spivak-invitation_D11040	106	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-106_312_1056_954_410.jpg	Free_square :=  Comm_square := 	Free_square :=  Comm_square := 	Free_square :=  Comm_square := 

fong-spivak-invitation_D11_09	109	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-109_222_132_1048_638.jpg			
fong-spivak-invitation_D11_09	109	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-109_174_342_1098_934.jpg			
fong-spivak-invitation_D11_09	109	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-109_148_230_1937_822.jpg			
fong-spivak-invitation_D11_10	110	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-110_200_342_1068_770.jpg			
fong-spivak-invitation_D11_10	110	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-110_276_1192_1801_344.jpg			
fong-spivak-invitation_D11_14	114	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-114_150_450_388_400.jpg			
fong-spivak-invitation_D11_14	114	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-114_150_434_388_1042.jpg			

fong-spivak-invitation_D1_115	115	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-115_204_1112_944_380.jpg	$\text{Gr} := \begin{array}{c} \text{Arrow} \xrightarrow{\text{source}} \text{Vertex} \\ \text{target} \\ \text{no equations} \end{array} \quad \text{DDS} := \begin{array}{c} \text{State} \\ \bullet \\ \updownarrow \\ \text{next} \\ \text{no equations} \end{array}$	$\text{Gr} := \begin{array}{c} \text{Arrow} \xrightarrow{\text{source}} \text{Vertex} \\ \text{target} \\ \text{no equations} \end{array} \quad \text{DDS} := \begin{array}{c} \text{State} \\ \bullet \\ \updownarrow \\ \text{next} \\ \text{no equations} \end{array}$	$\text{Gr} := \begin{array}{c} \text{Arrow} \xrightarrow{\text{source}} \text{Vertex} \\ \text{target} \\ \text{no equations} \end{array} \quad \text{DDS} := \begin{array}{c} \text{State} \\ \bullet \\ \updownarrow \\ \text{next} \\ \text{no equations} \end{array}$
fong-spivak-invitation_D1_116	116	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-116_328_542_272_670.jpg	$\begin{array}{c} \text{Arrow} \\ \text{source} \downarrow \text{target} \\ \text{Vertex} \end{array} \xrightarrow{F} \begin{array}{c} \text{State} \\ \bullet \\ \updownarrow \\ \text{next} \end{array}$	$\begin{array}{c} \text{Arrow} \\ \text{source} \downarrow \text{target} \\ \text{Vertex} \end{array} \xrightarrow{F} \begin{array}{c} \text{State} \\ \bullet \\ \updownarrow \\ \text{next} \end{array}$	$\begin{array}{c} \text{Arrow} \\ \text{source} \downarrow \text{target} \\ \text{Vertex} \end{array} \xrightarrow{F} \begin{array}{c} \text{State} \\ \bullet \\ \updownarrow \\ \text{next} \end{array}$
fong-spivak-invitation_D1_116	116	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-116_374_768_1869_556.jpg			
fong-spivak-invitation_D1_117	117	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-117_160_458_1038_452.jpg	$\mathcal{C} \xrightarrow{F} \mathcal{D} \begin{array}{c} \xrightarrow{I} \\ \Downarrow \alpha \\ \xrightarrow{J} \end{array} \mathbf{Set}$	$\mathcal{C} \xrightarrow{F} \mathcal{D} \begin{array}{c} \xrightarrow{I} \\ \Downarrow \alpha \\ \xrightarrow{J} \end{array} \mathbf{Set}$	$\mathcal{C} \xrightarrow{F} \mathcal{D} \begin{array}{c} \xrightarrow{I} \\ \Downarrow \alpha \\ \xrightarrow{J} \end{array} \mathbf{Set}$
fong-spivak-invitation_D1_117	117	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-117_160_256_1038_1168.jpg	$\mathcal{C} \begin{array}{c} \xrightarrow{F \circ I} \\ \Downarrow \alpha_F \\ \xrightarrow{F \circ J} \end{array} \mathbf{Set}$	$\mathcal{C} \begin{array}{c} \xrightarrow{F \circ I} \\ \Downarrow \alpha_F \\ \xrightarrow{F \circ J} \end{array} \mathbf{Set}$	$\mathcal{C} \begin{array}{c} \xrightarrow{F \circ I} \\ \Downarrow \alpha_F \\ \xrightarrow{F \circ J} \end{array} \mathbf{Set}$
fong-spivak-invitation_D1_118	118	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-118_206_480_2059_712.jpg	$\begin{array}{ccc} \mathcal{C}(c, Rd) & \xrightarrow{\alpha_{c,d}} & \mathcal{D}(Lc, d) \\ \mathcal{C}(f, Rg) \downarrow & & \downarrow \mathcal{D}(Lf, g) \\ \mathcal{C}(c', Rd') & \xrightarrow{\alpha_{c',d'}} & \mathcal{D}(Lc', d') \end{array}$	$\begin{array}{ccc} \mathcal{C}(c, Rd) & \xrightarrow{\alpha_{c,d}} & \mathcal{D}(Lc, d) \\ \mathcal{C}(f, Rg) \downarrow & & \downarrow \mathcal{D}(Lf, g) \\ \mathcal{C}(c', Rd') & \xrightarrow{\alpha_{c',d'}} & \mathcal{D}(Lc', d') \end{array}$	$\begin{array}{ccc} \mathcal{C}(c, Rd) & \xrightarrow{\alpha_{c,d}} & \mathcal{D}(Lc, d) \\ \mathcal{C}(f, Rg) \downarrow & & \downarrow \mathcal{D}(Lf, g) \\ \mathcal{C}(c', Rd') & \xrightarrow{\alpha_{c',d'}} & \mathcal{D}(Lc', d') \end{array}$
fong-spivak-invitation_D1_119	119	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-119_172_498_2115_690.jpg	$\mathcal{C}\text{-Inst} \begin{array}{c} \xrightarrow{\Sigma_F} \\ \xleftrightarrow{\Delta_F} \\ \xleftarrow{\Pi_F} \end{array} \mathcal{D}\text{-Inst}$	$\mathcal{C}\text{-Inst} \begin{array}{c} \xrightarrow{\Sigma_F} \\ \xleftrightarrow{\Delta_F} \\ \xleftarrow{\Pi_F} \end{array} \mathcal{D}\text{-Inst}$	$\mathcal{C}\text{-Inst} \begin{array}{c} \xrightarrow{\Sigma_F} \\ \xleftrightarrow{\Delta_F} \\ \xleftarrow{\Pi_F} \end{array} \mathcal{D}\text{-Inst}$
fong-spivak-invitation_D1_120	120	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-120_368_1215_1217_334.jpg	$A := \begin{array}{c} \text{Economy} \xrightarrow{\text{price}} \text{\$} \xleftarrow{\text{price}} \text{First class} \\ \text{position} \searrow \text{string} \swarrow \text{position} \end{array} \xrightarrow{F} \begin{array}{c} \text{\$} \\ \uparrow \text{price} \\ \text{Airline seat} \\ \downarrow \text{position} \\ \text{string} \end{array} =: B$	$A := \begin{array}{c} \text{Economy} \xrightarrow{\text{price}} \text{\$} \xleftarrow{\text{price}} \text{First class} \\ \text{position} \searrow \text{string} \swarrow \text{position} \end{array} \xrightarrow{F} \begin{array}{c} \text{\$} \\ \uparrow \text{price} \\ \text{Airline seat} \\ \downarrow \text{position} \\ \text{string} \end{array} =: B$	$A := \begin{array}{c} \text{Economy} \xrightarrow{\text{price}} \text{\$} \xleftarrow{\text{price}} \text{First class} \\ \text{position} \searrow \text{string} \swarrow \text{position} \end{array} \xrightarrow{F} \begin{array}{c} \text{\$} \\ \uparrow \text{price} \\ \text{Airline seat} \\ \downarrow \text{position} \\ \text{string} \end{array} =: B$
fong-spivak-invitation_D1_124	124	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-124_184_350_712_766.jpg	$\begin{array}{ccccc} & f & C & g & \\ X & \swarrow & \vdots & \searrow & Y \\ & p_X & \langle f, g \rangle & p_Y & \\ & & X \times Y & & \end{array}$	$\begin{array}{ccccc} & f & C & g & \\ X & \swarrow & \vdots & \searrow & Y \\ & p_X & \langle f, g \rangle & p_Y & \\ & & X \times Y & & \end{array}$	$\begin{array}{ccccc} & f & C & g & \\ X & \swarrow & \vdots & \searrow & Y \\ & p_X & \langle f, g \rangle & p_Y & \\ & & X \times Y & & \end{array}$

fong-spivak-invitation_D11_244	124	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-124_562_760_1665_560.jpg			
fong-spivak-invitation_D11_245	125	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-125_220_354_1893_762.jpg			
fong-spivak-invitation_D11_246	126	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-126_118_318_1357_782.jpg			
fong-spivak-invitation_D11_246	126	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-126_268_524_1625_370.jpg			
fong-spivak-invitation_D11_246	126	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-126_266_522_1627_988.jpg			
fong-spivak-invitation_D11_248	128	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-128_206_226_654_560.jpg			
fong-spivak-invitation_D11_248	128	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-128_208_294_652_1026.jpg			

fong-spivak-invitation_D11_0146	128	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-128_512_786_1180_546.jpg			
fong-spivak-invitation_D11_0147	132	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-132_462_1347_652_206.jpg			
fong-spivak-invitation_D11_0148	134	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-134_156_512_1072_560.jpg			
fong-spivak-invitation_D11_0149	134	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-134_152_200_1076_1116.jpg			
fong-spivak-invitation_D11_0150	136	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-136_486_1038_310_422.jpg			
fong-spivak-invitation_D11_0151	137	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-137_366_1014_460_398.jpg			

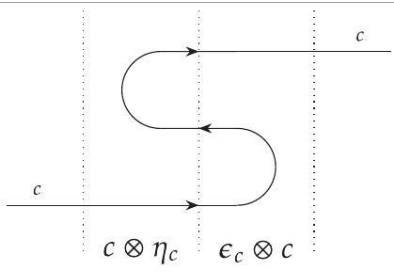
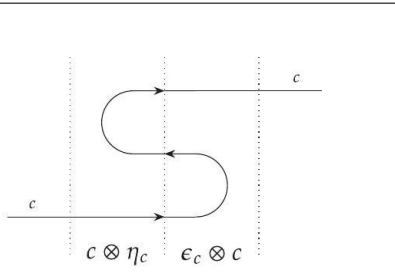
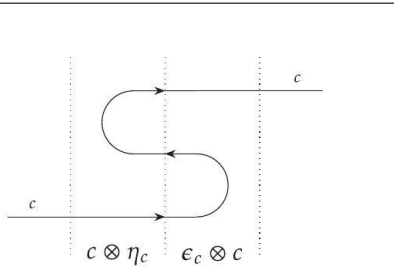
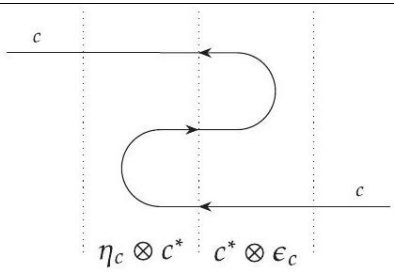
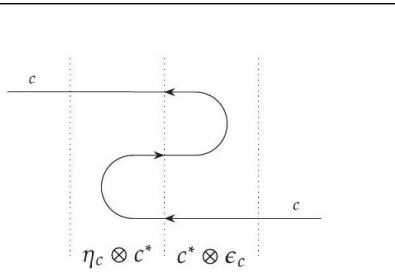
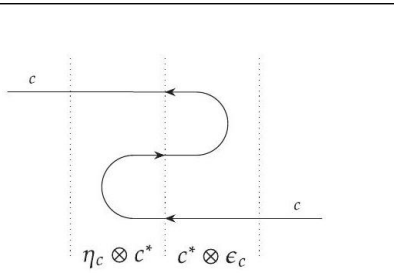
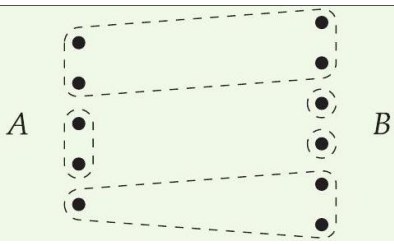
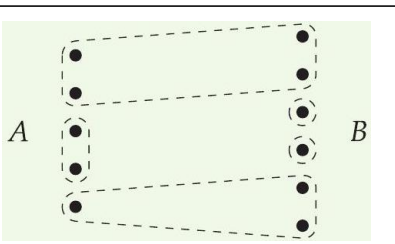
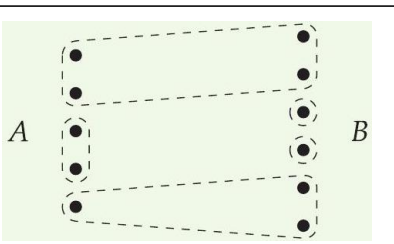
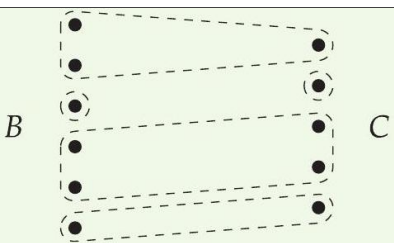
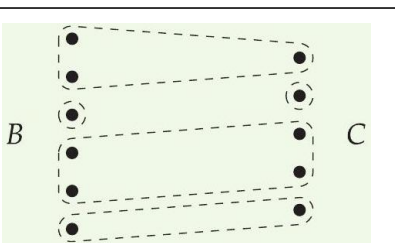
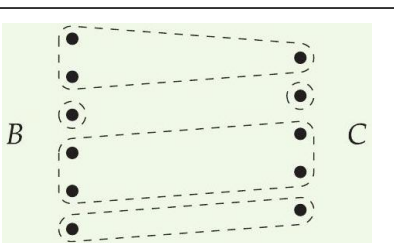
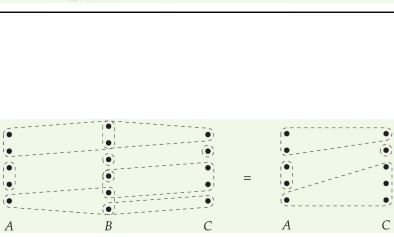
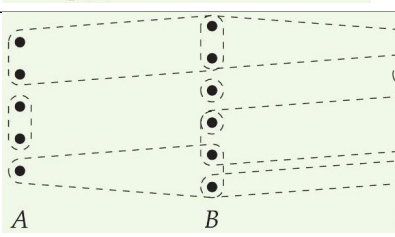
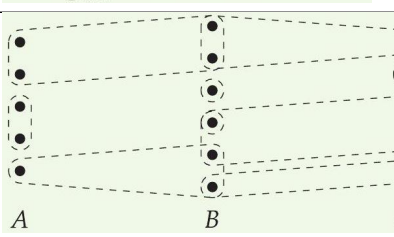
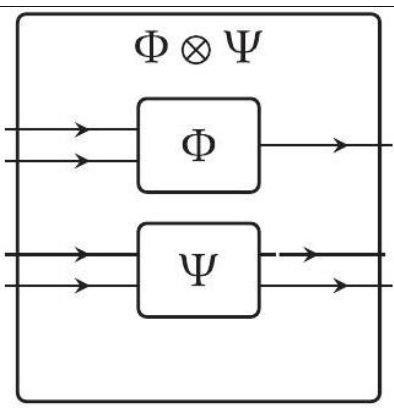
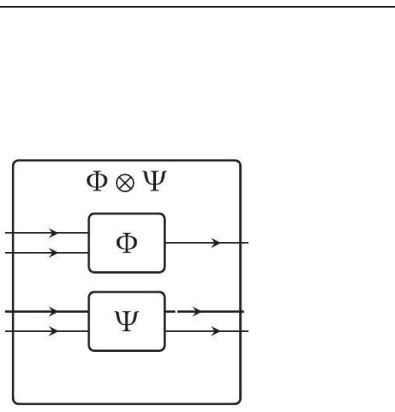
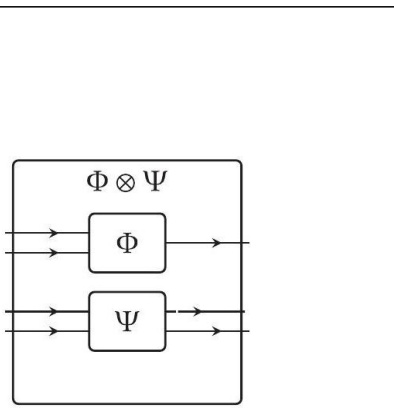
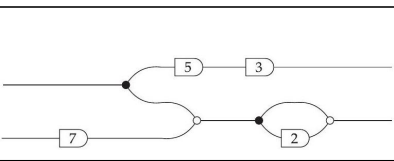
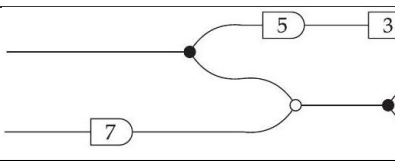
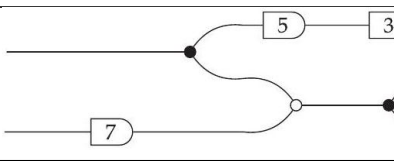
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fong-spivak-invitation_D1_018	138	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-138_290_318_1979_450.jpg			
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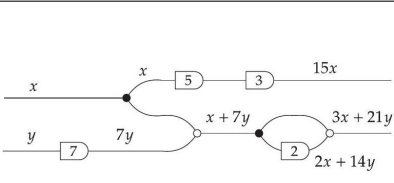
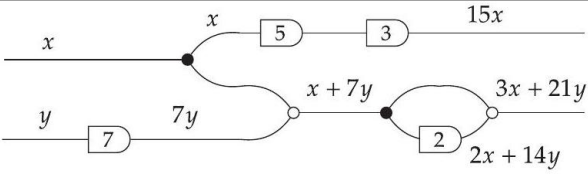
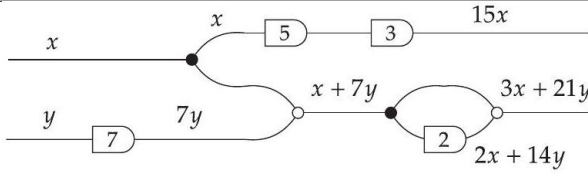
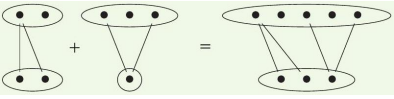
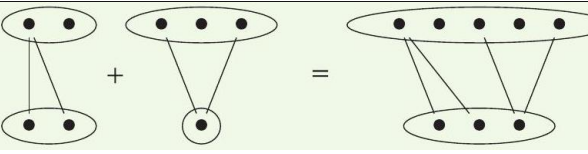
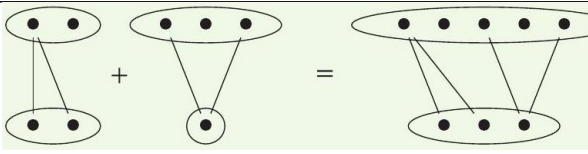
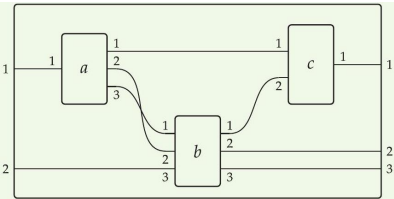
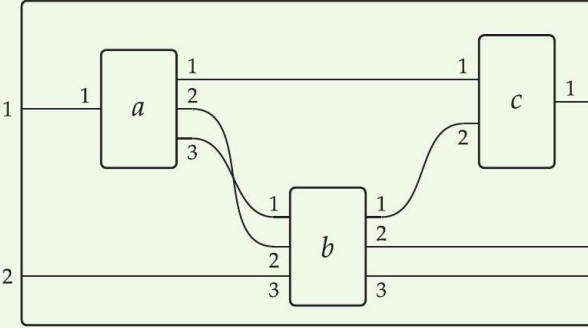
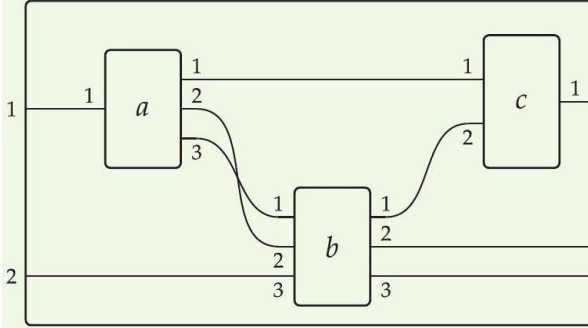
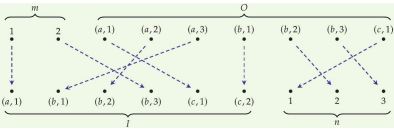
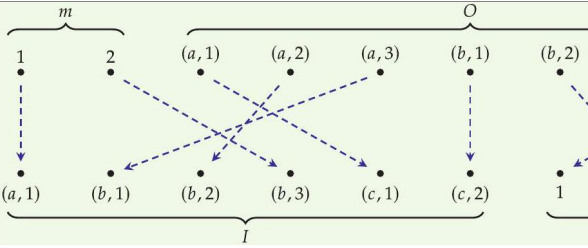
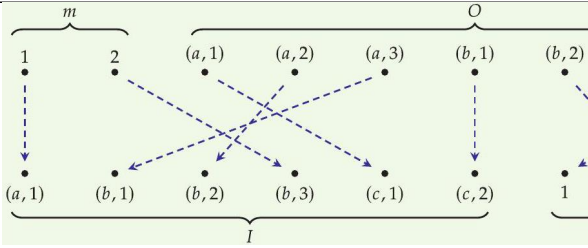
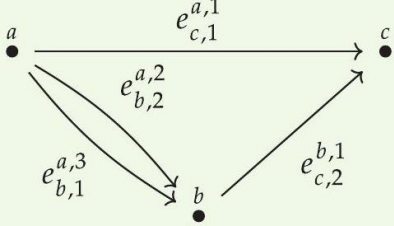
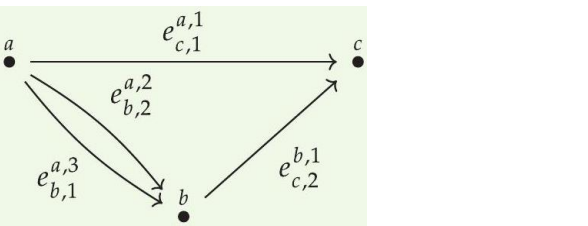
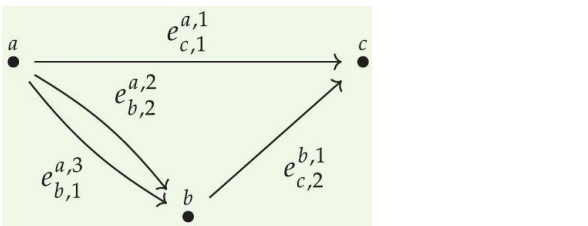
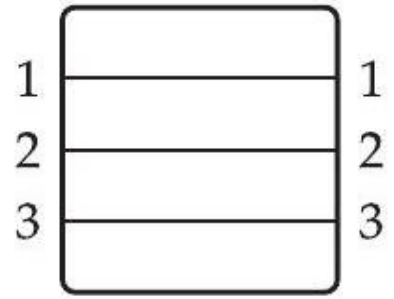
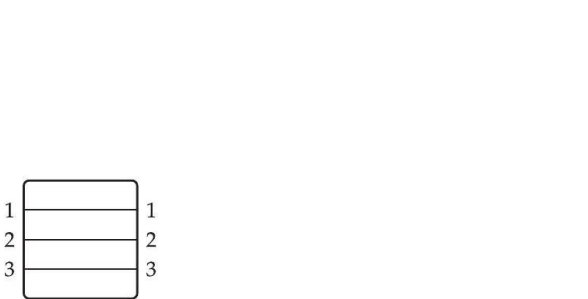
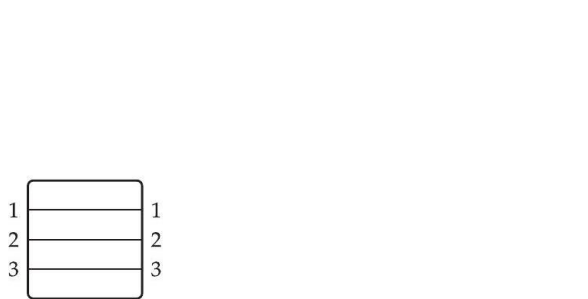
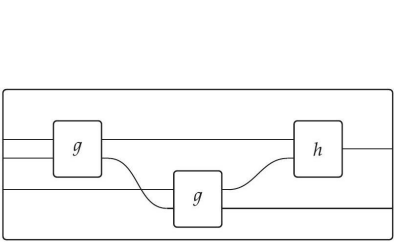
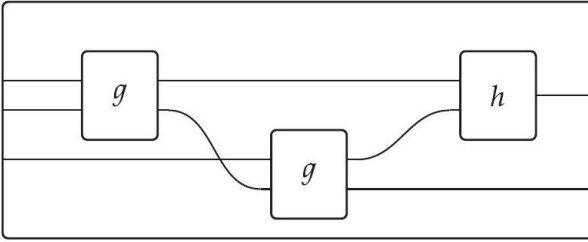
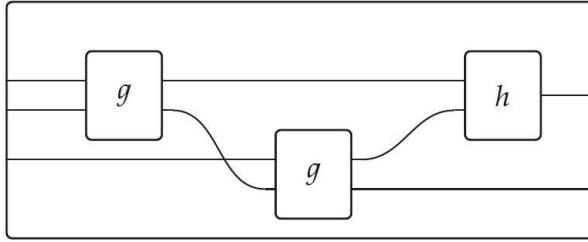
fong-spivak-invitation_D11_138	138	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-138_354_214_1949_1213.jpg			
fong-spivak-invitation_D11_139	139	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-139_422_992_364_440.jpg			
fong-spivak-invitation_D11_140	140	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-140_406_1090_258_362.jpg			
fong-spivak-invitation_D11_141	141	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-141_350_1116_686_380.jpg			
fong-spivak-invitation_D11_142	142	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-142_344_1225_744_330.jpg			
fong-spivak-invitation_D11_143	143	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-143_104_700_1144_588.jpg			

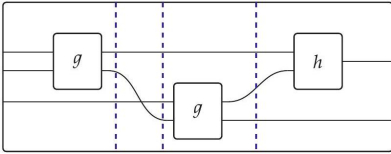
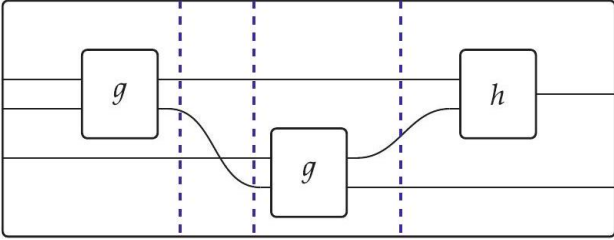
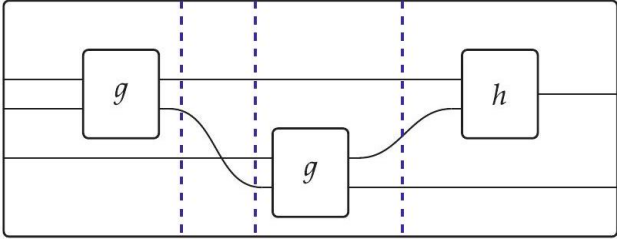
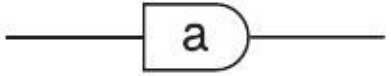
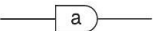
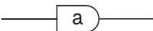
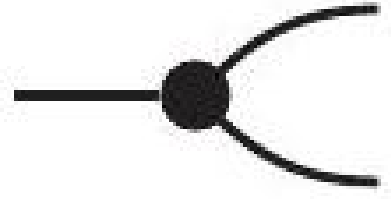
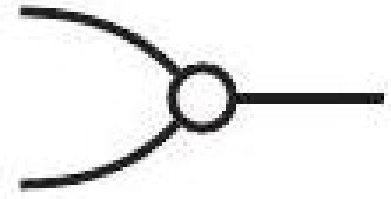
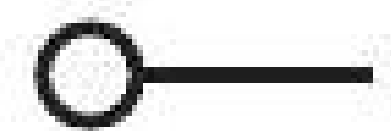
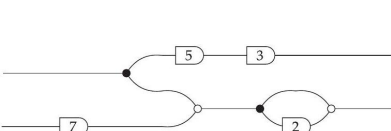
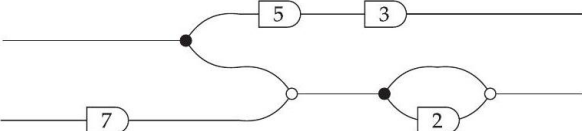
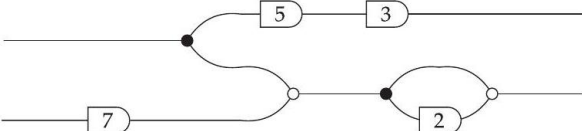
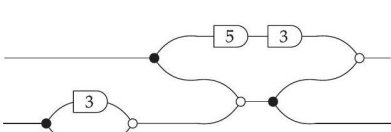
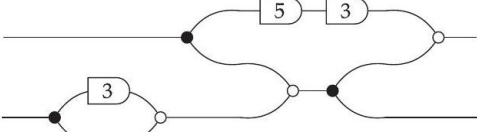
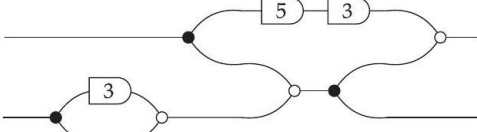
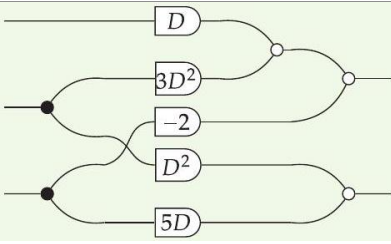
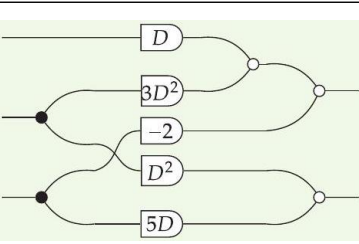
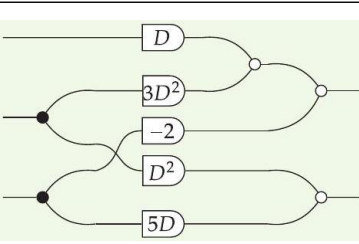
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fong-spivak-invitation_D11_416	146	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-146_312_758_792_562.jpg			
fong-spivak-invitation_D11_416	146	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-146_342_944_1969_468.jpg			
fong-spivak-invitation_D11_417	147	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-147_586_860_1369_508.jpg			
fong-spivak-invitation_D11_418	148	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-148_90_142_972_870.jpg			
fong-spivak-invitation_D11_418	148	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-148_88_376_1200_752.jpg			

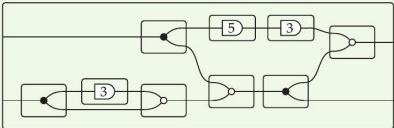
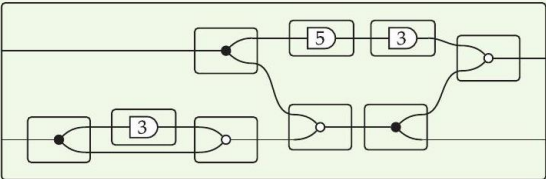
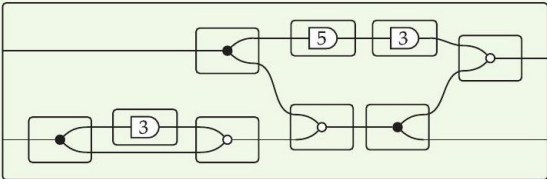
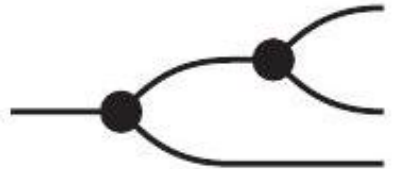
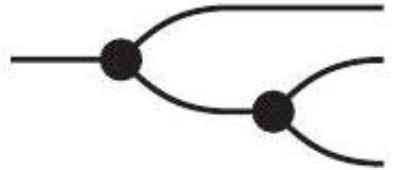
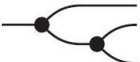
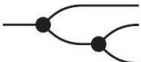
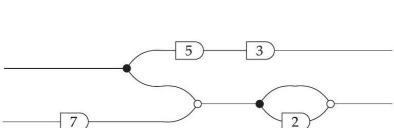
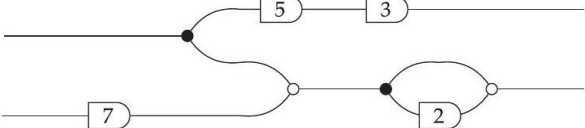
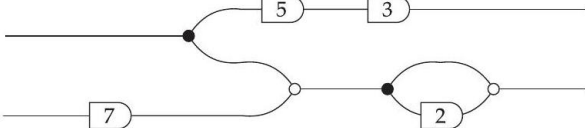
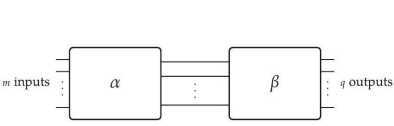
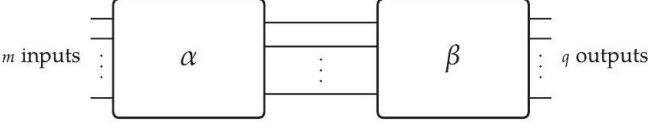
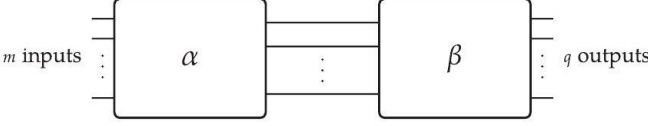
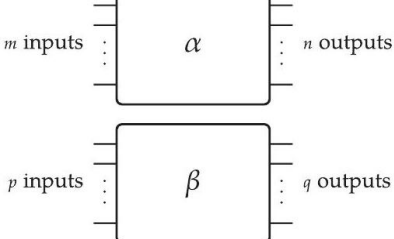
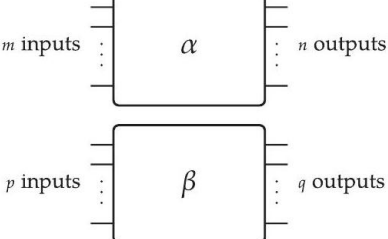
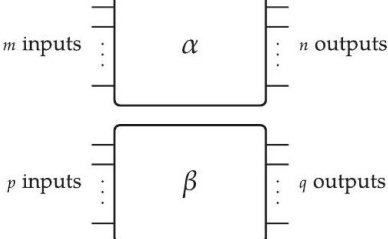
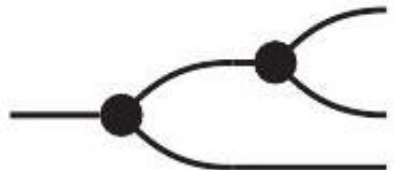
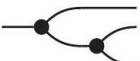
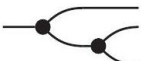
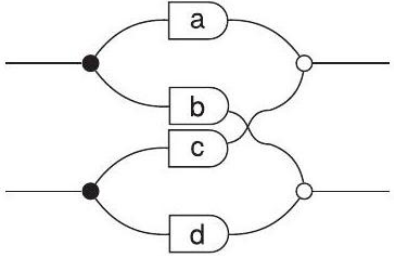
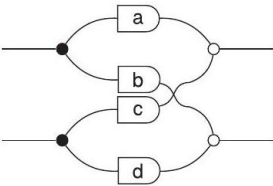
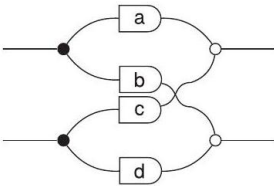
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fong-spivak-invitation_D11_416	149	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-149_64_104_476_542.jpg			
fong-spivak-invitation_D11_416	149	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-149_64_102_476_888.jpg			
fong-spivak-invitation_D11_416	149	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-149_30_104_492_1231.jpg			
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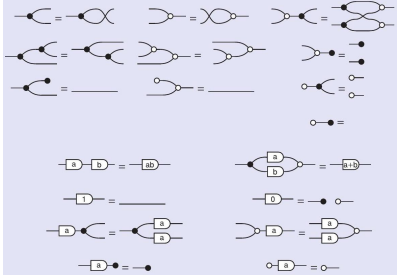
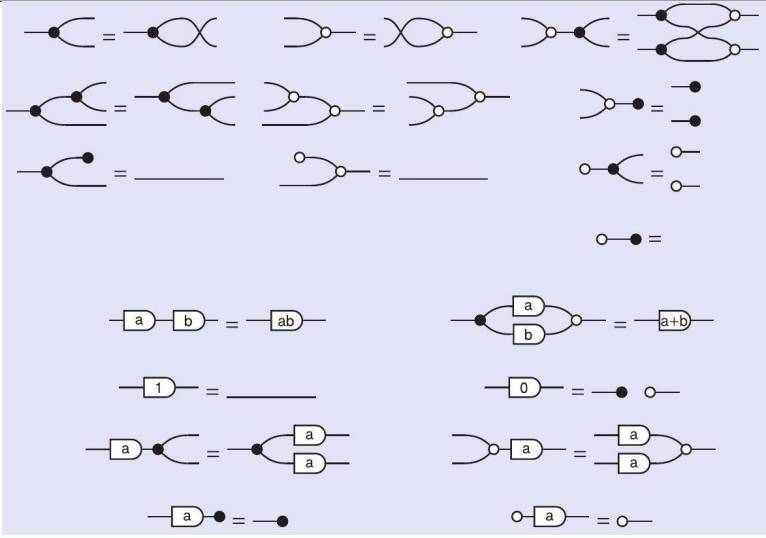
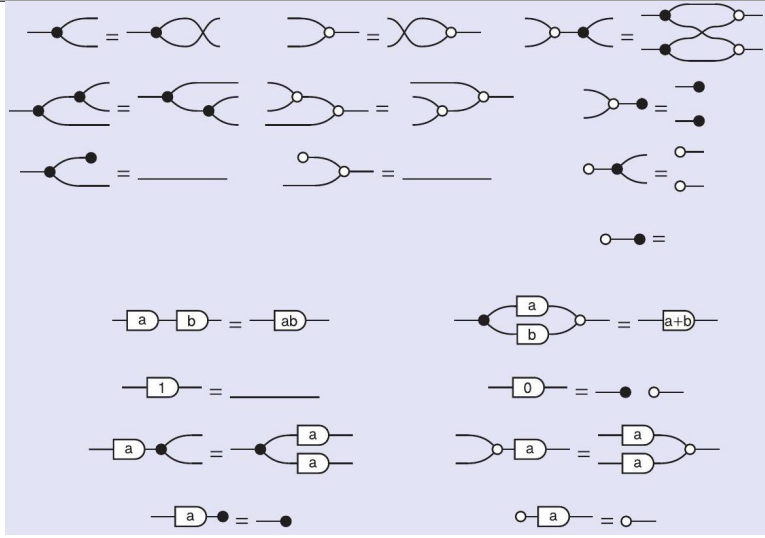
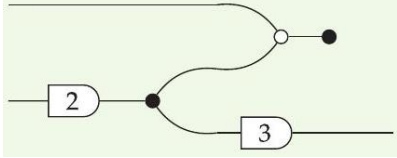
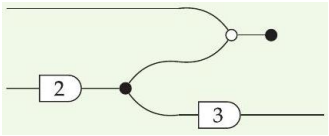
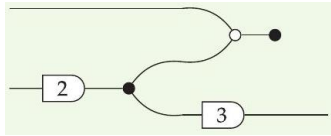
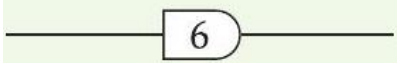
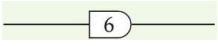
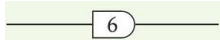
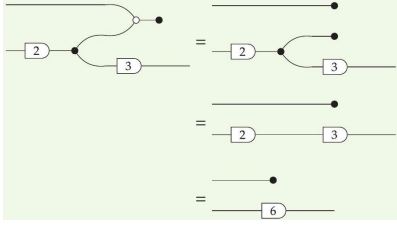
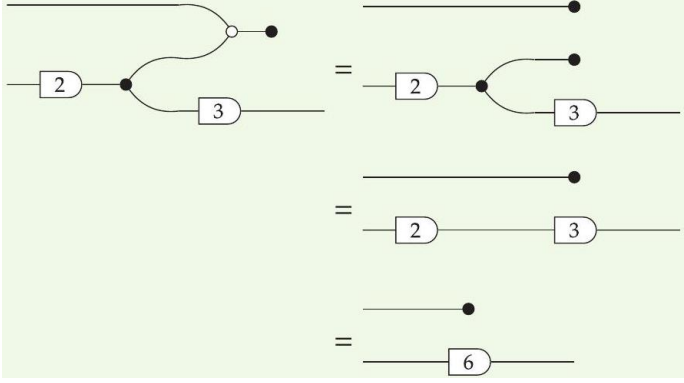
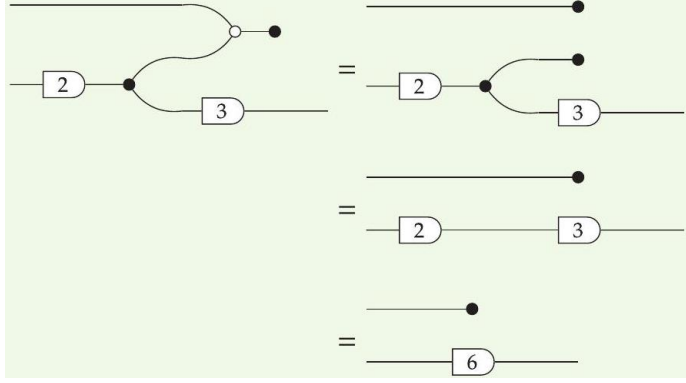
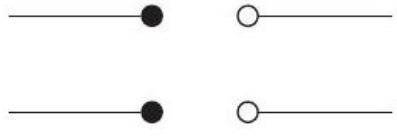
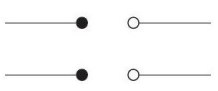
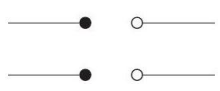
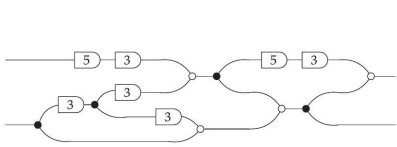
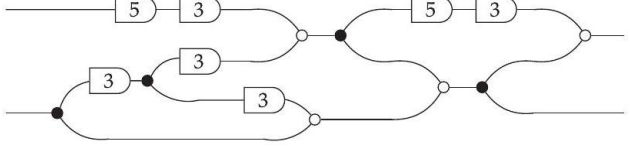
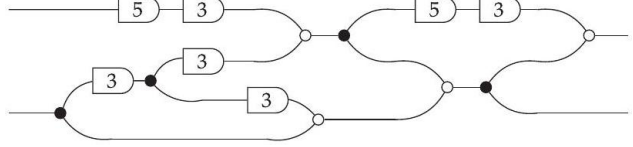
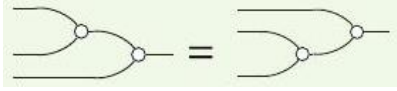
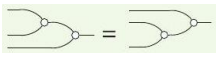
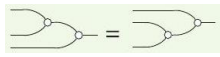
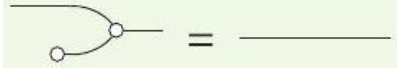
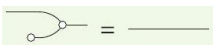
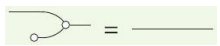
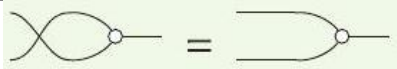
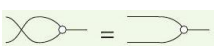
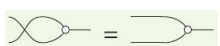
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fong-spivak-invitation_D1_156	156	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-156_284_420_360_400.jpg			
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fong-spivak-invitation_D1_161	161	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-161_186_767_1826_556.jpg			

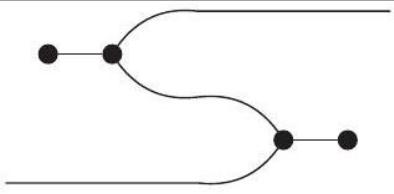
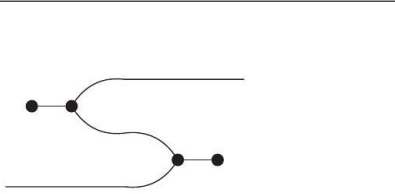
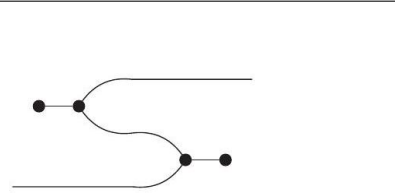
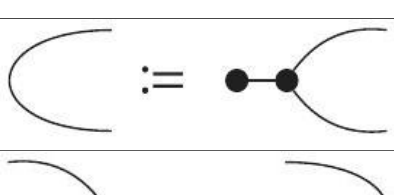
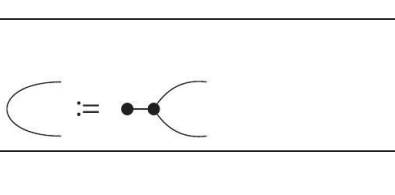
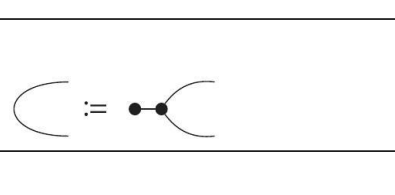
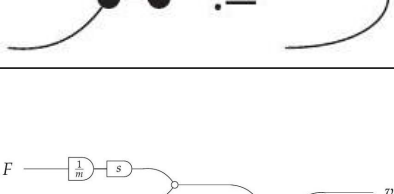
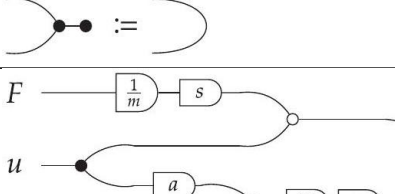
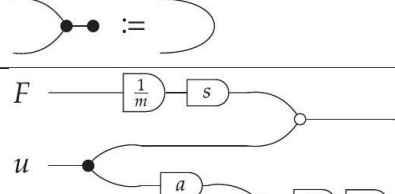
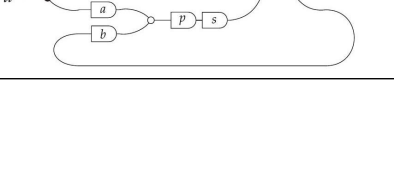
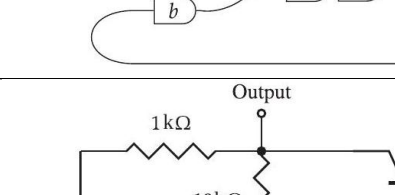
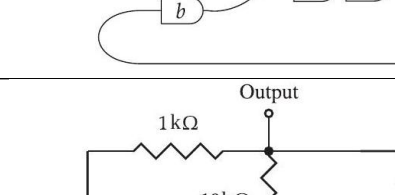
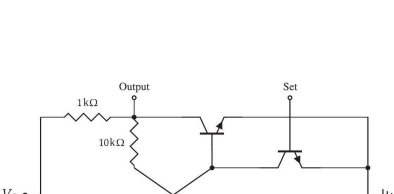
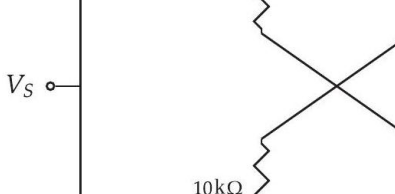
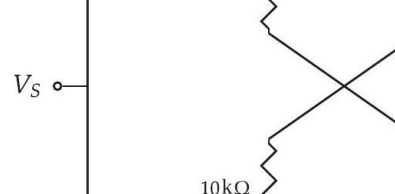
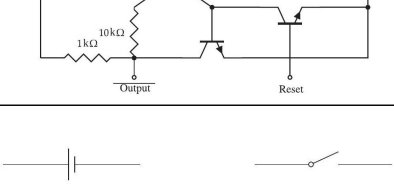
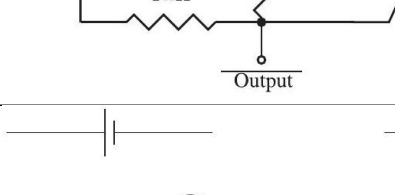
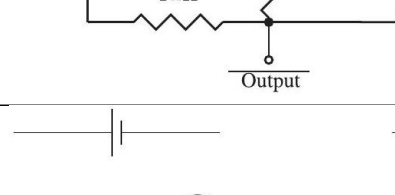
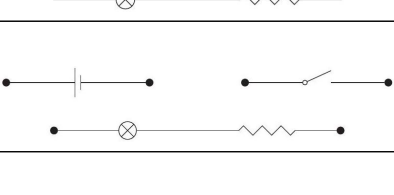
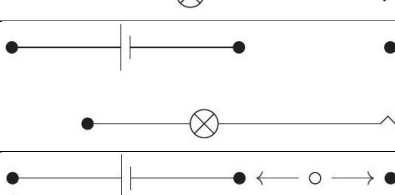
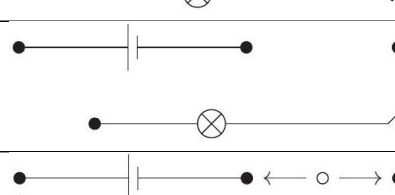
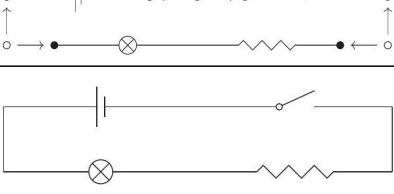
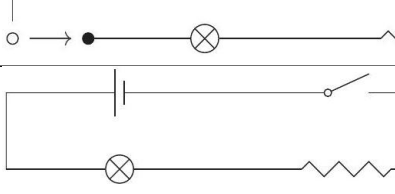
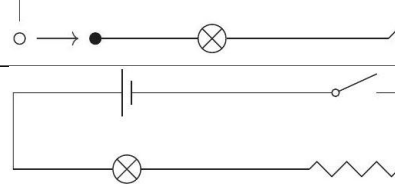
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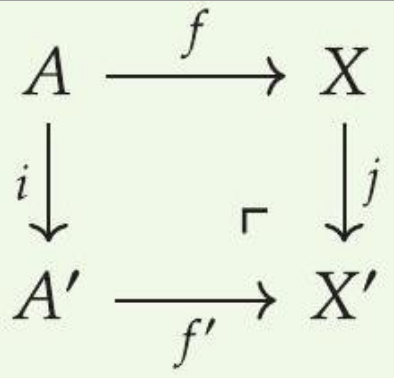
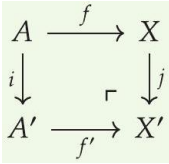
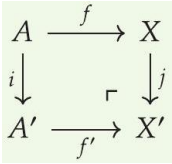
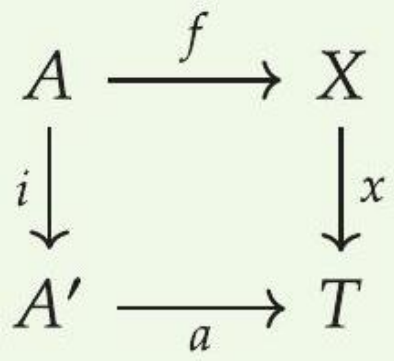
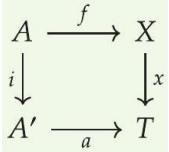
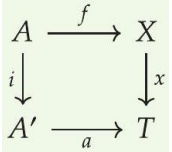
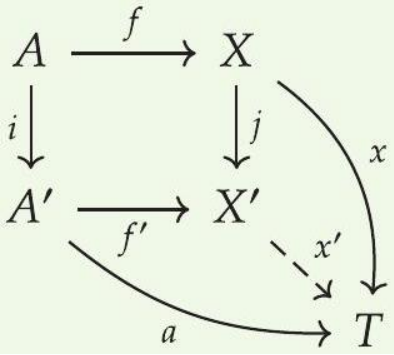
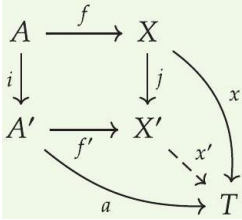
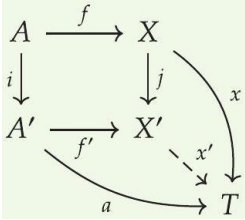
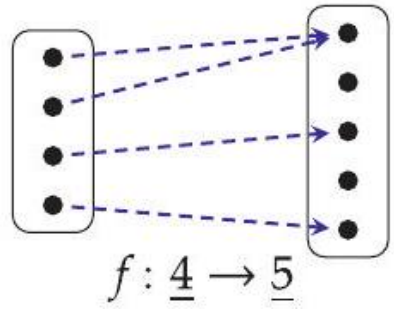
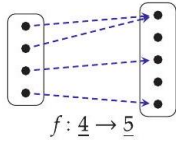
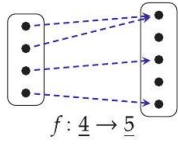
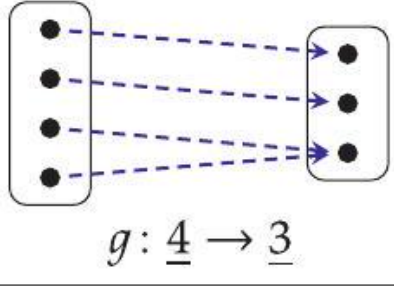
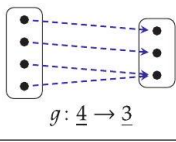
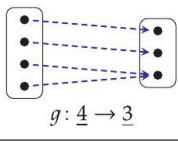
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fong-spivak-invitation_D11_097	185	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-185_50_280_492_1070.jpg			
fong-spivak-invitation_D11_098	185	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-185_498_888_678_494.jpg			
fong-spivak-invitation_D11_099	185	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-185_94_276_1781_332.jpg			
fong-spivak-invitation_D11_100	185	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-185_198_814_1729_738.jpg			
fong-spivak-invitation_D11_101	186	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-186_62_278_2075_394.jpg			
fong-spivak-invitation_D11_102	186	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-186_50_274_2157_396.jpg			
fong-spivak-invitation_D11_103	186	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-186_50_276_2223_396.jpg			

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fong-spivak-invitation_D11_919	189	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-189_204_628_564_624.jpg			
fong-spivak-invitation_D11_920	190	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-190_146_612_624_636.jpg			
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fong-spivak-invitation_D11_922	191	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-191_146_320_336_800.jpg			
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fong-spivak-invitation_D11_0226	195	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-195_158_772_662_552.jpg			
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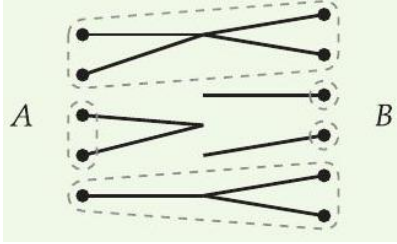
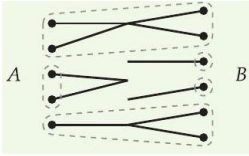
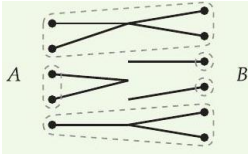
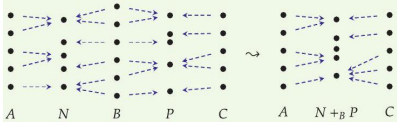
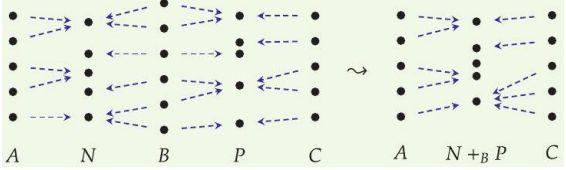
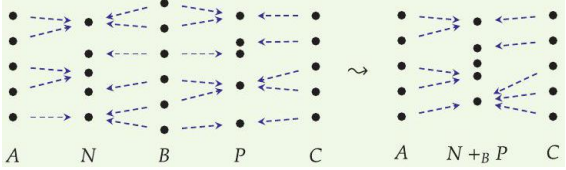
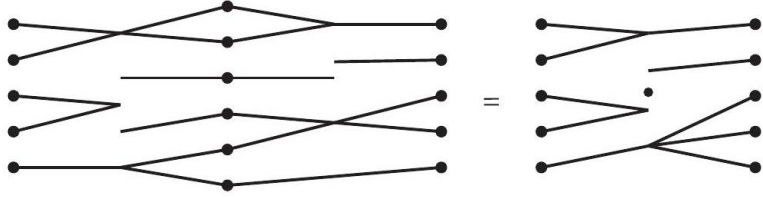
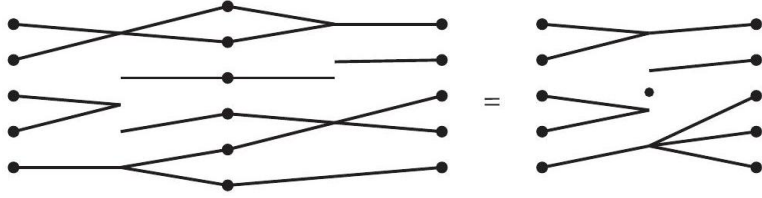
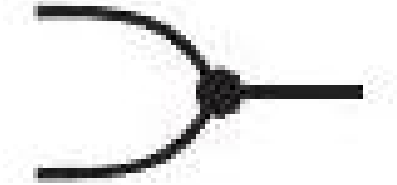
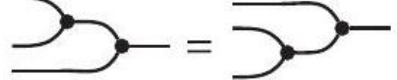
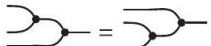
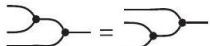
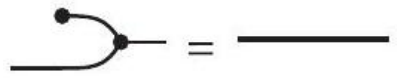
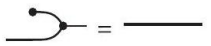
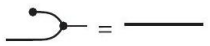
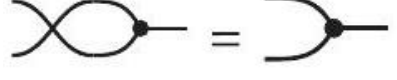
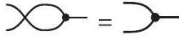
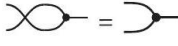
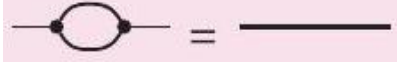
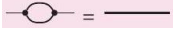
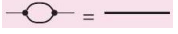
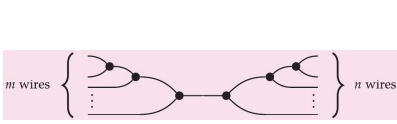
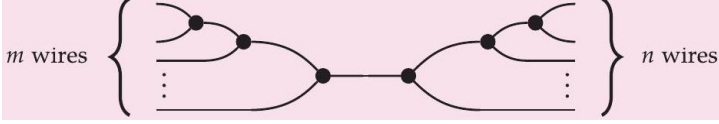
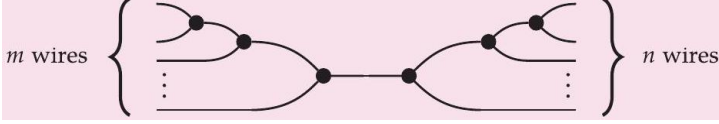
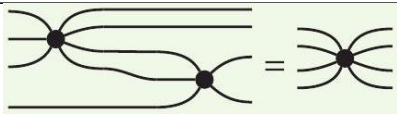
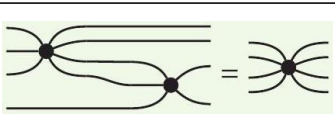
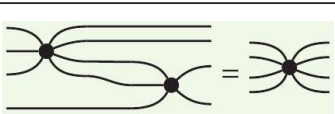
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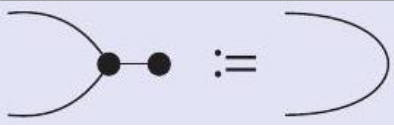
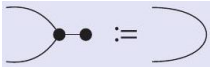
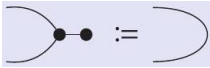
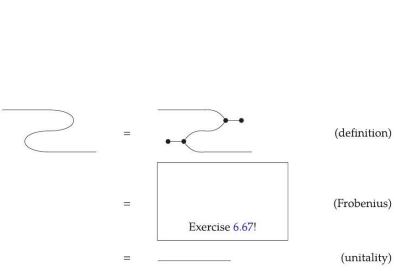
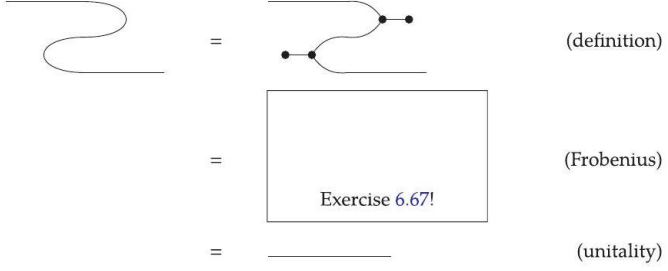
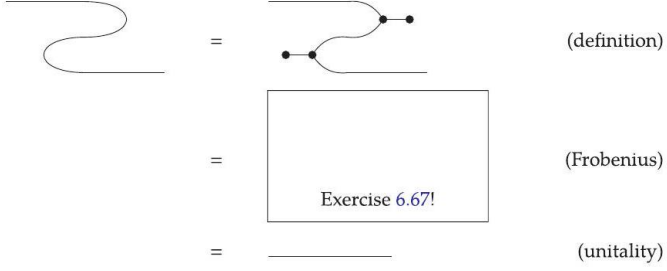
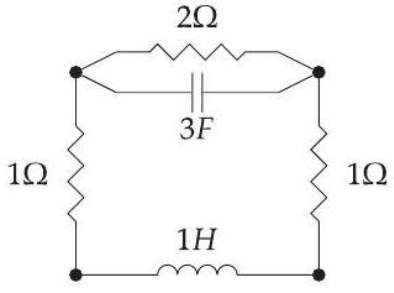
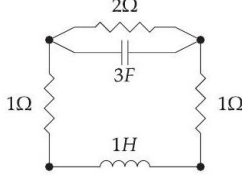
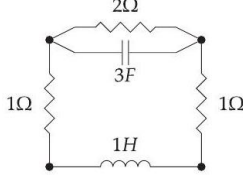
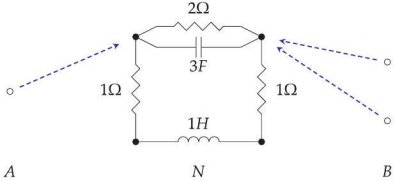
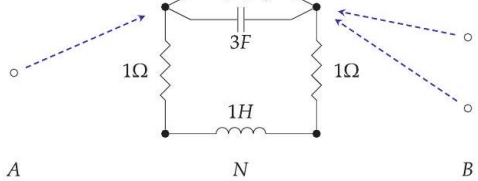
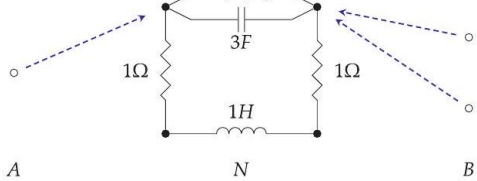
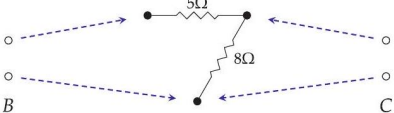
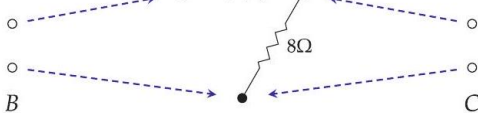
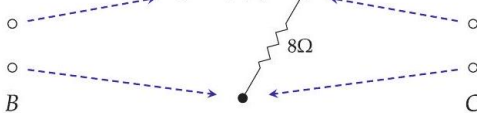
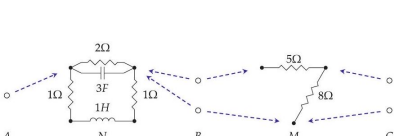
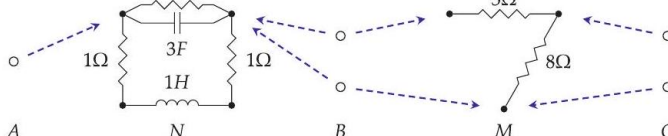
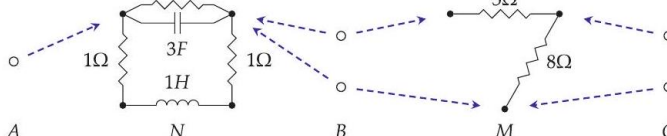
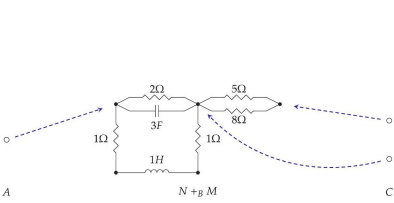
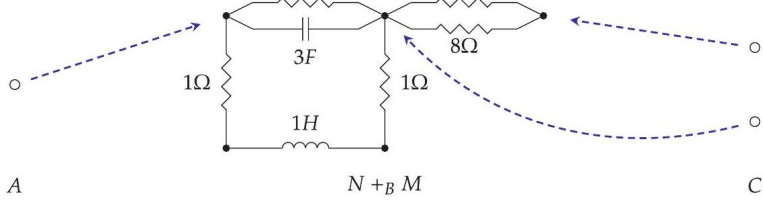
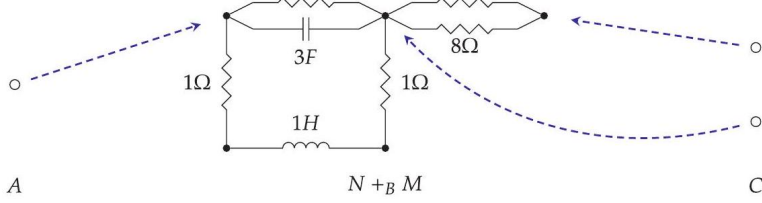
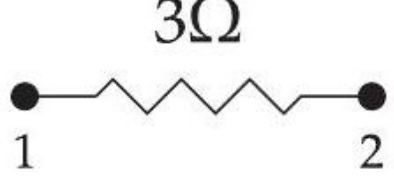
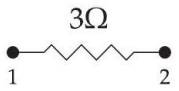
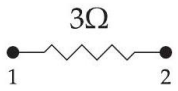
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fong-spivak-invitation_D12_023	203	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-203_184_230_1024_582.jpg			
fong-spivak-invitation_D12_023	203	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-203_164_228_1040_1064.jpg			

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fong-spivak-invitation_D1_023	203	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-203_204_214_1649_1088.jpg	$\begin{array}{ccc} \emptyset & \xrightarrow{f} & X \\ g \downarrow & & \downarrow x \\ Y & \xrightarrow{y} & T \end{array}$	$\begin{array}{ccc} \emptyset & \xrightarrow{f} & X \\ g \downarrow & & \downarrow x \\ Y & \xrightarrow{y} & T \end{array}$	$\begin{array}{ccc} \emptyset & \xrightarrow{f} & X \\ g \downarrow & & \downarrow x \\ Y & \xrightarrow{y} & T \end{array}$
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fong-spivak-invitation_D1_024	204	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-204_208_406_1907_436.jpg			
fong-spivak-invitation_D1_024	204	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-204_208_408_1905_1038.jpg			
fong-spivak-invitation_D1_025	205	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-205_196_234_1705_822.jpg	$\begin{array}{ccc} & B & \longrightarrow Z \\ & \downarrow & \\ A & \longrightarrow & C \\ \downarrow & & \\ D & & \end{array}$	$\begin{array}{ccc} & B & \longrightarrow Z \\ & \downarrow & \\ A & \longrightarrow & C \\ \downarrow & & \\ D & & \end{array}$	$\begin{array}{ccc} & B & \longrightarrow Z \\ & \downarrow & \\ A & \longrightarrow & C \\ \downarrow & & \\ D & & \end{array}$
fong-spivak-invitation_D1_025	205	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-205_200_238_2065_582.jpg	$\begin{array}{ccccc} & B & \longrightarrow & Z & \\ & \downarrow & & \downarrow & \\ A & \longrightarrow & Y & \xrightarrow{r} & R \\ \downarrow & & \downarrow & & \\ X & \xrightarrow{r} & Q & & \end{array}$	$\begin{array}{ccccc} & B & \longrightarrow & Z & \\ & \downarrow & & \downarrow & \\ A & \longrightarrow & Y & \xrightarrow{r} & R \\ \downarrow & & \downarrow & & \\ X & \xrightarrow{r} & Q & & \end{array}$	$\begin{array}{ccccc} & B & \longrightarrow & Z & \\ & \downarrow & & \downarrow & \\ A & \longrightarrow & Y & \xrightarrow{r} & R \\ \downarrow & & \downarrow & & \\ X & \xrightarrow{r} & Q & & \end{array}$

fong-spivak-invitation_D12_026	205	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-205_198_236_2067_1060.jpg			
fong-spivak-invitation_D12_027	207	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-207_156_816_1329_532.jpg			
fong-spivak-invitation_D12_028	208	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-208_134_864_598_508.jpg			
fong-spivak-invitation_D12_028	208	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-208_132_652_898_616.jpg			
fong-spivak-invitation_D12_028	208	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-208_216_1186_1180_346.jpg			
fong-spivak-invitation_D12_028	208	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-208_106_176_2013_644.jpg			
fong-spivak-invitation_D12_028	208	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-208_106_176_2013_1060.jpg			
fong-spivak-invitation_D12_029	209	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-209_224_224_1437_576.jpg			

fong-spivak-invitation_D12_120	209	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-209_198_322_1455_980.jpg			
fong-spivak-invitation_D12_120	210	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-210_224_732_310_576.jpg			
fong-spivak-invitation_D12_120	210	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-210_260_998_1076_408.jpg			
fong-spivak-invitation_D12_121	211	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-211_44_80_686_779.jpg			
fong-spivak-invitation_D12_121	211	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-211_19_51_699_1081.jpg			
fong-spivak-invitation_D12_121	211	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-211_70_272_895_452.jpg			
fong-spivak-invitation_D12_121	211	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-211_53_266_902_806.jpg			
fong-spivak-invitation_D12_121	211	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-211_51_238_904_1157.jpg			
fong-spivak-invitation_D12_121	211	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-211_70_352_1999_635.jpg			
fong-spivak-invitation_D12_121	211	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-211_40_222_2014_1054.jpg			
fong-spivak-invitation_D12_122	212	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-212_164_936_620_472.jpg			
fong-spivak-invitation_D12_122	212	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-212_92_416_968_732.jpg			
fong-spivak-invitation_D12_122	212	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-212_120_430_1921_726.jpg			

fong-spivak-invitation_D12_128	213	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-213_72_114_324_904.jpg			
fong-spivak-invitation_D12_128	213	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-213_72_184_480_868.jpg			
fong-spivak-invitation_D12_128	213	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-213_116_232_634_844.jpg			
fong-spivak-invitation_D12_128	213	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-213_70_232_836_844.jpg			
fong-spivak-invitation_D12_128	213	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-213_132_232_986_844.jpg			
fong-spivak-invitation_D12_128	213	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-213_160_320_1245_800.jpg			
fong-spivak-invitation_D12_128	214	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-214_304_894_1096_494.jpg			
fong-spivak-invitation_D12_128	215	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-215_324_924_420_478.jpg			
fong-spivak-invitation_D12_128	216	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-216_84_272_954_488.jpg			

fong-spivak-invitation_D12_126	216	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-216_84_270_954_1122.jpg			
fong-spivak-invitation_D12_126	216	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-216_352_866_1200_510.jpg			
fong-spivak-invitation_D12_127	217	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-217_210_582_1673_648.jpg	$\begin{array}{ccc} P(S) \times P(T) & \xrightarrow{\varphi_{S,T}} & P(S \times T) \\ \text{im}_f \times \text{im}_g \downarrow & & \downarrow \text{im}_f \times g \\ P(S') \times P(T') & \xrightarrow{\varphi_{S',T'}} & P(S' \times T') \end{array}$	$\begin{array}{ccc} P(S) \times P(T) & \xrightarrow{\varphi_{S,T}} & P(S \times T) \\ \text{im}_f \times \text{im}_g \downarrow & & \downarrow \text{im}_f \times g \\ P(S') \times P(T') & \xrightarrow{\varphi_{S',T'}} & P(S' \times T') \end{array}$	$\begin{array}{ccc} P(S) \times P(T) & \xrightarrow{\varphi_{S,T}} & P(S \times T) \\ \text{im}_f \times \text{im}_g \downarrow & & \downarrow \text{im}_f \times g \\ P(S') \times P(T') & \xrightarrow{\varphi_{S',T'}} & P(S' \times T') \end{array}$
fong-spivak-invitation_D12_128	218	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-218_234_320_366_780.jpg			
fong-spivak-invitation_D12_128	218	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-218_286_618_962_632.jpg			
fong-spivak-invitation_D12_128	218	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-218_194_626_1625_628.jpg			
fong-spivak-invitation_D12_128	218	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-218_226_878_2021_502.jpg			
fong-spivak-invitation_D12_129	219	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-219_336_1138_396_370.jpg			
fong-spivak-invitation_D12_120	220	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-220_110_224_1927_828.jpg			

fong-spivak-invitation_D14_221	221	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-221_108_520_1068_678.jpg			
fong-spivak-invitation_D14_221	221	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-221_192_520_1269_678.jpg			
fong-spivak-invitation_D14_221	221	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-221_70_810_2049_534.jpg	$b := \text{---} \text{---}$ and $s := \text{---} / \text{---}$	$b := \text{---} \text{---}$ and $s := \text{---} / \text{---}$	$b := \text{---} \text{---}$ and $s := \text{---} / \text{---}$
fong-spivak-invitation_D14_222	222	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-222_26_86_672_898.jpg			
fong-spivak-invitation_D14_222	222	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-222_94_420_1068_728.jpg			
fong-spivak-invitation_D14_222	222	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-222_94_420_1703_728.jpg			
fong-spivak-invitation_D14_222	222	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-222_96_420_2147_396.jpg			
fong-spivak-invitation_D14_222	222	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-222_96_420_2149_1064.jpg			
fong-spivak-invitation_D14_223	223	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-223_260_814_394_530.jpg			
fong-spivak-invitation_D14_223	223	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-223_94_420_764_728.jpg			
fong-spivak-invitation_D14_223	223	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-223_70_616_1068_632.jpg			
fong-spivak-invitation_D14_223	223	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-223_68_420_1393_728.jpg			
fong-spivak-invitation_D14_223	223	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-223_94_618_1579_630.jpg			
fong-spivak-invitation_D14_224	224	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-224_96_616_382_632.jpg			
fong-spivak-invitation_D14_224	224	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-224_96_616_552_632.jpg			
fong-spivak-invitation_D14_224	224	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-224_194_618_746_632.jpg			
fong-spivak-invitation_D14_224	224	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-224_128_360_1204_460.jpg	$\eta := \emptyset \quad \boxed{\bullet}$	$\eta := \emptyset \quad \boxed{\bullet}$	$\eta := \emptyset \quad \boxed{\bullet}$

fong-spivak-invitation_D12_224.jpg	224	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-224_122_364_1207_1056.jpg			
fong-spivak-invitation_D12_225.jpg	225	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-225_410_1223_1519_330.jpg			
fong-spivak-invitation_D12_226.jpg	226	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-226_382_592_258_644.jpg			
fong-spivak-invitation_D12_227.jpg	227	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-227_222_1130_884_374.jpg			
fong-spivak-invitation_D12_228.jpg	228	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-228_380_592_1893_644.jpg			
fong-spivak-invitation_D12_229.jpg	229	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-229_208_208_1024_856.jpg			
fong-spivak-invitation_D12_230.jpg	230	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-229_184_312_1477_804.jpg			

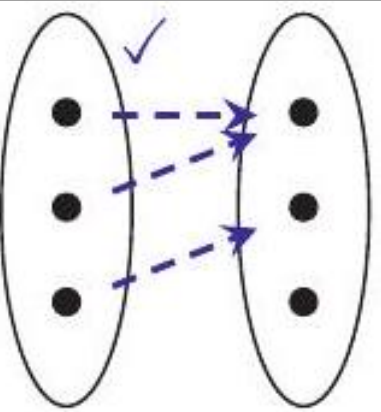
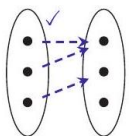
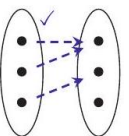
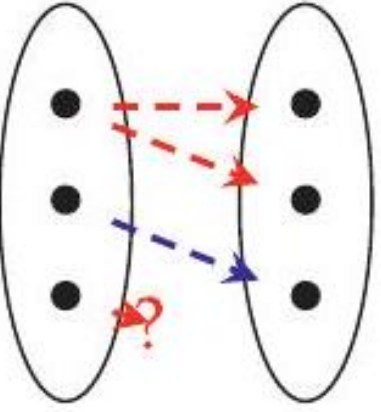
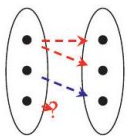
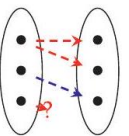
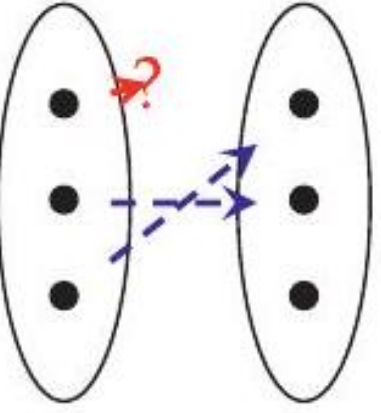
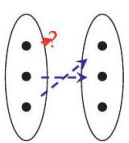
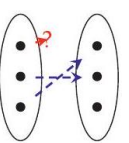
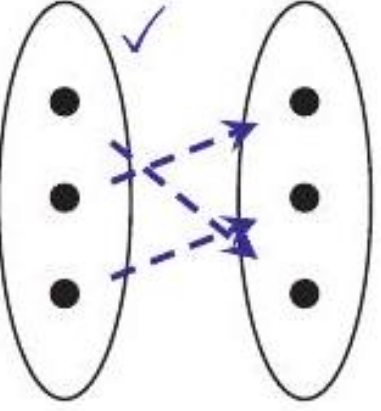
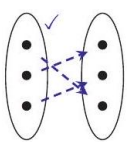
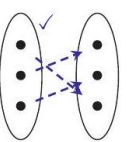
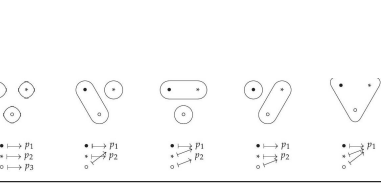
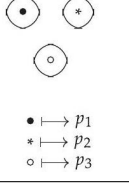
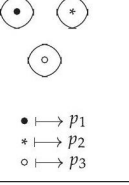
fong-spivak-invitation_D12_0813	231	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-231_240_452_298_712.jpg			
fong-spivak-invitation_D12_0814	231	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-231_158_814_864_532.jpg			
fong-spivak-invitation_D12_0815	231	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-231_156_608_1116_638.jpg			
fong-spivak-invitation_D12_0816	233	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-233_148_770_1661_552.jpg			
fong-spivak-invitation_D12_0817	234	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-234_214_560_710_660.jpg			
fong-spivak-invitation_D12_0818	235	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-235_138_772_1475_550.jpg			
fong-spivak-invitation_D12_0819	237	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-237_112_232_652_612.jpg			
fong-spivak-invitation_D12_0820	237	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-237_112_232_652_1034.jpg			
fong-spivak-invitation_D12_0821	237	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-237_114_250_1128_814.jpg			
fong-spivak-invitation_D12_0822	237	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-237_208_604_1945_636.jpg			

fong-spivak-invitation_D12_088	238	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-238_212_220_744_852.jpg	$\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ i' \downarrow \cong & \lrcorner & \cong \downarrow i \\ A & \xrightarrow{f} & B \end{array}$	$\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ i' \downarrow \cong & \lrcorner & \cong \downarrow i \\ A & \xrightarrow{f} & B \end{array}$	$\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ i' \downarrow \cong & \lrcorner & \cong \downarrow i \\ A & \xrightarrow{f} & B \end{array}$
fong-spivak-invitation_D12_088	238	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-238_208_188_1084_868.jpg	$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \parallel & \lrcorner & \parallel \\ A & \xrightarrow{f} & B \end{array}$	$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \parallel & \lrcorner & \parallel \\ A & \xrightarrow{f} & B \end{array}$	$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \parallel & \lrcorner & \parallel \\ A & \xrightarrow{f} & B \end{array}$
fong-spivak-invitation_D12_088	238	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-238_194_232_1489_824.jpg	$\begin{array}{ccc} A' & \xrightarrow{g} & A \\ f' \downarrow & \lrcorner & \downarrow f \\ B' & \xrightarrow{h} & B \end{array}$	$\begin{array}{ccc} A' & \xrightarrow{g} & A \\ f' \downarrow & \lrcorner & \downarrow f \\ B' & \xrightarrow{h} & B \end{array}$	$\begin{array}{ccc} A' & \xrightarrow{g} & A \\ f' \downarrow & \lrcorner & \downarrow f \\ B' & \xrightarrow{h} & B \end{array}$
fong-spivak-invitation_D12_088	239	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-239_186_276_370_800.jpg			
fong-spivak-invitation_D12_456	240	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-240_232_312_1076_474.jpg	$\begin{array}{ccc} X & \xrightarrow{!} & 1 \\ m \downarrow & \lrcorner & \downarrow \text{true} \\ Y & \xrightarrow{\ulcorner m \urcorner} & \Omega \end{array}$	$\begin{array}{ccc} X & \xrightarrow{!} & 1 \\ m \downarrow & \lrcorner & \downarrow \text{true} \\ Y & \xrightarrow{\ulcorner m \urcorner} & \Omega \end{array}$	$\begin{array}{ccc} X & \xrightarrow{!} & 1 \\ m \downarrow & \lrcorner & \downarrow \text{true} \\ Y & \xrightarrow{\ulcorner m \urcorner} & \Omega \end{array}$

fong-spivak-invitation_D14_450	240	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-240_236_376_1070_1028.jpg			
fong-spivak-invitation_D14_450	245	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-245_154_238_426_820.jpg			
fong-spivak-invitation_D14_450	245	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-245_134_592_806_640.jpg			
fong-spivak-invitation_D14_450	245	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-245_202_304_1036_786.jpg			
fong-spivak-invitation_D14_450	245	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-245_192_968_1421_456.jpg			
fong-spivak-invitation_D14_450	247	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-247_400_998_694_442.jpg			
fong-spivak-invitation_D14_450	250	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-250_434_730_1138_592.jpg			

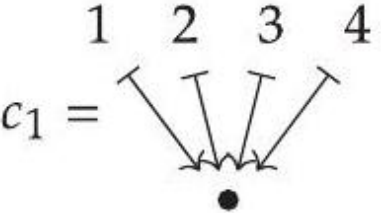
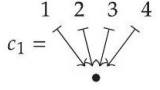
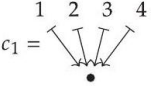
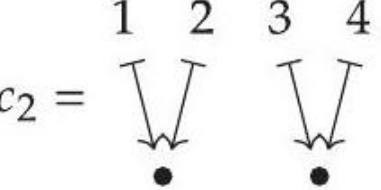
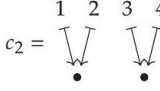
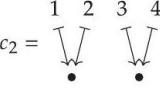
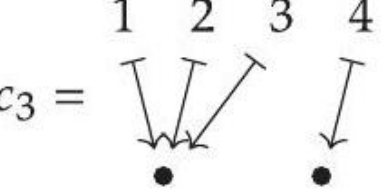
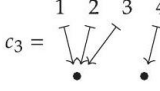
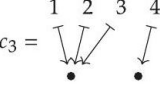
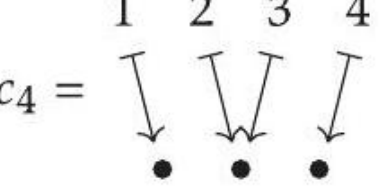
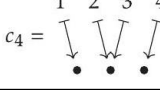
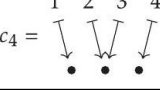
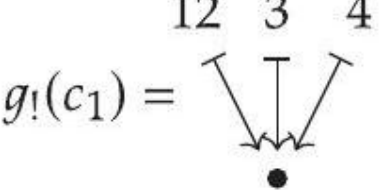
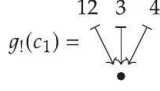
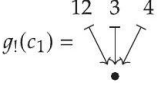
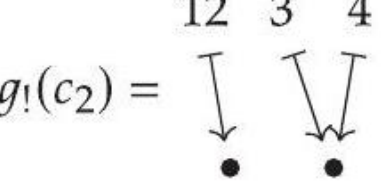
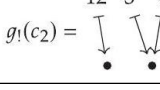
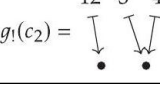
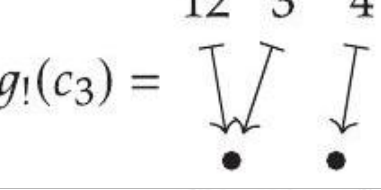
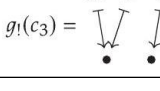
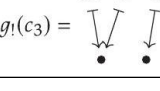
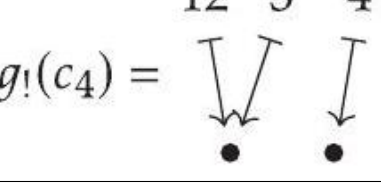
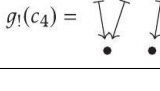
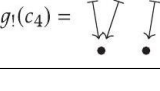
fong-spivak-invitation_D12_251.jpg	251	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-251_296_1016_768_388.jpg			
fong-spivak-invitation_D12_251.jpg	251	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-251_242_626_1787_624.jpg			
fong-spivak-invitation_D12_252.jpg	252	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-252_530_516_564_700.jpg			
fong-spivak-invitation_D12_252.jpg	252	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-252_484_264_1317_808.jpg			
fong-spivak-invitation_D12_253.jpg	253	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-253_354_502_1529_688.jpg			

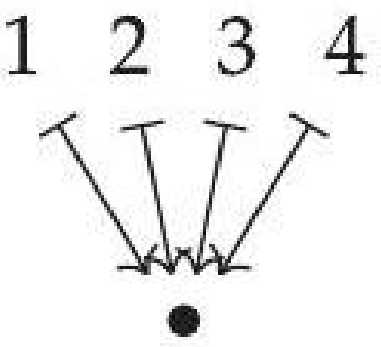
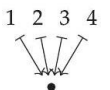
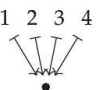
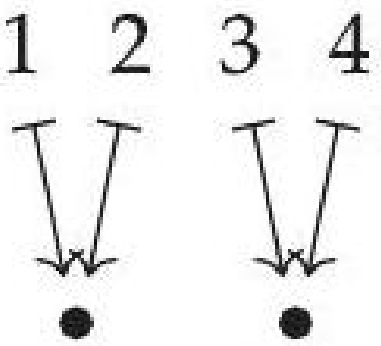
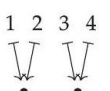
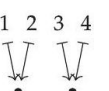
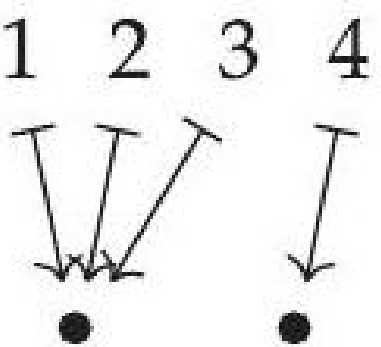
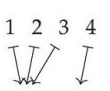
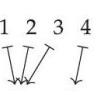
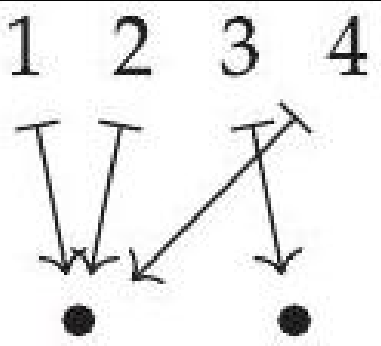
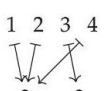
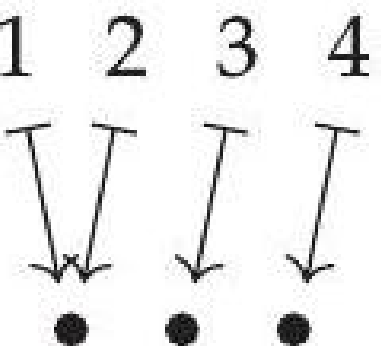
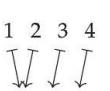
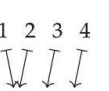
fong-spivak-invitation_D12_0590	257	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-257_262_696_482_590.jpg	$\Omega_{\text{Grph}} =$	$\Omega_{\text{Grph}} =$	$\Omega_{\text{Grph}} =$
fong-spivak-invitation_D12_0591	257	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-257_124_1010_1413_434.jpg	$A \longrightarrow B \quad C \subseteq A \xrightarrow{f} B \xleftarrow{g} B \xrightarrow{h} C \xrightarrow{i} D$	$A \longrightarrow B \quad C \subseteq A \xrightarrow{f} B \xleftarrow{g} B \xrightarrow{h} C \xrightarrow{i} D$	$A \longrightarrow B \quad C \subseteq A \xrightarrow{f} B \xleftarrow{g} B \xrightarrow{h} C \xrightarrow{i} D$
fong-spivak-invitation_D12_0622	262	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-262_206_304_378_788.jpg	$\forall_t p \longrightarrow 1 \quad S \xrightarrow{p'} \Omega^T$	$\forall_t p \longrightarrow 1 \quad S \xrightarrow{p'} \Omega^T$	$\forall_t p \longrightarrow 1 \quad S \xrightarrow{p'} \Omega^T$
fong-spivak-invitation_D12_0622	262	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-262_176_384_1643_748.jpg	$\{S \times T \mid p\} \xrightarrow{\quad} S \times T \quad \downarrow \quad \exists_t p \xrightarrow{\quad} S$	$\{S \times T \mid p\} \xrightarrow{\quad} S \times T \quad \downarrow \quad \exists_t p \xrightarrow{\quad} S$	$\{S \times T \mid p\} \xrightarrow{\quad} S \times T \quad \downarrow \quad \exists_t p \xrightarrow{\quad} S$
fong-spivak-invitation_D12_0630	270	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-270_250_1219_1635_334.jpg			
fong-spivak-invitation_D12_0630	270	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-270_180_236_2065_844.jpg	$(12) = \bullet \quad (1)(2) = \bullet \quad \bullet$	$(12) = \bullet \quad (1)(2) = \bullet \quad \bullet$	$(12) = \bullet \quad (1)(2) = \bullet \quad \bullet$
fong-spivak-invitation_D12_0736	271	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-271_512_1070_332_422.jpg			

fong-spivak-invitation_D14_0320	272	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-272_172_164_2071_436.jpg			
fong-spivak-invitation_D14_0320	272	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-272_174_162_2071_732.jpg			
fong-spivak-invitation_D14_0320	272	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-272_174_162_2071_1028.jpg			
fong-spivak-invitation_D14_0320	272	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-272_174_162_2071_1323.jpg			
fong-spivak-invitation_D14_0320	273	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-273_286_1202_794_338.jpg			

fong-spivak-invitation_D1_0361	274	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-274_354_558_866_660.jpg			
fong-spivak-invitation_D1_0361	274	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-274_58_64_1661_682.jpg			
fong-spivak-invitation_D1_0361	274	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-274_140_80_1621_852.jpg			
fong-spivak-invitation_D1_0361	274	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-274_218_156_1581_1040.jpg			

fong-spivak-invitation_D12_736	275	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-275_360_252_382_336.jpg			
fong-spivak-invitation_D12_736	275	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-275_458_894_384_642.jpg			
fong-spivak-invitation_D12_736	275	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-275_324_732_1096_572.jpg			
fong-spivak-invitation_D12_736	275	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-275_244_784_1573_548.jpg			
fong-spivak-invitation_D12_736	276	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-276_248_776_370_572.jpg			
fong-spivak-invitation_D12_736	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_152_472_616_724.jpg			
fong-spivak-invitation_D12_736	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_156_472_1044_724.jpg			

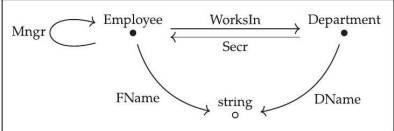
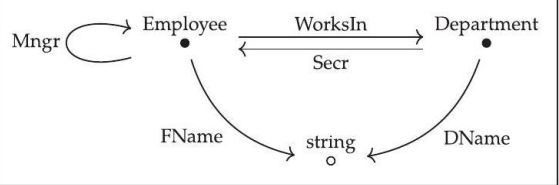
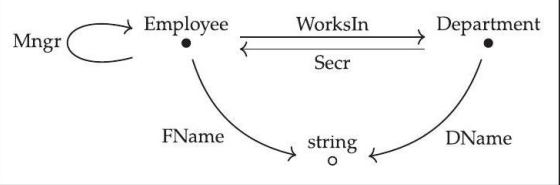
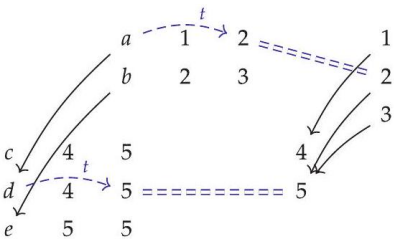
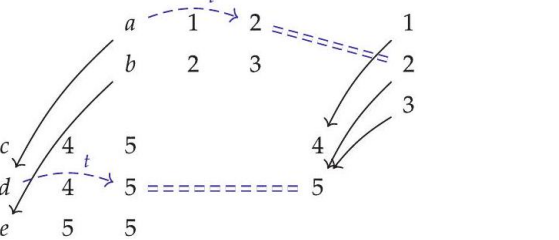
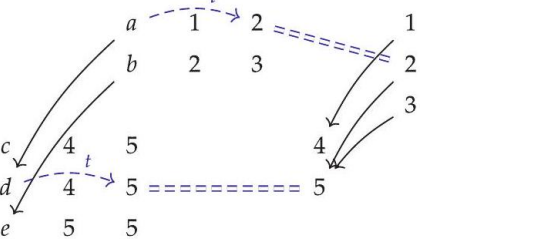
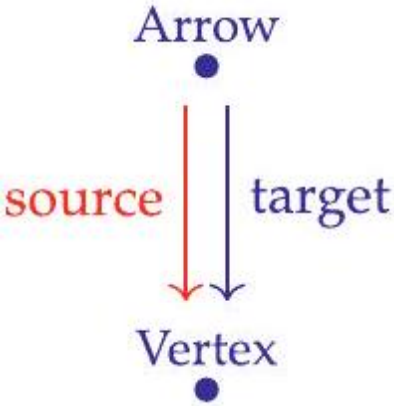
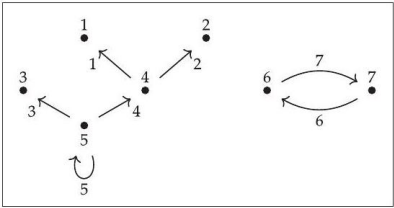
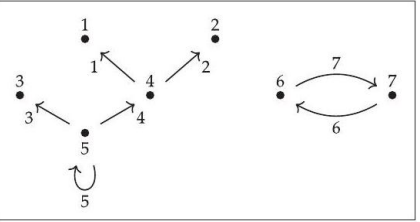
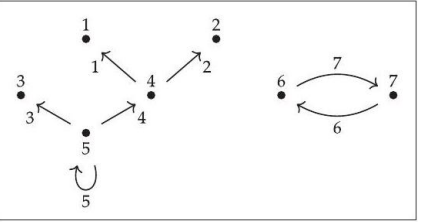
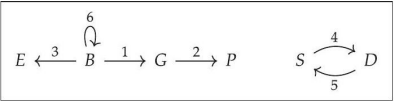
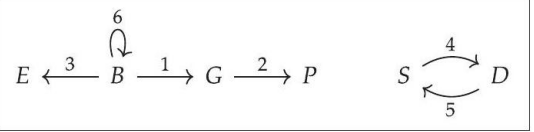
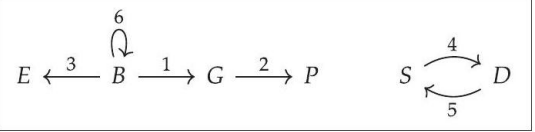
fong-spivak-invitation_D12_438	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_118_200_1877_382.jpg	$c_1 =$ 	$c_1 =$ 	$c_1 =$ 
fong-spivak-invitation_D12_438	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_120_220_1877_668.jpg	$c_2 =$ 	$c_2 =$ 	$c_2 =$ 
fong-spivak-invitation_D12_438	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_118_218_1879_974.jpg	$c_3 =$ 	$c_3 =$ 	$c_3 =$ 
fong-spivak-invitation_D12_438	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_116_220_1881_1279.jpg	$c_4 =$ 	$c_4 =$ 	$c_4 =$ 
fong-spivak-invitation_D12_438	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_116_210_2111_376.jpg	$g_!(c_1) =$ 	$g_!(c_1) =$ 	$g_!(c_1) =$ 
fong-spivak-invitation_D12_438	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_116_222_2113_670.jpg	$g_!(c_2) =$ 	$g_!(c_2) =$ 	$g_!(c_2) =$ 
fong-spivak-invitation_D12_438	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_116_220_2113_976.jpg	$g_!(c_3) =$ 	$g_!(c_3) =$ 	$g_!(c_3) =$ 
fong-spivak-invitation_D12_438	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_116_222_2113_1281.jpg	$g_!(c_4) =$ 	$g_!(c_4) =$ 	$g_!(c_4) =$ 

fong-spivak-invitation_D14_#382	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_118_128_426_456.jpg			
fong-spivak-invitation_D14_#382	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_116_126_428_666.jpg			
fong-spivak-invitation_D14_#382	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_116_126_428_876.jpg			
fong-spivak-invitation_D14_#382	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_116_128_430_1084.jpg			
fong-spivak-invitation_D14_#382	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_114_124_430_1297.jpg			

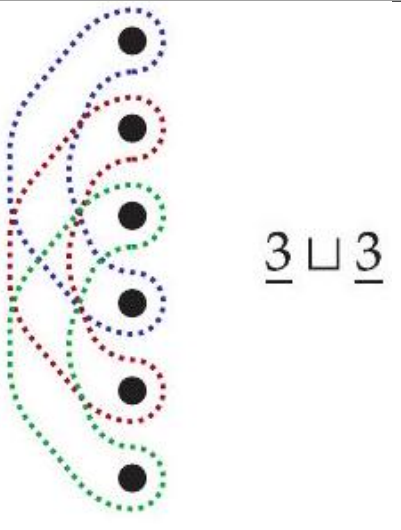
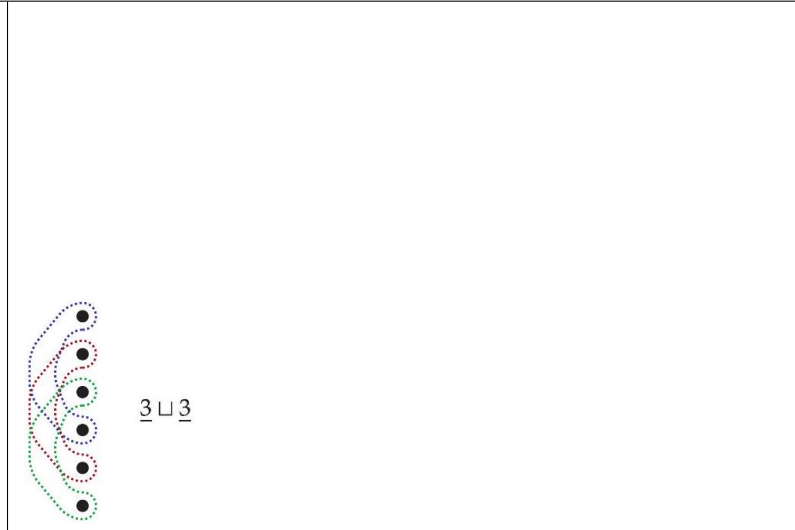
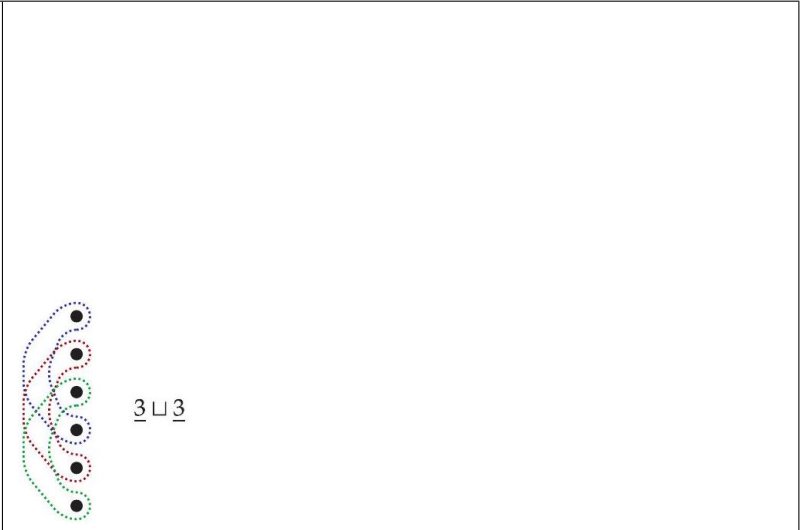
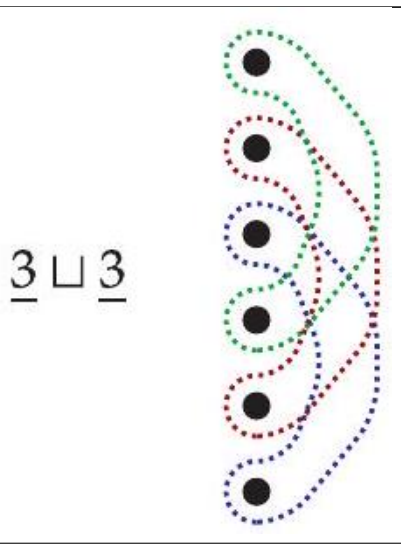
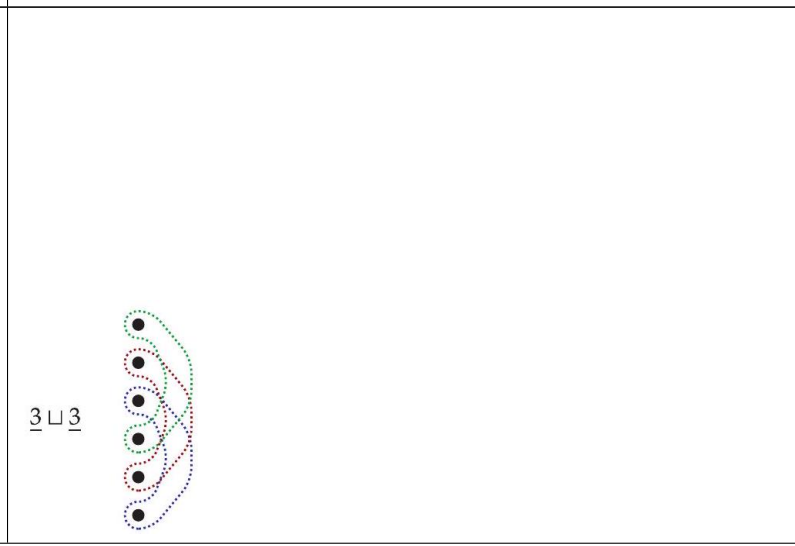
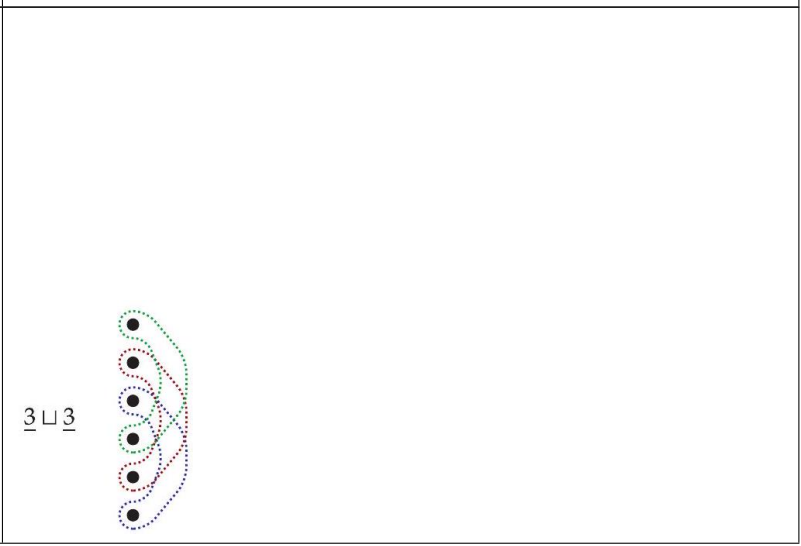
fong-spivak-invitation_D12_#388	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_114_206_804_644.jpg	$c =$	$c =$	$c =$
fong-spivak-invitation_D12_#389	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_116_230_804_1044.jpg	$g_!(c) =$	$g_!(c) =$	$g_!(c) =$
fong-spivak-invitation_D12_#390	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_112_174_1056_872.jpg	$d =$	$d =$	$d =$
fong-spivak-invitation_D12_#391	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_114_186_1309_866.jpg	$e =$	$e =$	$e =$
fong-spivak-invitation_D12_#392	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_114_232_1539_668.jpg	$g^*(d) =$	$g^*(d) =$	$g^*(d) =$
fong-spivak-invitation_D12_#393	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_116_250_1539_998.jpg	$g^*(e) =$	$g^*(e) =$	$g^*(e) =$
fong-spivak-invitation_D12_#394	280	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-280_320_788_1156_548.jpg	$P :=$ $\xleftrightarrow{g} \xleftrightarrow{f} =: Q$	$P :=$ $\xleftrightarrow{g} \xleftrightarrow{f} =: Q$	$P :=$ $\xleftrightarrow{g} \xleftrightarrow{f} =: Q$
fong-spivak-invitation_D12_#395	281	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-281_290_366_374_754.jpg	$X :=$ $\xrightarrow{f}$ $Y :=$	$X :=$ $\xrightarrow{f}$ $Y :=$	$X :=$ $\xrightarrow{f}$ $Y :=$

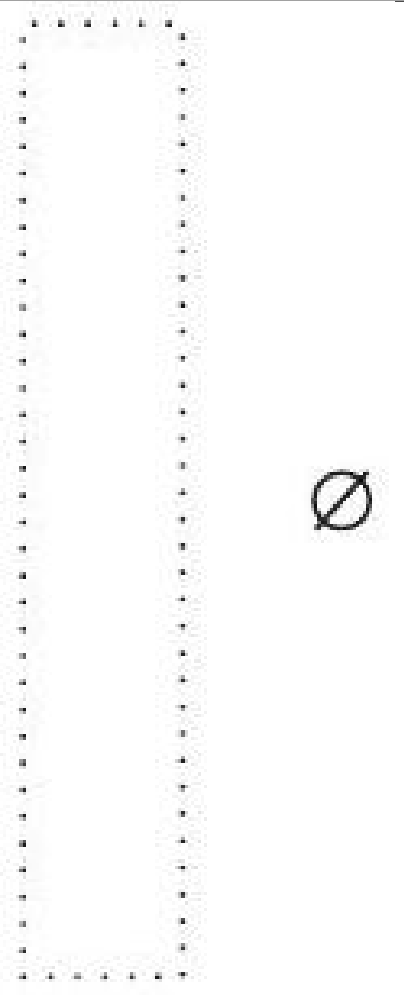
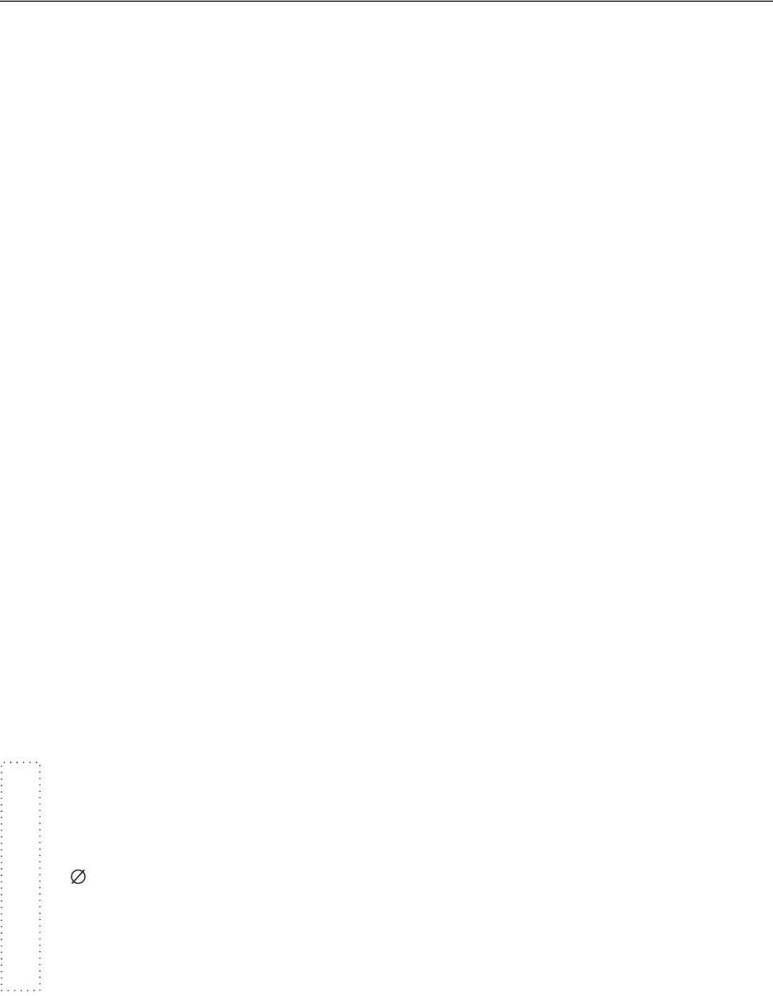
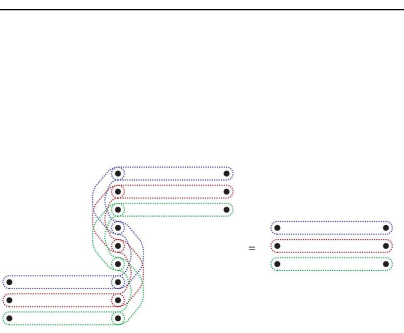
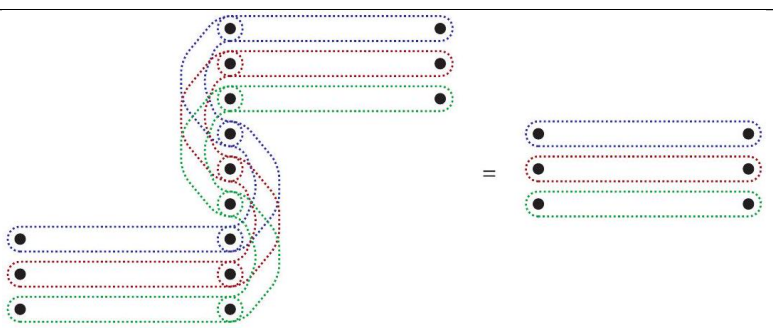
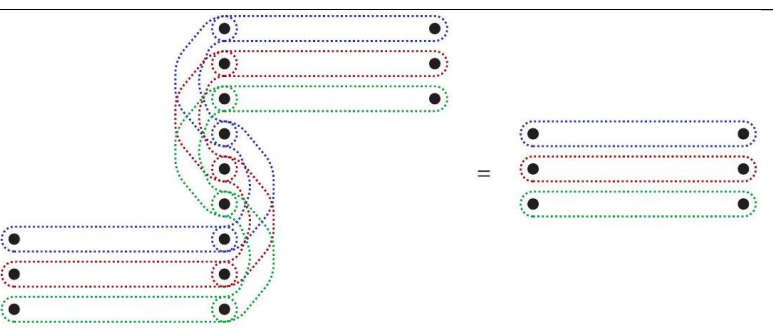
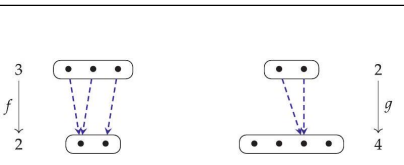
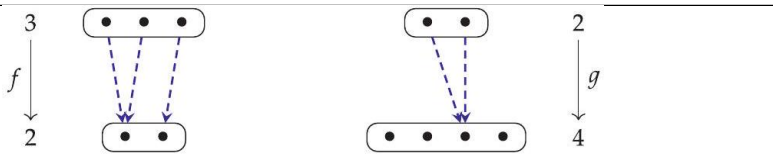
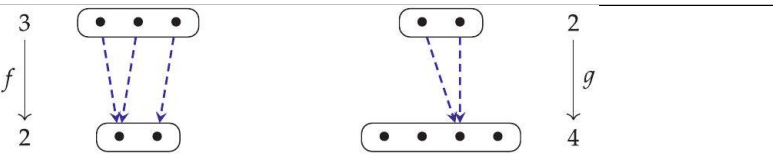
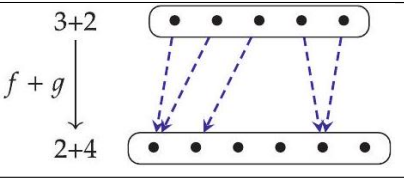
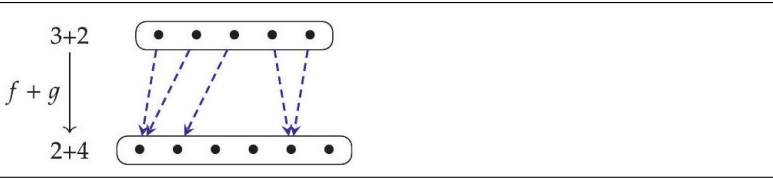
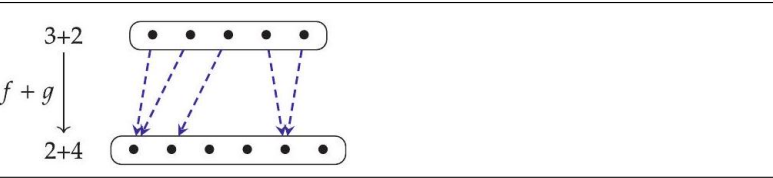
fong-spivak-invitation_D12_084	284	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-281_416_1132_1509_372.jpg			
fong-spivak-invitation_D12_085	285	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-285_326_548_1627_684.jpg			
fong-spivak-invitation_D12_086	286	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-286_228_314_718_804.jpg			
fong-spivak-invitation_D12_086	286	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-286_94_382_1757_750.jpg			
fong-spivak-invitation_D12_088	291	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-291_274_428_480_722.jpg			
fong-spivak-invitation_D12_092	292	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-292_306_258_1797_336.jpg			

fong-spivak-invitation_D14_292	292	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-292_306_258_1797_810.jpg			
fong-spivak-invitation_D14_292	292	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-292_306_258_1797_1287.jpg			
fong-spivak-invitation_D14_294	294	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-294_206_138_386_386.jpg			
fong-spivak-invitation_D14_294	294	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-294_176_304_404_804.jpg			
fong-spivak-invitation_D14_294	294	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-294_124_418_884_752.jpg			

fong-spivak-invitation_D12_294	294	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-294_188_448_2013_736.jpg	$\begin{array}{ccccc} F(c) & \xrightarrow{\alpha_c} & G(c) & \xrightarrow{\beta_c} & H(c) \\ F(f)\downarrow & & G(f)\downarrow & & H(f)\downarrow \\ F(c') & \xrightarrow{\alpha_c} & G(c') & \xrightarrow{\beta_c} & H(c') \end{array}$	$\begin{array}{ccccc} F(c) & \xrightarrow{\alpha_c} & G(c) & \xrightarrow{\beta_c} & H(c) \\ F(f)\downarrow & & G(f)\downarrow & & H(f)\downarrow \\ F(c') & \xrightarrow{\alpha_c} & G(c') & \xrightarrow{\beta_c} & H(c') \end{array}$	$\begin{array}{ccccc} F(c) & \xrightarrow{\alpha_c} & G(c) & \xrightarrow{\beta_c} & H(c) \\ F(f)\downarrow & & G(f)\downarrow & & H(f)\downarrow \\ F(c') & \xrightarrow{\alpha_c} & G(c') & \xrightarrow{\beta_c} & H(c') \end{array}$
fong-spivak-invitation_D12_295	295	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-295_192_292_398_812.jpg	$\begin{array}{ccc} F(c) & \xrightarrow{\text{id}_{F(c)}} & F(c) \\ F(f)\downarrow & & \downarrow F(f) \\ F(c') & \xrightarrow{\text{id}_{F(c')}} & F(c') \end{array}$	$\begin{array}{ccc} F(c) & \xrightarrow{\text{id}_{F(c)}} & F(c) \\ F(f)\downarrow & & \downarrow F(f) \\ F(c') & \xrightarrow{\text{id}_{F(c')}} & F(c') \end{array}$	$\begin{array}{ccc} F(c) & \xrightarrow{\text{id}_{F(c)}} & F(c) \\ F(f)\downarrow & & \downarrow F(f) \\ F(c') & \xrightarrow{\text{id}_{F(c')}} & F(c') \end{array}$
fong-spivak-invitation_D12_295	295	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-295_252_734_1569_570.jpg			
fong-spivak-invitation_D12_296	296	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-296_328_550_788_686.jpg			
fong-spivak-invitation_D12_296	296	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-296_214_198_1515_552.jpg			
fong-spivak-invitation_D12_296	296	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-296_300_548_1909_996.jpg			
fong-spivak-invitation_D12_297	297	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-297_180_700_1741_590.jpg			

fong-spivak-invitation_D12_299.jpg	299	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-299_372_672_1325_622.jpg			
fong-spivak-invitation_D12_302.jpg	302	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-302_326_852_414_514.jpg			
fong-spivak-invitation_D12_303.jpg	303	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-303_280_662_1024_608.jpg			
fong-spivak-invitation_D12_304.jpg	304	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-304_310_124_1523_692.jpg			

fong-spivak-invitation_D12_044	304	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-304_296_222_1533_1004.jpg			
fong-spivak-invitation_D12_044	304	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-304_302_226_1937_694.jpg			

fong-spivak-invitation_D12_044	304	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-304_308_122_1937_1106.jpg			
fong-spivak-invitation_D12_045	305	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-305_446_1074_338_422.jpg			
fong-spivak-invitation_D12_046	306	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-306_196_784_1345_568.jpg			
fong-spivak-invitation_D12_046	306	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-306_192_460_1637_732.jpg			

fong-spivak-invitation_D12_046	306	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-306_192_242_2051_410.jpg			
fong-spivak-invitation_D12_046	306	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-306_192_532_2051_976.jpg			
fong-spivak-invitation_D12_046	307	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-307_194_544_346_688.jpg			
fong-spivak-invitation_D12_046	308	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-308_166_560_1719_388.jpg			
fong-spivak-invitation_D12_046	308	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-308_146_494_1737_1000.jpg			
fong-spivak-invitation_D12_046	308	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-308_170_928_1997_476.jpg			
fong-spivak-invitation_D12_046	309	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-309_136_262_518_546.jpg			
fong-spivak-invitation_D12_046	309	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-309_234_278_420_1052.jpg			
fong-spivak-invitation_D12_046	311	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-311_186_648_898_614.jpg			
fong-spivak-invitation_D12_046	312	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-312_202_630_376_626.jpg			

fong-spivak-invitation_D10_142	312	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-312_82_184_1162_582.jpg			
fong-spivak-invitation_D10_142	312	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-312_84_182_1321_584.jpg			
fong-spivak-invitation_D10_142	312	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-312_144_424_1617_752.jpg	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$
fong-spivak-invitation_D10_142	312	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-312_182_446_1815_742.jpg	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$
fong-spivak-invitation_D10_142	312	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-312_86_144_2075_544.jpg	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$
fong-spivak-invitation_D10_142	312	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-312_268_532_1977_848.jpg			
fong-spivak-invitation_D10_143	313	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-313_140_514_1649_730.jpg	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$
fong-spivak-invitation_D10_143	313	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-313_106_528_1847_724.jpg	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$
fong-spivak-invitation_D10_143	313	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-313_154_912_2091_528.jpg	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \sim$	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \sim$	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \sim$
fong-spivak-invitation_D10_144	314	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-314_156_1006_420_484.jpg			
fong-spivak-invitation_D10_144	314	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-314_70_876_650_548.jpg			

fong-spivak-invitation_D19_144b	314	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-314_544_954_800_510.jpg			
fong-spivak-invitation_D19_144b	314	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-314_92_274_1573_374.jpg			
fong-spivak-invitation_D19_144b	314	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-314_192_800_1521_750.jpg			
fong-spivak-invitation_D19_144b	314	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-314_42_342_1887_790.jpg			
fong-spivak-invitation_D19_144b	314	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-314_98_148_2147_888.jpg			
fong-spivak-invitation_D19_144b	315	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-315_184_388_740_400.jpg			
fong-spivak-invitation_D19_144b	315	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-315_182_320_742_864.jpg			
fong-spivak-invitation_D19_144b	315	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-315_152_248_760_1269.jpg			
fong-spivak-invitation_D19_144b	315	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-315_184_380_1056_404.jpg			

fong-spivak-invitation_D12_145	315	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-315_182_322_1058_860.jpg			
fong-spivak-invitation_D12_145	315	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-315_152_248_1076_1265.jpg			
fong-spivak-invitation_D12_145	317	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-317_74_94_1443_1028.jpg			
fong-spivak-invitation_D12_145	317	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-317_72_114_1445_1166.jpg			
fong-spivak-invitation_D12_145	317	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-317_76_110_1445_1327.jpg			
fong-spivak-invitation_D12_145	319	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-319_306_394_978_760.jpg			
fong-spivak-invitation_D12_145	319	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-319_194_316_1455_800.jpg			

fong-spivak-invitation_010_240	320	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-320_284_598_724_662.jpg			
fong-spivak-invitation_010_241	321	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-321_524_616_436_630.jpg			
fong-spivak-invitation_010_241	321	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-321_196_244_1164_816.jpg			
fong-spivak-invitation_010_242	321	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-321_172_180_1909_870.jpg			
fong-spivak-invitation_010_242	322	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-322_192_230_410_588.jpg			

fong-spivak-invitation_D19_2466	322	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-322_198_234_406_1060.jpg			
fong-spivak-invitation_D19_2466	322	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-322_214_256_756_812.jpg			
fong-spivak-invitation_D19_2466	322	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-322_480_368_1703_756.jpg			
fong-spivak-invitation_D19_2466	323	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-323_300_1221_1144_328.jpg			
fong-spivak-invitation_D19_2466	323	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-323_172_274_1739_802.jpg			
fong-spivak-invitation_D19_2466	323	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-323_172_266_2031_806.jpg			

fong-spivak-invitation_D19_2476	324	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-324_242_310_470_786.jpg			
fong-spivak-invitation_D19_2476	324	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-324_92_352_984_764.jpg			
fong-spivak-invitation_D19_2476	325	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-325_94_420_766_728.jpg			
fong-spivak-invitation_D19_2476	325	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-325_68_618_1084_630.jpg			
fong-spivak-invitation_D19_2476	326	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-326_410_932_492_476.jpg			
fong-spivak-invitation_D19_2476	326	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-326_412_970_1016_456.jpg			
fong-spivak-invitation_D19_2476	326	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-326_176_828_1641_548.jpg			
fong-spivak-invitation_D19_2476	326	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-326_192_836_2057_544.jpg			
fong-spivak-invitation_D19_2476	327	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-327_210_922_358_498.jpg			

fong-spivak-invitation_D19_2488	328	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-328_234_264_2011_808.jpg			
fong-spivak-invitation_D19_2489	329	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-329_180_184_764_868.jpg			
fong-spivak-invitation_D19_2490	329	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-329_180_156_1249_880.jpg			
fong-spivak-invitation_D19_2491	329	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-329_280_334_1773_486.jpg			
fong-spivak-invitation_D19_2492	329	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-329_292_334_1761_1058.jpg			

fong-spivak-invitation_010_046	334	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-334_466_452_1499_464.jpg			
fong-spivak-invitation_010_046	334	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-334_424_228_1541_1241.jpg			
fong-spivak-invitation_010_046	335	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-335_230_1118_1721_378.jpg			